Supplementary Material: Localized Matched-Subspace Detection for Functional Ultrasound and Ultrafast Doppler Imaging

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Supplementary material. This document contains appendices for the main paper (derivations, diagnostic panels, and reproduction commands).

References

[1] Imbault, M., et al. Intraoperative functional ultrasound imaging of human brain activity. *Scientific Reports* 7, 7304 (2017).

[2] Macé, E., Montaldo, G., Pernot, M., Cohen, I., and Tanter, M. Functional ultrasound imaging of the brain. *NeuroImage* 54(4), 2720–2731 (2011).

[3] Urban, A., et al. In vivo whole brain microvascular imaging in mice using ultrafast ultrasound. *EBioMedicine* 76, 103861 (2022).

[4] Alexandrov, A. V., et al. Velocity measurements in the middle cerebral arteries of healthy adults by transcranial Doppler. *Journal of Neuroimaging* 1, 166–172 (1991).

[5] Alexandrov, A. V., et al. Optimal values of flow velocity on transcranial Doppler in patients with intracranial arterial stenosis. *Stroke* 33, 1757–1761 (2002).

[6] Macé, E., et al. Functional ultrasound imaging of the brain: theory and basic principles. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 60(3), 492–506 (2013).

[7] Rabut, C., et al. Functional ultrasound imaging of human brain activity through the skull. *Science Translational Medicine* 12(572), eaav4392 (2020).

[8] Rabut, C., Norman, S. L., Griggs, W. S., Russin, J. J., Jann, K., et al. Functional ultrasound imaging of human brain activity through an acoustically transparent cranial window. *Science Translational Medicine* 16, eadj3143 (2024).

[9] Deffieux, T., Demené, C., and Tanter, M. Functional ultrasound imaging: A new imaging modality for neuroscience. *Neuroscience* 474, 110–121 (2021).

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[10] Demeulenaere, O., Bertolo, A., Pezet, S., Osmanski, B.-F., Tanter, M., and Deffieux, T. In vivo whole brain microvascular imaging in mice using transcranial 3D ultrasound localization microscopy. *EBioMedicine* 79, 103995 (2022).

[11] Demené, C., et al. Spatiotemporal clutter filtering of ultrafast ultrasound data highly increases Doppler and fUltrasound sensitivity. *IEEE Transactions on Medical Imaging* (2015). DOI: `https://doi.org/10.1109/TMI.2015.2428634`.

[12] Baranger, J., et al. Adaptive spatiotemporal SVD clutter filtering for ultrafast Doppler imaging using similarity of spatial singular vectors. *IEEE Transactions on Medical Imaging* (2018). DOI: `https://doi.org/10.1109/TMI.2018.2789499`.

[13] Treeby, B. E., and Cox, B. T. k-Wave: MATLAB toolbox for the simulation and reconstruction of photoacoustic wave fields. *Journal of Biomedical Optics* 15(2), 021314 (2010). DOI: `https://doi.org/10.1117/1.3360308`.

[14] Yu, A. C. H., and Løvstakken, L. Eigen-based clutter filter design for ultrasound color flow imaging: a review. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2010). DOI: `https://doi.org/10.1109/TUFFC.2010.1521`.

[15] Song, P., et al. Ultrasound small vessel imaging with block-wise adaptive local clutter filtering. *IEEE Transactions on Medical Imaging* (2017). DOI: `https://doi.org/10.1109/TMI.2016.2605819`.

[16] Kasai, C., Namekawa, K., Koyano, A., and Omoto, R. Real-time two-dimensional blood flow imaging using an autocorrelation technique. *IEEE Transactions on Sonics and Ultrasonics* 32(3), 458–464 (1985). DOI: `https://doi.org/10.1109/T-SU.1985.31615`.

[17] Chee, A. J. Y., Yiu, B. Y. S., and Yu, A. C. H. A GPU-parallelized eigen-based clutter filter framework for ultrasound color flow imaging. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2017). DOI: `https://doi.org/10.1109/TUFFC.2016.2606598`.

[18] Lok, U.-W., Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Real time SVD-based clutter filtering using randomized SVD and spatial downsampling for microvessel imaging. *Ultrasonics* 107, 106163 (2020). DOI: `https://doi.org/10.1016/j.ultras.2020.106163`.

[19] Demene, C., et al. Improved color Doppler for cerebral blood flow axial velocity estimation (iCD-fUS). *IEEE Transactions on Medical Imaging* 40(5), 1498–1510 (2021).

[20] Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Improved color Doppler for cerebral blood flow axial velocity imaging. *IEEE Transactions on Medical Imaging* 40(2), 758–764 (2021).

[21] Demené, C., et al. Functional ultrasound imaging of brain activity in human newborns. *Science Translational Medicine* 9(411), eaah6756 (2017). DOI: `https://doi.org/10.1126/scitranslmed.aah6756`.

[22] Baranger, J., et al. Bedside functional monitoring of the dynamic brain connectivity in human neonates. *Nature Communications* 12, 1080 (2021). DOI: `https://doi.org/10.1038/s41467-021-21387-x`.

[23] Soloukey, S., et al. Mobile human brain imaging using functional ultrasound. *Science Advances* 11(25), eadu9133 (2025). DOI: https://doi.org/10.1126/sciadv.adu9133.

[24] Lambert, T., Brunner, C., Urban, A., et al. Functional ultrasound imaging and neuronal activity: How accurate is the spatiotemporal match? *Imaging Neuroscience* 3, IMAG.a.34 (2025). DOI: https://doi.org/10.1162/IMAG.a.34.

[25] Edelman, B. J., et al. The COMBO window: A chronic cranial implant for multiscale circuit interrogation in mice. *PLOS Biology* 22(6), e3002664 (2024). DOI: https://doi.org/10.1371/journal.pbio.3002664.

[26] Chen, Z., et al. Functional ultrasound imaging of auditory responses in comatose patients. *Research* 8, 0709 (2025). DOI: https://doi.org/10.34133/research.0709.

[27] Nunez-Elizalde, A. O., et al. Neural correlates of blood flow measured by ultrasound. *Neuron* 110(10), 1631–1640.e4 (2022). DOI: https://doi.org/10.1016/j.neuron.2022.02.012.

[28] Aaslid, R., Markwalder, T.-M., and Nornes, H. Noninvasive transcranial Doppler ultrasound recording of flow velocity in basal cerebral arteries. *Journal of Neurosurgery* 57(6), 769–774 (1982).

[29] Gao, S., Lam, W. W.-K., Chan, Y.-H., Liu, J., Wong, K.-P., Leung, T. W.-S., et al. Optimal values of flow velocity on transcranial Doppler in grading middle cerebral artery stenosis in comparison with magnetic resonance angiography. *Journal of Neuroimaging* 12(3), 213–218 (2002).

[30] Estrada, H., Gottschalk, S., Reiss, M., Neuschmelting, V., Razansky, D., et al. Observation of guided acoustic waves in a human skull. *Ultrasound in Medicine & Biology* 44(7), 1438–1447 (2018).

[31] Candès, E. J., Li, X., Ma, Y., and Wright, J. Robust principal component analysis? *Journal of the ACM* 58(3), 11 (2011).

[32] Solomon, O., et al. Deep unfolded robust PCA with application to clutter suppression in ultrasound. *IEEE Transactions on Medical Imaging* 39(4), 1051–1063 (2020). DOI: https://doi.org/10.1109/TMI.2019.2941271.

[33] Lin, Z., Chen, M., and Ma, Y. The augmented Lagrange multiplier method for exact recovery of corrupted low-rank matrices. *arXiv preprint* arXiv:1009.5055 (2010).

[34] Halko, N., Martinsson, P. G., and Tropp, J. A. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review* 53(2), 217–288 (2011).

[35] Soloukey, S., Vincent, A. J. P. E., Smits, M., Slump, C. H., Dirven, C. M. F., et al. Functional ultrasound (fUS) during awake brain surgery: the clinical potential for intra-operative functional and vascular brain mapping. *Frontiers in Neuroscience* 13, 1384 (2019). DOI: https://doi.org/10.3389/fnins.2019.01384.

[36] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. Zenodo dataset, DOI 10.5281/zenodo.10711806 (2024).

[37] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. *Nature Communications* 15, 3299 (2024). DOI: `https://doi.org/10.1038/s41467-024-47154-2`.

[38] Chavignon, A., Heiles, B., Hingot, V., Lopez, P., Teston, E., and Couture, O. In vivo rat brain for Ultrasound Localization Microscopy: raw and beamformed data. Zenodo dataset, DOI 10.5281/zenodo.7883227 (2023).

[39] Heiles, B., Chavignon, A., Hingot, V., Lopez, P., Teston, E., and Couture, O. Performance benchmarking of microbubble-localization algorithms for ultrasound localization microscopy. *Nature Biomedical Engineering* (2021). DOI: `https://doi.org/10.1038/s41551-021-00824-8`.

[40] Errico, C., et al. Ultrafast ultrasound localization microscopy for deep super-resolution vascular imaging. *Nature* 527(7579), 499–502 (2015). DOI: `https://doi.org/10.1038/nature16066`.

[41] Hingot, V., et al. Subwavelength motion-correction for ultrafast ultrasound localization microscopy. *Ultrasonics* 77, 17–21 (2017). DOI: `https://doi.org/10.1016/j.ultras.2017.01.008`.

[42] Insana, M. F., Dai, B., Babaei, S., and Abbey, C. K. Combining spatial registration with clutter filtering for power-Doppler imaging in peripheral perfusion applications. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 69(12), 3243–3254 (2022). DOI: `https://doi.org/10.1109/TUFFC.2022.3211469`.

[43] Leonov, D. V., Reshetnikov, R. V., Kulberg, N. S., Nasibullina, A. A., and Gromov, A. I. Doppler Twinkling Artifact Observations: an Open-Access Database of Raw Ultrasonic Signals. *Digital Diagnostics* (2021). DOI: `https://doi.org/10.17816/DD76511`.

[44] Ozgun, K., and Byram, B. C. Multidimensional clutter filtering of aperture domain data for improved blood flow sensitivity. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 68(8), 2645–2656 (2021). DOI: `https://doi.org/10.1109/TUFFC.2021.3073292`.

[45] Huang, Z., et al. Adaptive higher-order singular value decomposition clutter filter for ultrafast Doppler imaging of coronary flow under non-negligible tissue motion. *Ultrasonics* (2024). DOI: `https://doi.org/10.1016/j.ultras.2024.107307`.

[46] Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part I: theory and examples. *Computer Methods and Programs in Biomedicine* 218, 106726 (2022). DOI: `https://doi.org/10.1016/j.cmpb.2022.106726`.

[47] Cigier, A., Varray, F., and Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part II: comparison with four simulators. *Computer Methods and Programs in Biomedicine* 220, 106774 (2022). DOI: `https://doi.org/10.1016/j.cmpb.2022.106774`.

[48] Urban, A., et al. Real-time imaging of brain activity in freely moving rodents using functional ultrasound. *Nature Methods* 12(9), 873–878 (2015). DOI: `https://doi.org/10.1038/nmeth.3482`.

[49] Demené, C., et al. 4D microvascular imaging based on ultrafast Doppler tomography. *NeuroImage* 127, 472–483 (2016). DOI: `https://doi.org/10.1016/j.neuroimage.2015.11.014`.

[50] Slepian, D. Prolate spheroidal wave functions, Fourier analysis, and uncertainty—V: The discrete case. *Bell System Technical Journal* 57(5), 1371–1430 (1978). DOI: `https://doi.org/10.1002/j.1538-7305.1978.tb02104.x`.

[51] Thomson, D. J. Spectrum estimation and harmonic analysis. *Proceedings of the IEEE* 70(9), 1055–1096 (1982). DOI: `https://doi.org/10.1109/PROC.1982.12433`.

[52] Percival, D. B., and Walden, A. T. *Spectral Analysis for Physical Applications: Multitaper and Conventional Univariate Techniques*. Cambridge University Press (1993).

[53] Guerci, J. R. Artech House monograph (2003).

[54] Wicks, M. C., Rangaswamy, M., Adve, R., and Hale, T. B. *IEEE Signal Processing Magazine* 23(1), 51–65 (2006). DOI: `https://doi.org/10.1109/MSP.2006.1593337`.

[55] Melvin, W. L., and Guerci, J. R. *IEEE Transactions on Aerospace and Electronic Systems* 42(3), 983–996 (2006). DOI: `https://doi.org/10.1109/TAES.2006.248215`.

[56] Gurram, P. R., and Goodman, N. A. Spectral-domain covariance estimation with a priori knowledge. *IEEE Transactions on Aerospace and Electronic Systems* 42, 1010–1020 (2006). DOI: `https://doi.org/10.1109/TAES.2006.248209`.

[57] Van Veen, B. D., and Buckley, K. M. Beamforming: a versatile approach to spatial filtering. *IEEE ASSP Magazine* 5(2), 4–24 (1988). DOI: `https://doi.org/10.1109/53.665`.

[58] Kuglin, C. D., and Hines, D. C. The phase correlation image alignment method. In *Proceedings of the IEEE International Conference on Cybernetics and Society* (1975).

[59] Stone, H. S., Orchard, M. T., Chang, E.-C., and Martucci, S. A. A fast direct Fourier-based algorithm for subpixel registration of images. *IEEE Transactions on Geoscience and Remote Sensing* 39(10), 2235–2243 (2001). DOI: `https://doi.org/10.1109/36.957286`.

[60] Foroosh, H., Zerubia, J. B., and Berthod, M. Extension of phase correlation to subpixel registration. *IEEE Transactions on Image Processing* 11(3), 188–200 (2002). DOI: `https://doi.org/10.1109/83.988953`.

[61] Capon, J. High-resolution frequency-wavenumber spectrum analysis. *Proceedings of the IEEE* 57(8), 1408–1418 (1969).

[62] Frost, O. L. An algorithm for linearly constrained adaptive array processing. *Proceedings of the IEEE* 60(8), 926–935 (1972).

[63] Kay, S. M. *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory.* Prentice Hall (1998).

[64] Scharf, L. L., and Friedlander, B. Matched subspace detectors. *IEEE Transactions on Signal Processing* 42(8), 2146–2157 (1994).

[65] Ward, J. MIT Lincoln Laboratory, Tech. Rep. 1015 (1994).

A Appendix guide and terminology translation

The appendices collect formal detector derivations, robustness audits, and reproducibility details. Because some parts of the method borrow terminology from radar and detection theory, Table 1 gives a plain-language translation before the formal development. Readers interested mainly in clinical interpretation can use this table as a guide and then read only the appendices most relevant to their question.

Term in the paper	Plain-language meaning in ultrasound terms	Where it is developed
Fully whitened detector	Tile-local slow-time covariance whitening applied before matched-subspace scoring.	Appendix B
Adaptive guard rule	A conservative switch that keeps the simple detector by default and turns on whitening only where guard-band clutter evidence is strong.	Main text Section **??**; Supplementary Material
Shrink-only regularizer	A conservative score penalty map driven by band-energy cues and vessel-proxy protection masks; it can reduce scores but never increase them.	Appendix B; Supplementary Material
Empirically tail-calibrated thresholds	Background-tail threshold setting based on empirical negative-score distributions rather than analytic scanner-time formulas.	Main text Section **??**; Supplementary Material
Safeguards	Automatic checks that disable the optional penalty layer or force conservative behavior when metadata, band geometry, or proxy information are unreliable.	Appendix D

Table 1: Plain-language appendix guide. This table is intended as a reader map for fUS, neuroscience, and clinically oriented readers before the appendices turn to formal notation.

Symbol / name	What it measures	Used for	Data requirement
S_{base}	baseline ROC score map on X_{base} (PD power; higher = more flow-like)	baseline ROC / low-FPR tables	IQ
$S_{\text{det,pre}}$	pre-regularization matched-subspace detector score map	detector ROC / low-FPR tables	IQ + local covariance
$S_{\text{det,post}}$	post-regularization shrink-only score map $wS_{\text{det,pre}}$	regularizer behavior audits	IQ + feature map
$\text{PD}^{\uparrow}$ (display PD), masks	PD display/proxy map and derived masks	visualization and deterministic preprocessing (gating/protected sets)	IQ (or PD-only proxy mode)

Table 2: Detector-score and display-map taxonomy used throughout. ROC score maps are kept separate from display/proxy maps used only for visualization and deterministic preprocessing. Display/proxy maps are never used to compute ROC/TPR/FPR, only to define gating or protected-set masks that are audited separately.

B Shrink-Only Regularizer (Modality–Agnostic)

In this part we describe regime conditions and operator-level sufficient structure for a deterministic shrink-only score-space transform layered on top of a baseline slow-time detector. The analysis is formulated in terms of abstract band projectors (P_o, P_f, P_g, P_a), alias features, and score distributions, and does not depend on any particular imaging modality. Only in later sections do we instantiate these conditions for functional ultrasound (fUS) brain imaging with concrete band choices, acquisition parameters, and baselines.

How to read this appendix

A plain-language glossary for the appendix labels appears first in Appendix A, and the main-text method overview is shown in Figure **??**. This appendix then states the formal preconditions under which the optional shrink-only regularizer is allowed to modify scores; if any precondition fails, the layer is disabled ($w \equiv 1$).

B.1 Shrink-only score mapping used in the main manuscript

The main paper keeps the optional shrink-only regularizer to a brief summary because the primary efficacy claims do not depend on it. For completeness, we collect the core manuscript-level definitions here. The regularizer is driven by deterministic side-information features computed from the same tile-local slow-time vector z_i and Doppler-band projectors (P_o, P_f, P_g, P_a) used by the detector:

$$E_f(i) = z_i^{\mathrm{H}} P_f z_i, \qquad E_g(i) = z_i^{\mathrm{H}} P_g z_i, \qquad E_a(i) = z_i^{\mathrm{H}} P_a z_i,$$

with total in-band energy $E_{\mathrm{tot}}(i) = z_i^{\mathrm{H}} z_i + \epsilon$. The main risk statistic is the alias-to-flow ratio

$$m_{\mathrm{alias}}(i) = \log \frac{E_a(i) + \epsilon}{E_f(i) + \epsilon},$$

and we also track the guard-band fraction

$$r_g(i) = \frac{E_g(i)}{E_{\mathrm{tot}}(i)}.$$

A deterministic flow-support proxy $c_f(i) \in [0, 1]$ defines conservative background-like and flow-like proxy sets

$$\mathcal{T}_{\mathrm{bg}} := \{i : c_f(i) \le c_{\mathrm{bg}}\}, \qquad \mathcal{T}_{\mathrm{flow}} := \{i : c_f(i) \ge c_{\mathrm{flow}}\}.$$

These features are used only to decide whether shrinkage is warranted and where it may be applied; they are not used as class labels for ROC reporting.

The score-space mapping itself is bounded and monotone:

$$S_{\mathrm{det,post}}(i) = w(i)\, S_{\mathrm{det,pre}}(i), \qquad w(i) \in (0, 1].$$

The full construction of $w(i)$, including protected sets, regime logic, deterministic sentinels, and the reciprocal scale-map convention used in some exported audit files, is given by the appendix sections below.

Lemma 1 (Shrink-only monotonicity)**.** *Let $w(i) \in (0,1]$ and define $S_{\text{det,post}}(i) = w(i)\, S_{\text{det,pre}}(i)$. For any threshold τ,*

$$\{S_{\text{det,post}} \geq \tau\} \subseteq \{S_{\text{det,pre}} \geq \tau\}.$$

Consequently, on any subset A (e.g. H_0), the tail probability cannot increase at fixed τ:

$$\mathbb{P}\{S_{\text{det,post}} \geq \tau \mid A\} \leq \mathbb{P}\{S_{\text{det,pre}} \geq \tau \mid A\}.$$

Proof. If $S_{\text{det,post}}(i) \geq \tau$, then $S_{\text{det,pre}}(i) = S_{\text{det,post}}(i)/w(i) \geq \tau$ since $w(i) \leq 1$. □

When reporting strict low-FPR operating points where τ is reselected after regularization to satisfy $\mathbb{P}\{S_{\text{det,post}} \geq \tau \mid H_0\} = \alpha$, any TPR change reflects background-tail compression despite scores never increasing.

B.2 Regime Preconditions (Data–Level, Non–Negotiable)

The most important empirical lesson is: even if the shrink-only operator Q and band design (P_f, P_a) are perfect in an abstract sense, *uplift requires the data themselves to be in a compatible regime.* This section codifies that.

We write $S(f)$ for the (multi–taper) PSD of a baseline–whitened tile, and define

$$E_f := \sum_{f \in \mathcal{B}_f} S(f), \quad E_a := \sum_{f \in \mathcal{B}_a} S(f), \quad E_g := \sum_{f \in \mathcal{B}_g} S(f),$$

with $\mathcal{B}_f, \mathcal{B}_a, \mathcal{B}_g$ the flow, alias, and guard bands in frequency.

B.3 R–1: Band Occupancy Separation

We require that the fixed bands $(\mathcal{B}_f, \mathcal{B}_a)$ are physically meaningful for the current PRF, slow–time length T, and DPSS parameters, in the sense that label statistics show actual occupancy.

On a held–out labeled sample:

$$\mathbb{P}\{f_{\text{peak}} \in \mathcal{B}_f \mid H_1\} \geq p_{f,\min}, \qquad p_{f,\min} \in [0.5, 0.7], \tag{1}$$

$$\mathbb{P}\{f_{\text{peak}} \in \mathcal{B}_a \mid H_0\} \geq p_{a,\min}, \qquad p_{a,\min} \in [0.2, 0.3], \tag{2}$$

where f_{peak} is the frequency at which $S(f)$ attains its maximum and H_1/H_0 denote flow/background tiles.

Interpretation: a majority of flow tiles have their dominant PSD peak *inside* the flow band, and a nontrivial fraction of background tiles have their dominant peak in the alias band. This ensures that the band projectors (P_f, P_a) actually correspond to distinct physical processes.

Implementation in labeled brain simulation profiles. For the main labeled brain simulation regimes, the raw k-Wave RF tensor $x(t, a, e)$ (angle index a, element index e) lives on an angle and element grid that does not align pixelwise with the exported detector masks ($N_y \times N_x$). We therefore implement R–1 using the whitened multi–taper PSD and band-ratio feature statistics computed on the *same* spatial grid as the exported PD scoring maps (the right-tail score S_{PD} stored as `score_pd_*.npy`). We take f_{peak} from the whitened PSD peaks used in the band-ratio recorder and report the fraction of H_1 tiles whose peak lies in $\mathcal{B}_f$ and the fraction of H_0 tiles whose peak lies in $\mathcal{B}_a$ from those tilewise peak statistics. When the RF tensor is spatially aligned with the masks (e.g. 1D pilots), we instead use a direct FFT on $x(t, \cdot, \cdot)$ as in the definition above.

B.4 R–2: Alias Score Class Separation

We generalize the alias metric used for localization.

Alias score. Let $m_{\text{alias}}(z)$ be any scalar score derived from the baseline PSD and/or covariance satisfying:

- m_{alias} increases when E_a grows relative to E_f,
- m_{alias} is larger when Pa energy is more coherent or concentrated (e.g. a strong tone),
- m_{alias} is penalized when guard/edge energy E_g is large.

Examples (not exhaustive):

- Log band ratio: $m_{\text{alias}}(z) = \log \frac{E_a+\epsilon}{E_f+\epsilon}$,
- Combination of band fractions: $m_{\text{alias}}(z) = \frac{E_a}{E_f+E_a+E_g} \cdot c_a$ where c_a is an alias–band coherence metric (e.g. leading eigenvalue fraction of a band–restricted covariance),
- Peak–location based scores (peak in Pa, high tone prominence).

For the shrink-only regularizer to be meaningfully guided by alias evidence, we require *class–differential separation* of m_{alias}:

$$\text{median}(m_{\text{alias}} \mid H_0) - \text{median}(m_{\text{alias}} \mid H_1) \;\geq\; \Delta_{\min}, \tag{3}$$

with $\Delta_{\min}$ set so that alias dominance differs by at least a factor $\approx e^{\Delta_{\min}}$ in expectation (for log ratio, $\Delta_{\min} \approx 0.3$ corresponds to a $\approx 35\%$ relative difference in E_a/E_f medians).

If (3) fails (alias score almost identical on H_0 and H_1), the alias-localization rule (Appendix K.11) has no reliable signal and regularizer uplift cannot be expected.

B.5 R–3: Baseline Headroom

Let S^- be the baseline score distribution on negatives, and $\text{pAUC}_{\text{base}}(\alpha)$ be the partial AUC up to FPR α.

We require:

- Sufficient *quantile density* at the target FPR α (as already assumed implicitly in SL–2): the density $f_-(t_\alpha)$ at the α–quantile t_α is bounded away from 0.
- Baseline not already numerically saturated: for a chosen $\alpha_{\max}$ (e.g. 10^{-3}),

$$\text{pAUC}_{\text{base}}(\alpha \leq \alpha_{\max}) \leq \text{pAUC}_{\max} - \varepsilon_{\text{AUC}}, \qquad \varepsilon_{\text{AUC}} > 0,$$

  where $\text{pAUC}_{\max}$ is the ideal partial AUC under perfect separation.

If the baseline already achieves near–maximal pAUC at the FPR of interest, even a perfect shrink-only transform cannot provide significant gains; in this regime the regularizer can serve as a safety mechanism, but not as a source of measurable uplift.

B.6 Operator–Level Sufficient Structure (Non–Negotiable Core)

This section describes conditions on Q alone under which the shrink-only regularizer behaves like a band–wise scalar rescaling of scores, up to small perturbations.

B.7 OC–1: Approximate Band Commutation

We require Q to be nearly block–diagonal with respect to (P_o, P_f, P_g, P_a):

$$\|[Q, P_f]\|_2 \le \varepsilon_{\text{mix}}, \qquad \|[Q, P_a]\|_2 \le \varepsilon_{\text{mix}},$$

for some small $\varepsilon_{\text{mix}} \in [0, 1)$ (typical target ≤ 0.05).

Interpretation: Q preserves band subspaces up to small mixing; it does not rotate energy from Pf into Pa or vice versa in any substantial way. This rules out the failure mode where the regularizer accidentally feeds alias into the flow numerator.

Runtime label–free check. Let $\widehat{Q} := R_\beta^{-1/2}\widehat{R}^{1/2}$ be the realized shrink-only transform. Estimate

$$\widehat{\varepsilon}_{\text{mix}} := \max\left\{\frac{\left\|P_f\widehat{Q}P_a\right\|_F}{\left\|P_f\widehat{Q}P_f\right\|_F}, \frac{\left\|P_a\widehat{Q}P_f\right\|_F}{\left\|P_a\widehat{Q}P_a\right\|_F}\right\},$$

and require $\widehat{\varepsilon}_{\text{mix}} \le \varepsilon_{\text{mix}}$.

B.8 OC–2: Differential Band Gains with Spectral Bounds

We require that $Q^{\text{H}}Q$ is well approximated by bandwise scalars on the ranges of P_f, P_a, P_o, i.e., there exist $s_f, s_a, s_o > 0$ and tolerances $\delta_f, \delta_a, \delta_o \in [0, 1)$ such that

$$\begin{aligned}
s_f^2(1-\delta_f)\,P_f &\preceq P_f Q^{\text{H}} Q P_f \preceq s_f^2(1+\delta_f)\,P_f,\\
s_a^2(1-\delta_a)\,P_a &\preceq P_a Q^{\text{H}} Q P_a \preceq s_a^2(1+\delta_a)\,P_a,\\
s_o^2(1-\delta_o)\,P_o &\preceq P_o Q^{\text{H}} Q P_o \preceq s_o^2(1+\delta_o)\,P_o,
\end{aligned}$$

with *strict separation*

$$\frac{s_a}{s_f} \ge \Lambda > 1 + c\,\varepsilon_{\text{mix}}, \qquad c \in [2, 4].$$

Interpretation: within each band, $Q^{\text{H}}Q$ behaves like a scalar (s_f^2, s_a^2, s_o^2) up to modest anisotropy $\delta_\bullet$, and the alias band gains are strictly larger than flow band gains.

Runtime label–free check. Compute eigenvalues of the band–projected operators:

$$\lambda_{\min}^f, \lambda_{\max}^f := \lambda_{\min\,/\,\max}(Q^{\text{H}}Q|_{P_f}),$$

and similarly for a, o. Set

$$\widehat{s}_f^2 := \tfrac{1}{2}(\lambda_{\min}^f + \lambda_{\max}^f), \quad \widehat{\delta}_f := \frac{\lambda_{\max}^f - \lambda_{\min}^f}{\lambda_{\max}^f + \lambda_{\min}^f},$$

and analogously for $\widehat{s}_a, \widehat{s}_o$ and $\widehat{\delta}_a, \widehat{\delta}_o$. Require $\widehat{\delta}_\bullet \le \delta_\bullet$ and $\widehat{s}_a/\widehat{s}_f \ge \Lambda$.

B.9 Consequence: Approximate Multiplicative Effect on Scores

Under OC–1–OC–2, for any z one has

$$\mathrm{BR}(Qz) = \frac{z^{\mathrm{H}} Q^{\mathrm{H}} P_f Q z}{z^{\mathrm{H}} Q^{\mathrm{H}} P_a Q z} \approx \frac{s_f^2}{s_a^2} \cdot \mathrm{BR}(z) \cdot (1 + \Delta_{\mathrm{BR}}(z))$$

with $|\Delta_{\mathrm{BR}}(z)| \le C_1(\delta_f + \delta_a) + C_2 \varepsilon_{\mathrm{mix}}$, for constants C_1, C_2 depending only on band dimensions. A similar approximation holds for $\mathrm{GLRT}_\rho(z)$ with denominators involving s_a^2, s_o^2.

Thus, if $\Lambda = s_a/s_f > 1$ and tolerances are small, BR/GLRT on a tile are rescaled approximately by a *bandwise constant* factor, with predictable sign (e.g. downwards when $s_a > s_f$).

B.10 Band Geometry & Leakage Budget

This section remains as in the original doc: its role is to ensure that the projectors P_f, P_a correspond to well–separated spectral bands given the PRF, T, and DPSS parameters, using DPSS tapers and multi-taper spectral estimates in the sense of Slepian and Thomson [50–52].

B.11 BG–1: Leakage Separation Margin

Let $G(f, g)$ be the multi–taper spectral leakage kernel associated with your DPSS design (known from design). Require

$$\max_{f \in \mathcal{B}_f} \sum_{g \in \mathcal{B}_a} G(f, g) \le \ell_{\max}, \qquad \ell_{\max} \le 0.15.$$

Interpretation: at most 15% of energy from an on–grid flow bin leaks into the alias band, and vice versa. This supports keeping the band mixing in OC–1 small after whitening.

B.12 BG–2: Band–Edge Guard

Define a guard band $\mathcal{B}_g$ separating $\mathcal{B}_f$ and $\mathcal{B}_a$ such that

$$\mathrm{dist}(\mathcal{B}_f, \mathcal{B}_a) \ge b_g,$$

for some guard width b_g in frequency (or bins). Require that any energy within b_g of $\mathcal{B}_f$ leaks only modestly into $\mathcal{B}_a$:

$$\sum_{g:\, \mathrm{dist}(g, \mathcal{B}_f) \le b_g} G(f, g) \le \ell_{\mathrm{edge}}, \qquad \ell_{\mathrm{edge}} \le 0.1.$$

Interpretation: avoid designing $\mathcal{B}_f, \mathcal{B}_a$ so tightly that off–grid peaks straddle the boundary; the guard ensures band–edge leakage is limited.

B.13 BG–3: Within–Band Condition Numbers

From the baseline whitened covariance in each band, require

$$\kappa_f := \frac{\lambda_{\max}(\widehat{R}|_{P_f})}{\lambda_{\min}(\widehat{R}|_{P_f})} \le \kappa_f^{\max}, \qquad \kappa_a := \frac{\lambda_{\max}(\widehat{R}|_{P_a})}{\lambda_{\min}(\widehat{R}|_{P_a})} \le \kappa_a^{\max},$$

with $\kappa^{\max} \in [10, 40]$ depending on T and sample support.

Interpretation: this makes it feasible to construct R_β that achieves OC–2 with moderate β, δ without overly crushing the flow band.

B.14 Covariance blending and localization (summary)

The shrink-only regularizer reshapes covariances only within bands. We use either a commuting band-scalar transform with target gains (s_f, s_a, s_o) or a trace-normalized blend R_β with small diagonal lift; both enforce bandwise structure (no off-block priors). Localization restricts the regularizer to alias-dominated, low-coverage tiles using alias scores, guard ratios, optional flow coherence/coverage checks, baseline score windows, depth/PD/registration cues, and is validated via coverage rates p_-, p_+. Full localization definitions are provided in Appendix K.11 (PD-only instantiation in Appendix H).

B.15 Score–Level Guarantees

This section is unchanged in structure; we only make the dependence on p_-, p_+ explicit.

B.16 SL–1: Per–Tile Multiplicative Effect

Under OC–1–OC–2, for any z,

$$\mathrm{BR}_{\mathrm{reg}}(z) = \mathrm{BR}(Qz) \approx \frac{s_f^2}{s_a^2} \cdot \mathrm{BR}_{\mathrm{base}}(z) \cdot (1 + O(\delta_\bullet + \varepsilon_{\mathrm{mix}})).$$

Similarly,

$$\mathrm{GLRT}_{\rho,\mathrm{reg}}(z) \approx \frac{s_f^2}{\alpha_a(z)\, s_a^2 + \alpha_o(z)\, s_o^2} \cdot \mathrm{GLRT}_{\rho,\mathrm{base}}(z),$$

where

$$\alpha_a(z) := \frac{z^{\mathrm{H}} P_a z}{z^{\mathrm{H}}(P_a + \rho I)z} \in [0,1], \quad \alpha_o(z) := \frac{\rho \, \|z\|^2}{z^{\mathrm{H}}(P_a + \rho I)z} \in [0,1].$$

On alias–dominant tiles (large $\alpha_a(z)$), both BR and GLRT are approximately multiplied by $\mu := s_f^2/s_a^2 < 1$.

B.17 SL–2: Tail Threshold Improvement at Fixed FPR

Let S^-_{base} be the baseline negative score distribution, with CDF F_-. Let t_α be the α–quantile: $F_-(t_\alpha) = 1-\alpha$. After shrink-only regularization with gating, the negative CDF becomes a mixture:

$$F'_-(t) = (1 - p_-)F_-(t) + p_- F_-(t/\mu),$$

ignoring higher–order mixing terms and assuming $\mu < 1$ is the multiplicative shrinkage factor on gated negatives. Under mild regularity and for α in the low–FPR regime,

$$t'_\alpha \approx t_\alpha - \frac{p_-(1-\mu)}{f_-(t_\alpha)},$$

so that $t'_\alpha < t_\alpha$ whenever $p_- > 0$ and $\mu < 1$. In the PD-mode experiments in this paper we report operating points using the exported PD flow-evidence score map S_{PD} (stored as `score_pd_*.npy`; higher = more flow-like): choose τ so that $\mathbb{P}\{S_{\mathrm{PD}} \geq \tau \mid \mathrm{bg}\} = \alpha$ and evaluate $\mathbb{P}\{S_{\mathrm{PD}} \geq \tau \mid \mathrm{flow}\}$. For compatibility, acceptance bundles may also include `pd_*.npy`, which we denote $\mathrm{PD}^{\downarrow}$ and relate deterministically by $S_{\mathrm{PD}} = -\mathrm{PD}^{\downarrow}$.

Interpretation (plain language). The shrink-only regularizer is designed to shrink scores only on a subset of artifact-dominated background tiles. This selectively compresses the extreme right tail of H_0 scores, which lowers the background threshold needed to hit a target FPR. Lowering that threshold can reveal additional true flow pixels without increasing the false-positive rate, because scores never increase on penalized pixels. Equivalently, scaling down a fraction p_- of top–tail negatives removes mass from the negative tail and shifts t_α left; uplift is only plausible when p_+ (positive gating rate) is small.

We emphasize that the SL–2 guarantee is conditioned on:

- Regime preconditions R–1/R–2/R–3,
- Gating coverage in the range $0.2 \le p_- \le 0.5$, $p_+ \le 0.1 p_-$.

B.18 SL–3: Positive Collateral Bound

Let p_+ be the positive gating rate and assume gated positives suffer at most a multiplicative drop $\nu := s_f^2/s_a^2 \cdot (1 + O(\delta_\bullet + \varepsilon_{\text{mix}}))$. For any threshold t'_α satisfying $F'_-(t'_\alpha) = 1 - \alpha$, the TPR under regularization satisfies

$$\text{TPR}_{\text{reg}}(t'_\alpha) \ge \text{TPR}_{\text{base}}(t_\alpha) - \eta\, \Phi(\nu),$$

where $0 \le \eta \lesssim \frac{p_+}{p_-}$ (up to geometry) and $\Phi(\nu)$ is the fraction of positives near t'_α that are moved below the threshold. For $p_+ \ll p_-$ and ν close to 1, the loss term is dominated by the gain from SL–2.

B.19 Evaluation Discipline

This section remains as in the original document: it covers GLRT parameter sharing, block/cluster–aware tail CIs, and effective negatives. These aspects ensure fair comparison but do not themselves cause uplift.

B.20 Feature Statistics and Pass/Fail Checklist

At runtime, label–free feature statistics must support the following checks:

(i) Band design: BG–1/BG–2/BG–3 satisfied for current DPSS/PRF/T.
(ii) Regularizer operator: $\widehat{\varepsilon}_{\text{mix}} \le \varepsilon_{\text{mix}}$ (OC–1); $\widehat{s}_a/\widehat{s}_f \ge \Lambda$ and $\widehat{\delta}_\bullet \le 0.1$ (OC–2).
(iii) Localization primitives: risk scores m_{alias} and m_{guard}, guard ratios, flow coverage, FPC, depth, PD, registration; localization indicator g constructed via G–1/G–2/G–2'/G–3/G–4.
(iv) Invariance sentinel: if $\widehat{s}_a/\widehat{s}_f \in [1-\epsilon, 1+\epsilon]$ with $\epsilon \le 0.02$, the regularizer is effectively a scalar; disable it (see Section B.22).

B.21 Shrink-only regularizer: score-space penalty

To keep the shrink-only regularizer conservative under heterogeneity, we implement a deterministic, feature-conditioned prior that acts in score space as a shrink-only penalty. Concretely, the post-regularizer score is defined as

$$S_{\text{post}}(i) = w(i)\, S_{\text{pre}}(i), \qquad w(i) \in (0, 1],$$

equivalently $T_{\text{post}} = T_{\text{pre}} + \log w$ when working with log-likelihood scores (MAP log-odds decomposition). By construction, $w \equiv 1$ (no regularizer effect) is an explicit, expected outcome whenever the feature statistics are uninformative or fail integrity checks.

B.21.1 Feature-space regimes (R0/R1/R2)

We partition the feature space into three regimes per volume (or per slice/plane in PD-only mode):

- $\mathcal{R}_0$ **(uninformative).** The prior is uninformative and $w \equiv 1$; output equals the baseline detector (matched-subspace detector on IQ; PD-only detector on PD-only). Feature statistics are still recorded.
- $\mathcal{R}_1$ **(guarded shrink).** A bounded, shrink-only penalty may be applied on a localized set of high-risk tiles while strictly protecting a high-confidence set. No uplift claim is implied.
- $\mathcal{R}_2$ **(alias-dominant; experimental).** Features strongly indicate alias contamination and additional label-free actionability evidence suggests that shrink preferentially targets false activations. This regime is used only as a conservative tail-hygiene mode to gate reporting of regularizer deltas; it is not a default deployment mode or the basis for primary performance claims.

For auditability, the implementation also logs a reason label (why $w \equiv 1$) and a *risk mode* (alias vs guard) so that $\mathcal{R}_0$ behavior is interpretable. (For backward compatibility, exported metadata may also include legacy string labels such as `C0_OFF`/`C1_SAFETY`/`C2_UPLIFT` mapped to $\mathcal{R}_0/\mathcal{R}_1/\mathcal{R}_2$.)

B.21.2 Feature primitives (band-energy statistics)

On each tile i we record band energies using fixed, disjoint projectors (P_o, P_f, P_g, P_a) with $P_o + P_f + P_g + P_a = I$:

$$E_o(i) = z_i^{\mathrm{H}} P_o z_i, \quad E_f(i) = z_i^{\mathrm{H}} P_f z_i, \quad E_g(i) = z_i^{\mathrm{H}} P_g z_i, \quad E_a(i) = z_i^{\mathrm{H}} P_a z_i,$$

where $z_i \in \mathbb{C}^T$ (conjugate-transpose $(\cdot)^{\mathrm{H}}$; in real PD-only instantiations $(\cdot)^{\mathrm{H}}$ reduces to $(\cdot)^\top$). We optionally track a DC subcomponent $E_{\mathrm{dc}}(i)$ (e.g. energy in a narrow DC tone within P_o) as an auxiliary feature statistic. We define

$$E_{\mathrm{tot}}(i) = E_o + E_f + E_g + E_a + \epsilon,$$

and form robust, label-free ratios:

$$m_{\mathrm{alias}}(i) = \log \frac{E_a(i) + \epsilon}{E_f(i) + \epsilon}, \quad r_g(i) = \frac{E_g(i)}{E_{\mathrm{tot}}(i)}, \quad r_f(i) = \frac{E_f(i)}{E_{\mathrm{tot}}(i)}.$$

We also use a flow-coverage proxy $c_f(i) \in [0, 1]$ (fraction of tile pixels in the flow mask), a depth proxy $d(i) \in [0, 1]$ when available, and a baseline detector score $S_{\mathrm{base}}(i)$ on which the regularizer operates (defined in a flow-evidence convention so that higher means "more flow-like"; in PD-mode this is $S_{\mathrm{base}} = S_{\mathrm{PD}} = \texttt{score_pd}$).

B.21.3 Proxy sets and actionability tests

Because class labels are unavailable at runtime, the regime selector uses two proxy sets:

$$\mathcal{T}_{\mathrm{bg}} := \{i : c_f(i) \le c_{\mathrm{bg}}\}, \qquad \mathcal{T}_{\mathrm{flow}} := \{i : c_f(i) \ge c_{\mathrm{flow}}\},$$

with frozen design values (typical $c_{\mathrm{bg}} \approx 0.05$, $c_{\mathrm{flow}} \approx 0.20$). These thresholds are intentionally conservative, order-of-magnitude safety bounds chosen a priori to avoid penalizing dense-flow tiles; they are not optimized to maximize ROC on the evaluated datasets. The uplift-eligibility checks are reframed as *label-free actionability*:

- **Background-vs-flow separation:** $\Delta_{bf} = \mathrm{median}(m_{\mathrm{alias}} \mid \mathcal{T}_{\mathrm{bg}}) - \mathrm{median}(m_{\mathrm{alias}} \mid \mathcal{T}_{\mathrm{flow}})$.
- **Tail association in background:** within $\mathcal{T}_{\mathrm{bg}}$, define a tail subset $\mathcal{T}_{\mathrm{tail}}$ as the top 1% of S_{base} and a mid subset $\mathcal{T}_{\mathrm{mid}}$ as a central quantile band; require that m_{alias} is higher in $\mathcal{T}_{\mathrm{tail}}$ than in $\mathcal{T}_{\mathrm{mid}}$ (risk-feature association explains "false-alarm-looking" background tiles).

When guard fractions dominate, the actionable regime $\mathcal{R}_2$ is disallowed but guarded shrink $\mathcal{R}_1$ may still be permitted. In this case we switch to a guard-risk mode using $m_{\mathrm{guard}}(i) = \log \frac{E_g(i)+\epsilon}{E_f(i)+\epsilon}$ (or equivalently $r_g(i)$) and apply the same tail association check using m_{guard}.

B.21.4 Protected set, candidate window, and shrink mapping

The shrink-only regularizer must not disturb high-confidence flow tiles. We define a protected set

$$\mathcal{T}_{\mathrm{prot}} := \{i : c_f(i) \geq c_{\mathrm{flow}}\} \;\cup\; \{i : S_{\mathrm{base}}(i) \geq Q_{q_{\mathrm{hi}}}(S_{\mathrm{base}})\} \;\cup\; \{i : r_f(i) \geq r_{f,\mathrm{prot}}\},$$

and a candidate window $\mathcal{T}_{\mathrm{cand}}$ consisting of non-protected tiles whose baseline scores lie in an "uncertain" quantile band $[Q_{q_{\mathrm{lo}}}, Q_{q_{\mathrm{hi}}}]$. On $\mathcal{T}_{\mathrm{bg}}$ we set an adaptive risk threshold $\tau_{\mathrm{risk}} = Q_{q_{\mathrm{risk}}}(m_{\mathrm{risk}} \mid \mathcal{T}_{\mathrm{bg}})$ where m_{risk} is m_{alias} or m_{guard} depending on risk mode. The localization indicator is

$$g(i) = \mathbf{1}\{i \in \mathcal{T}_{\mathrm{cand}},\; m_{\mathrm{risk}}(i) \geq \tau_{\mathrm{risk}}\}.$$

For penalized tiles (those with $g(i) = 1$) we compute a bounded shrink factor

$$w(i) = \max\left(w_{\min}, \exp(-k\, u(i))\right), \qquad u(i) = \max\left(0, \frac{m_{\mathrm{risk}}(i) - \tau_{\mathrm{risk}}}{\mathrm{MAD}(m_{\mathrm{risk}} \mid \mathcal{T}_{\mathrm{bg}}) + \epsilon}\right),$$

with $w_{\min} \in (0, 1]$ so the corresponding scale $\gamma(i) = 1/w(i)$ is finite, and set $w(i) = 1$ on non-penalized and protected tiles. In Doppler/IQ PD-mode bundles we export the per-pixel scale $\gamma(i) = 1/w(i) \geq 1$ (averaged over overlapping tiles) and apply it on the union of penalized tiles as

$$\mathrm{PD}^{\downarrow}_{\mathrm{post}}(i) = \gamma(i)\, \mathrm{PD}^{\downarrow}_{\mathrm{pre}}(i),$$

which inflates the legacy PD-mode map $\mathrm{PD}^{\downarrow}$ on penalized regions. Since $S_{\mathrm{PD}} = -\mathrm{PD}^{\downarrow}$, this is equivalent to $S_{\mathrm{post}} = w\, S_{\mathrm{pre}}$ and therefore is non-increasing in score space. For PD-only detectors such as PD-z that take both signs, we use hard-suppression/soft-shrink variants (Appendix H) so that shrink-only is literal for signed scores.

B.21.5 Sentinels, reason codes, and auditability

The regime selector defaults to $\mathcal{R}_0$ ($w \equiv 1$) when any of the following hold: insufficient background proxy support, flat risk statistics (e.g. $\mathrm{IQR}(m_{\mathrm{risk}} \mid \mathcal{T}_{\mathrm{bg}})$ too small), guard domination that is not actionable, coverage outside bounds (localization inert or global), invariance (penalized w nearly constant), or a monotonicity guard failure on near-protected tiles. When $w \not\equiv 1$, we record coverage and strength summaries (e.g. $p_{\mathrm{shrink}} = \mathbb{P}\{g = 1 \mid \mathcal{T}_{\mathrm{cand}}\}$ and quantiles of w) along with a short reason label (e.g. `alias_metric_flat`, `guard_dominates_not_actionable`, `coverage_out_of_bounds`, `monotonicity_guard_failed`). This makes regularizer behavior reviewable and supports the intended posture: $\mathcal{R}_0$ is an expected success regime, $\mathcal{R}_1$ is a conservative safety regime, and $\mathcal{R}_2$ is a research-only regime used to gate uplift reporting.

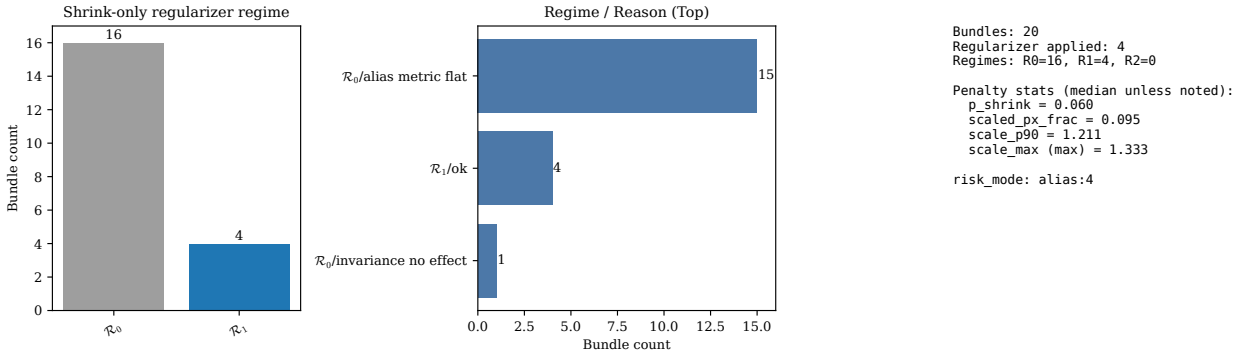

Figure 1: Shrink-only regularizer summary panel on a representative sweep. Left: regime counts. Middle: top (regime, reason) labels. Right: summary of shrink coverage and strength on runs where $w \not\equiv 1$.

B.21.6 PD-only instantiation

On PD-only datasets (Macé/Urban, fUS-BMI, cranial-window PD), the same regime selection is applied after replacing Doppler bands with hemodynamic bands and treating tile-averaged PD time series as the slow-time signal. The feature primitives (E_f, E_a, E_g) and the shrink-only map $w(i)$ are defined identically, and the regime label is produced per plane. This yields a uniform export schema across Doppler IQ and hemodynamic PD regimes.

Figure 1 shows a compact prior/regime summary panel over a representative sweep across regimes: most runs remain in $\mathcal{R}_0$ with an explicit disable reason, while a minority enter $\mathcal{R}_1$ and apply a mild, localized shrink in an alias- or guard-risk lane (never boosting scores).

B.22 Regime Sentinel and Safety Guard

Experience shows that even when OC/BG/CB/G conditions are roughly satisfied, there can exist regimes where the shrink-only regularizer is either inert or harmful (e.g. risk feature flat, localization never triggers or triggers everywhere, baseline ROC saturated). This section formalizes when the regularizer should be *explicitly disabled*.

B.23 RS–1: Regime Sentinel (Design–Time)

On a held–out labeled configuration, check:

- **RS–1a: Band occupancy.** If R–1 fails (e.g. most flow peaks near DC instead of Pf, or alias peaks rarely in Pa), the band design is not physically realized; do not deploy the regularizer.
- **RS–1b: Alias separation.** If R–2 fails (alias scores m_{alias} show negligible difference between classes), alias localization has no reliable signal; do not deploy the regularizer.
- **RS–1c: Localization coverage.** If, after choosing $q_{\text{alias}}, c_{f,\max}, c_{\text{pc}}$, the empirical localization rates satisfy $p_- \approx 0$ or $p_- \approx 1$, or $p_+ \approx p_-$, then localization is either inert, global, or non–differential; do not expect uplift and default to $\mathcal{R}_0$.
- **RS–1d: Baseline saturation.** If $\text{pAUC}_{\text{base}}$ in the target FPR range is already at or near maximum and the score ratio distribution under regularization has median close to 1 with tiny spread, you are in a ROC–invariant regime; the regularizer can be left off.

B.24 Safety–1: Score Monotonicity Guard (Runtime)

Even when regime preconditions and localization coverage hold, we require that the regularizer does not produce grossly pathological score transformations on *penalized flow tiles*. Let $S_{\text{base}}, S_{\text{reg}}$ denote baseline and regularized scores on the chosen metric (BR or GLRT).

For the set of penalized flow tiles, define

$$r_i := \frac{S_{\text{reg}}(i)}{\max(S_{\text{base}}(i), \varepsilon)}, \quad u_i := \log r_i,$$

with a small $\varepsilon > 0$.

Let

$$\text{median}_f := \text{median}(u_i), \quad \text{IQR}_f := Q_{0.75}(u_i) - Q_{0.25}(u_i),$$

and τ_f the Kendall τ rank correlation between S_{reg} and S_{base} on these pixels.

We require:

- **Sample size.** Apply the guard only if the number of gated flow pixels exceeds a threshold $N_{\min}$ (e.g. $N_{\min} = 200$). If not, mark the regularizer as "too sparse to evaluate" but do not disable it solely on this basis.
- **Flow monotonicity.** The regularizer is marked unsafe if either

$$\begin{aligned} &\text{IQR}_f > \log 10 \quad (\text{heterogeneity } > \times 10), \quad \text{or} \\ &\text{median}_f > \log 10 \quad \text{and} \quad |\tau_f| < 0.1, \end{aligned}$$

  i.e. if the regularizer systematically boosts flow scores by more than an order of magnitude while destroying rank consistency, or applies extremely heterogeneous multiplicative effects.

If Safety–1 fails, the regularizer is disabled at runtime for that configuration and the system falls back to the unregularized matched-subspace detector.

B.25 What the Regime Does Not Promise (Updated)

The shrink-only regularizer may *fail to provide uplift* or may be disabled by the regime sentinel / safety guard in the following cases:

- If $\Lambda \approx 1$ (band–constant scaling, invariance sentinel) or ε_{mix} is large, global band–wise transforms are nearly ROC–invariant or harmful; expect parity or degradation.
- If R–2 alias separation fails, i.e. m_{alias} shows no class–differential behavior, alias localization cannot preferentially capture alias–dominated negatives; the regularizer is out of regime.
- If localization is not class–differential (offline $p_+ \approx p_-$) or p_- is either tiny or nearly one, the regularizer is either inert or global; SL–2 no longer applies.
- If baseline ROC is already saturated in the target FPR range, even an ideal shrink-only transform cannot move the operating point significantly.
- If Safety–1 indicates that the regularizer acts in a grossly non–monotone way on penalized flow tiles (e.g. large heterogeneous boosts), it is disabled for that configuration.

In all these regimes, MC–SVD or matched-subspace-only scoring should be preferred; the shrink-only regularizer can be viewed as an optional add-on rather than a source of guaranteed uplift.

B.26 Safety and Regime Sentinel Summary

For deployed use it is helpful to summarize, in plain language, when the shrink-only regularizer is allowed to act and when it is automatically turned off.

- *When the regularizer is allowed to act.* The regularizer is enabled only in regimes that satisfy the structural preconditions (R–1/R–2/R–3), have useful but not global localization coverage (roughly $0.2 \leq p_- \leq 0.5$ with $p_+ \ll p_-$), and pass the invariance and Safety–1 checks. In this case it behaves as a controlled, band–aware add-on on top of the matched-subspace detector: it is permitted to shrink scores on alias–dominated negatives but must remain monotone and mild on flow tiles.
- *When the regularizer is turned off.* The regularizer defaults to $\mathcal{R}_0$ by design when band occupancy or alias separation fail (R–1/R–2), when localization becomes inert or non–differential, when the baseline ROC is already saturated in the target low–FPR band (R–3), or when Safety–1 detects non–monotone, highly heterogeneous score changes on penalized flow tiles. The invariance sentinel also forces $\mathcal{R}_0$ when the realized gains satisfy $\widehat{s}_a/\widehat{s}_f \approx 1$ so that the regularizer is effectively a scalar rescaling.
- *Behavior in skull/OR–like regimes.* In skull/OR realism regimes such as the structured-clutter simulation pilot (internal profile ID `Brain-SkullOR`), alias power deliberately leaks into H_1 and R–2 fails by construction. In these regimes the regularizer is expected to be off or ROC–neutral; the primary detector is the fixed matched-subspace profile, with MC–SVD/RPCA/HOSVD as reference baselines. This is a safety feature, not a failure of the method: whenever the regime preconditions are not met, the system falls back to matched-subspace-only scoring rather than forcing the regularizer to act.

C Supplementary Results

This appendix collects additional dataset audits and diagnostic panels that support the main paper's methodology, but are not required for the primary labeled low-FPR detection claims.

Evaluation discipline (summary). Quantitative ROC claims are restricted to regimes with independent labels (simulator truth from the labeled brain simulations; Gammex B-mode structural masks). Operating points are reported under prespecified settings by selecting thresholds on labeled (or structurally labeled) H_0 sets at target FPR and evaluating TPR on the corresponding H_1 set; the empirically supported floor $fpr_{\min} = 1/n_{\text{neg}}$ is reported. Uncertainty is quantified at the natural unit of variation (windows for the labeled brain simulations; frames for cine acquisitions) rather than treating pooled pixels as independent samples. Conditional execution is treated as a changed method and is reported only with explicit leakage controls. For Twinkling sequences, PRF is fixed once per sequence family using source-paper values where available and explicit frozen overrides otherwise; band-discretization sentinels force conservative behavior when discretized bands are invalid. When the shrink-only regularizer is enabled, it preserves a protected do-not-penalize set (scores unchanged on high-confidence flow locations); labeled ROC reporting uses the pre-regularizer matched-subspace score.

Supplement roadmap. The supplementary material is organized as supporting evidence for the main paper, with structurally labeled and real-IQ audits first, optional adaptive and shrink-only-

regularizer diagnostics next, and archival fairness or repository-style material later.

- **Supporting structural audits:** full ROC shapes for labeled regimes, cross-window calibration, whitened-head stability, relaxed-tail sanity checks, open-skull parameter sensitivity, and the Gammex matched-subspace-only / PRF sensitivity check (Figure 2, Figure 3, Figure 5, Figure 6).
- **Adaptive and shrink-only regularizer supporting audits:** finalized threshold sensitivity, conditional-execution leakage, and artifact-heavy calculi hygiene (Tables 6, 7, and 8).
- **Real-IQ supporting audits:** ULM motion variants and structural audit, Shin real-IQ proxy/motion/hygiene details, and the compact real-IQ same-residual score audit later in this appendix.
- **PD-only and whole-brain retrospectives:** the Macé/Urban alias-gating and atlas-based summary panels.
- **Supplementary simulation and held-out benchmark details:** detailed labeled brain tables, corrected SIMUS/PyMUST fairness checkpoints, the secondary held-out functional same-cube audit, and the use-case-motivated synthetic checkpoint tables (Tables 21, 23, and 24).
- **Baseline-family summaries:** the compact baseline-family summary (Table 22).

Supporting Structural Audits

This block collects supplementary labeled and structurally labeled checks that reinforce the main structural claims without adding new benchmark families.

C.1 Full ROC curves for labeled regimes

The main text reports strict low-FPR operating points. Here we provide representative ROC-style curves to show the overall curve shape under the same masks and scoring conventions.
ROC curves are reported with the same unit-of-variation discipline as the operating-point tables. For the labeled brain simulations, we compute ROC curves per 64-frame window and summarize them as median and IQR across **windows**; the supported floor $fpr_{\min} = 1/n_{\text{neg}}$ is fixed by the window's negative-set size. For Gammex, thresholds are calibrated on pooled pixels for stability, while uncertainty is quantified via a frame-level bootstrap (resampling **cine frames** as the unit of variation at fixed pooled thresholds).

C.2 Cross-window threshold calibration audit (labeled brain simulations)

The main labeled brain simulation results calibrate the operating threshold τ_α on the negative set of the *same* 64-frame window being evaluated. To quantify how sensitive these strict low-FPR operating points are to window-to-window distribution shift (and to avoid over-interpreting quantized tail points), we report two transfer audits for the whitening profile used in the detector component audit: (i) pairwise cross-window transfer, where τ_α is calibrated on one window's negatives and applied to the other windows, and (ii) a leave-one-window-out fixed calibration bank, where negatives from the other four windows are pooled and a single threshold is applied to the held-out window. Table 3 reports within-window TPR (as in the main reporting protocol) alongside both transfer variants and their realized FPRs. **Checksum.** The within-window column in Table 3

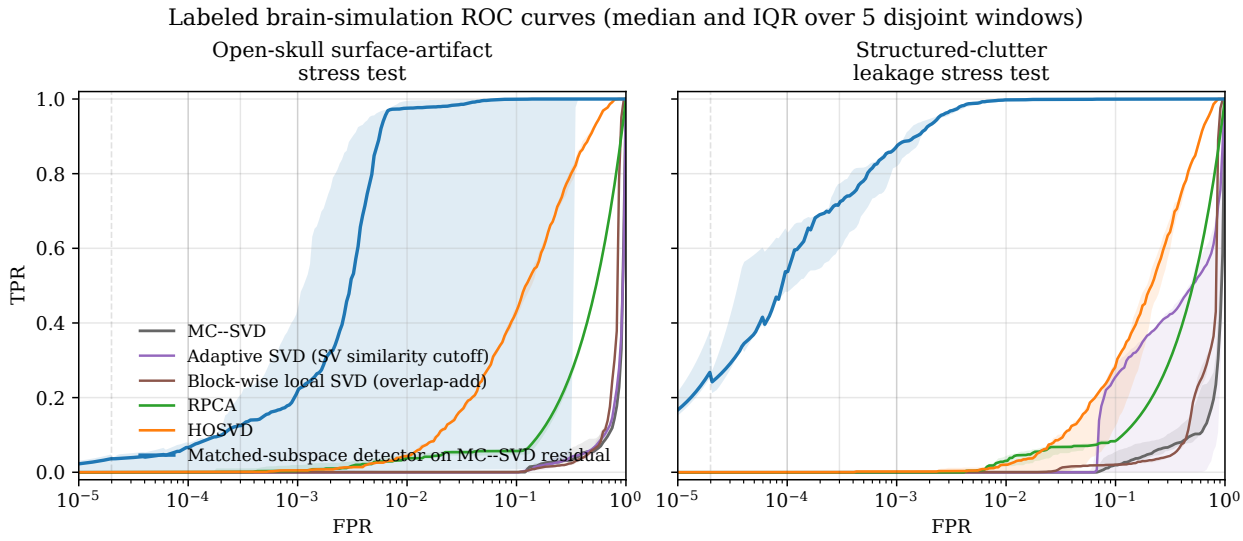

Figure 2: Labeled k-Wave brain simulation regimes: ROC-style curves (TPR vs FPR on a log-x axis) under the same 64-frame window protocol used for the low-FPR operating-point reporting in the main paper. Curves show the median and interquartile range over five disjoint windows. The vertical dashed line indicates the supported floor $fpr_{\min} = 1/n_{\text{neg}}$.

matches the matched-subspace operating-point values reported in the main paper build under the same score/mask protocol.

C.3 Robust-whitening stability audit

The frozen Huber-whitened advanced recipe uses Huber covariance estimation, stronger loading, and trimmed high-energy training support. To test whether its operating characteristics remain stable under modest changes to that trim, we reran the open-skull and structured-clutter stress tests while varying only the trim parameter over $q \in \{0.05, 0.08, 0.10\}$. Table 4 reports within-window TPR at the strictest operating point together with realized FPR under pairwise cross-window transfer and a leave-one-window-out fixed calibration bank. The values are numerically unchanged across the tested trim settings in both regimes. In particular, the open-skull realized FPR remains near target for the Huber-whitened advanced recipe under both pairwise cross-window transfer (median 1.10×10^{-4}, $q_{75} = 1.64 \times 10^{-4}$ at $\alpha = 10^{-4}$) and fixed-calibration transfer (median 1.20×10^{-4}). This is the main stability result for the finalized whitened variant.

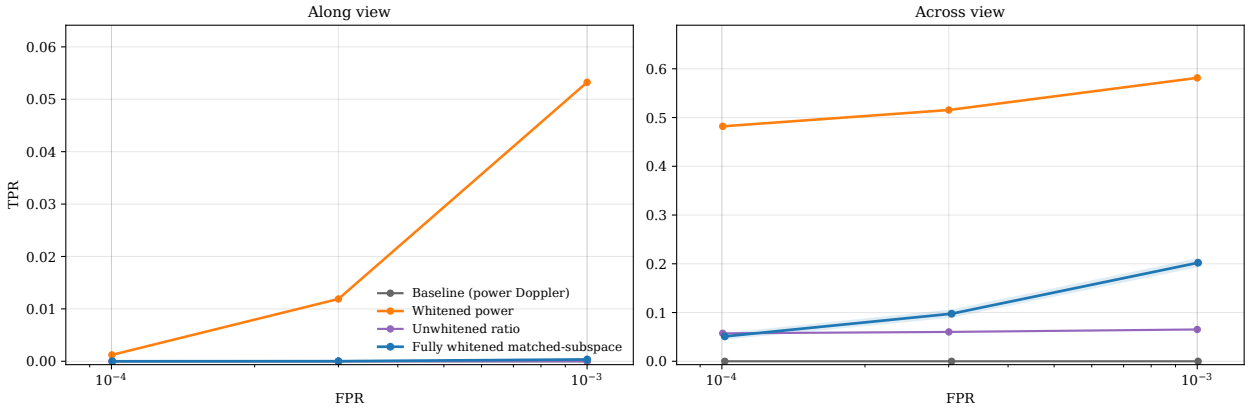

Figure 3: Gammex flow phantom structural labels: pooled ROC-style curves on along and across views, shown for baseline comparators and the pre-regularizer matched-subspace score. The vertical guides indicate the strict low-FPR operating points reported in the main paper.

Regime	α	within-window TPR med (IQR)	pairwise cross TPR med (IQR)	pairwise cross FPR med (IQR)	fixed-cal-set TPR med (IQR)	fixed-cal-set FPR med (IQR)
Open-skull	10^{-4}	0.0730 (0.0008,0.0865)	0.0621 (0.0000,0.1839)	1.20e-04 (0,3.29e-01)	0.0000 (0.0000,0.0000)	0 (0,1.99e-05)
Open-skull	3×10^{-4}	0.1288 (0.0012,0.2222)	0.1409 (0.0000,0.3285)	3.49e-04 (0,3.32e-01)	0.0000 (0.0000,0.0012)	0 (0,3.39e-04)
Open-skull	10^{-3}	0.2218 (0.0015,0.4368)	0.2785 (0.0000,0.4883)	1.06e-03 (0,3.34e-01)	0.0000 (0.0000,0.0027)	0 (0,3.19e-03)
Alias-dominant	10^{-4}	0.6098 (0.4790,0.6755)	0.6363 (0.4344,0.7441)	9.96e-05 (1.99e-05,6.72e-04)	0.3837 (0.2207,0.3960)	0 (0,1.99e-05)
Alias-dominant	3×10^{-4}	0.6867 (0.6090,0.7493)	0.7351 (0.5552,0.8146)	3.78e-04 (5.48e-05,9.06e-04)	0.6490 (0.4856,0.7236)	1.39e-04 (1.20e-04,1.79e-04)
Alias-dominant	10^{-3}	0.8301 (0.7847,0.8820)	0.8404 (0.7236,0.9022)	1.04e-03 (2.99e-04,2.49e-03)	0.8328 (0.7493,0.8358)	6.97e-04 (5.18e-04,1.95e-03)
Structured clutter	10^{-4}	0.5871 (0.5006,0.6682)	0.5827 (0.4474,0.7139)	1.29e-04 (1.99e-05,4.48e-04)	0.4752 (0.3618,0.6036)	5.98e-05 (3.98e-05,1.20e-04)
Structured clutter	3×10^{-4}	0.7205 (0.6048,0.7785)	0.6801 (0.5755,0.7937)	3.19e-04 (5.48e-05,6.92e-04)	0.6248 (0.5413,0.7559)	2.59e-04 (1.99e-04,3.59e-04)
Structured clutter	10^{-3}	0.8731 (0.8189,0.8758)	0.8516 (0.7726,0.9056)	1.02e-03 (6.52e-04,1.40e-03)	0.8147 (0.8131,0.8908)	1.20e-03 (8.37e-04,1.35e-03)

Table 3: Labeled brain simulation threshold-transfer audit for the whitening profile used in the detector component audit. "Within-window" matches the main reporting protocol (calibrate τ_α on a window's negatives and evaluate on that same window). "Pairwise cross" calibrates τ_α on one window and evaluates on a disjoint window ($n = 20$ ordered window pairs). "Fixed-cal-set" uses a leave-one-window-out calibration bank: for each held-out window, pool negatives from the other four windows, calibrate τ_α once, and evaluate on the held-out window ($n = 5$). Reported values are median (IQR) over windows/pairs.

Trim setting	Open-skull stress test			Structured-clutter stress test		
	Within TPR@10^{-4}	Cross FPR med. (q75)	Fixed-cal FPR med. (q75)	Within TPR@10^{-4}	Cross FPR med. (q75)	Fixed-cal FPR med. (q75)
$q = 0.05$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)
$q = 0.08$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)
$q = 0.10$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)

Table 4: Robust-whitening stability audit for the prespecified fully whitened detector on the labeled brain stress tests. The only changing parameter is the high-energy covariance-training trim q. Entries report within-window TPR at the strictest operating point ($\alpha = 10^{-4}$) together with realized FPR under pairwise cross-window transfer and a leave-one-window-out fixed calibration bank. The reported values are numerically unchanged across $q \in \{0.05, 0.08, 0.10\}$, indicating that the finalized advanced recipe is stable to modest covariance-training trim variation in these stress tests.

	Brain-OpenSkull			Brain-AliasContract			Brain-SkullOR		
Method / score	TPR@10^{-3}	TPR@10^{-2}	TPR@10^{-1}	TPR@10^{-3}	TPR@10^{-2}	TPR@10^{-1}	TPR@10^{-3}	TPR@10^{-2}	TPR@10^{-1}
MC–SVD	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0046 (0.0035,0.0050)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0188 (0.0154,0.0538)
Adaptive SVD (SV similarity cutoff)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0554 (0.0050,0.1496)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.2537 (0.0000,0.2860)
Block-wise local SVD (overlap-add)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0012)	0.0000 (0.0000,0.0000)	0.0004 (0.0000,0.0004)	0.0211 (0.0196,0.0242)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0200 (0.0185,0.0238)
RPCA	0.0042 (0.0042,0.0173)	0.0338 (0.0261,0.0419)	0.0538 (0.0415,0.0569)	0.0000 (0.0000,0.0000)	0.0342 (0.0138,0.0396)	1.0000 (1.0000,1.0000)	0.0000 (0.0000,0.0000)	0.0227 (0.0123,0.0334)	1.0000 (1.0000,1.0000)
HOSVD	0.0046 (0.0046,0.0077)	0.0442 (0.0431,0.0469)	0.4237 (0.4152,0.4460)	0.0031 (0.0015,0.0046)	0.0331 (0.0308,0.0377)	0.3214 (0.2718,0.3222)	0.0015 (0.0000,0.0031)	0.0235 (0.0000,0.0304)	0.2841 (0.1300,0.2910)
Matched-subspace detector on MC–SVD	0.2218 (0.0015,0.4368)	0.9754 (0.0054,0.9946)	0.9992 (0.0077,1.0000)	0.8301 (0.7847,0.8820)	0.9985 (0.9781,0.9996)	1.0000 (0.9996,1.0000)	0.8731 (0.8189,0.8758)	0.9977 (0.9958,0.9988)	0.9996 (0.9996,1.0000)

Table 5: Baseline sanity check on labeled Brain-* simulations at a strict-tail operating point ($\alpha = 10^{-3}$) versus relaxed operating points ($\alpha \in \{10^{-2}, 10^{-1}\}$). Entries report median (IQR) TPR over the same five disjoint 64-frame windows used in the main tables, with thresholds calibrated per window on that window's negative set. The relaxed columns are included only to verify that PD-after-filter baselines are not degenerate; strict-tail reporting remains the focus.

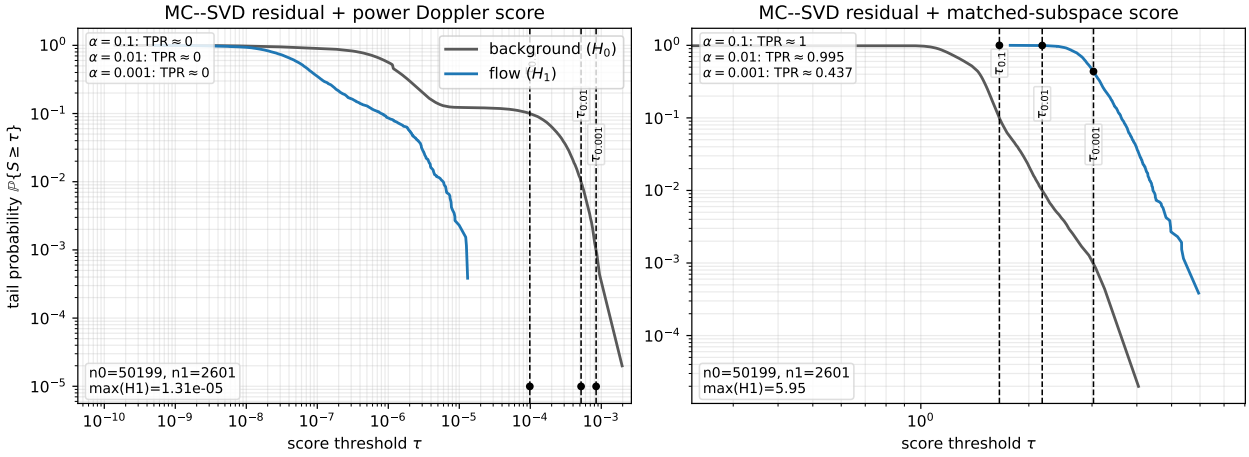

Figure 4: Representative open-skull labeled brain window illustrating strict-tail collapse of PD-after-filter baselines. Curves show empirical tail probabilities $\mathbb{P}\{S \geq \tau\}$ for background (H_0) and flow (H_1) score distributions versus score threshold τ, for (left) the baseline power Doppler score on the MC–SVD residual and (right) the pre-regularizer matched-subspace score computed on the same residual. Vertical lines mark background-calibrated thresholds τ_α at $\alpha \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ and markers show the corresponding $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$. For the baseline PD score, $\tau_{10^{-3}}$ lies above $\max(H_1)$, yielding TPR ≈ 0 by construction; the matched-subspace score retains nonzero sensitivity at the same strict-tail operating point.

C.4 Baseline sanity checks at relaxed operating points (labeled brain simulations)

Several baseline rows in the main labeled brain simulation strict low-FPR table take the value $\mathrm{TPR}(\alpha) \approx 0$ at $\alpha \leq 10^{-3}$. This can occur even when a baseline is not degenerate: if the background right tail is extreme, the background-calibrated threshold τ_α can lie above essentially all H_1 pixels at strict-tail α. To provide a reviewer-facing sanity anchor, Table 5 contrasts one strict-tail operating point ($\alpha = 10^{-3}$) with two relaxed operating points ($\alpha \in \{10^{-2}, 10^{-1}\}$) under the same window protocol and evaluation discipline; $\alpha = 10^{-2}$ is a moderate reference point and $\alpha = 10^{-1}$ is an intentionally relaxed non-degeneracy check.

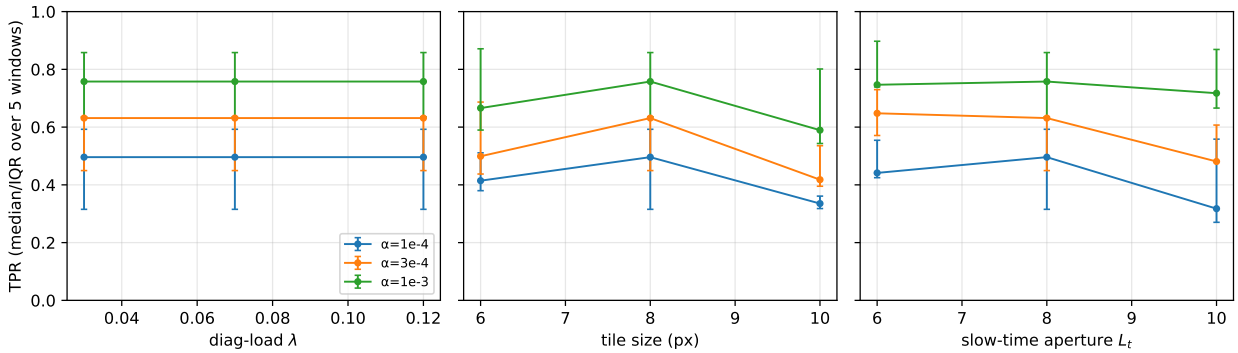

Figure 5: Open-skull pilot parameter-sensitivity sweep. Left: covariance diagonal loading λ (conditioning). Middle: tile size (stride chosen to approximately preserve overlap fraction relative to the default 8×8/stride-3 setting). Right: slow-time aperture L_t. Points show within-window TPR medians with IQR error bars over five disjoint 64-frame windows at strict-tail operating points $\alpha \in \{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$.

C.5 Parameter sensitivity (Open-Skull Pilot)

Because the detector relies on covariance whitening, a natural concern is sensitivity to conditioning and geometry choices. We therefore vary one parameter at a time around the prespecified open-skull setting: covariance diagonal loading λ (conditioning strength), tile size (with stride chosen to preserve overlap), and slow-time aperture L_t. Each point reports within-window TPR medians (IQR) over five disjoint 64-frame windows at fixed strict-tail FPR targets $\alpha \in \{10^{-4}, 3\times10^{-4}, 10^{-3}\}$ under the same baseline and mask protocol as the main labeled brain simulation tables. In this pilot, performance is effectively insensitive to λ over the tested range, shows modest dependence on tile geometry, and is strongly sensitive to under-sized L_t (e.g. $L_t = 6$ collapses strict-tail TPR), consistent with the need for adequate slow-time support to estimate and invert R_i.

Adaptive and shrink-only regularizer Supporting Audits

This block isolates threshold robustness, conditional execution, and shrink-only penalty behavior so that optional mechanisms are easy to find and do not interrupt the main structural or real-IQ narratives.

C.6 Sensitivity of the finalized adaptive and shrink-only regularizer thresholds

An obvious overfitting concern is that the adaptive guard threshold τ_g and the shrink-only regularizer flow-support proxy threshold c_{flow} might have been tuned so tightly that small perturbations would materially change the reported detector behavior. We therefore perturb each finalized threshold by $\pm10\%$ while leaving the rest of the pipeline fixed.

For the adaptive variant, Table 6 evaluates the frozen tile-mean guard threshold around the deployed value on representative regimes spanning labeled brain stress tests, a structurally labeled phantom, and the Shin real-IQ corroboration. For the shrink-only regularizer, Table 7 recomputes the shrink-only calculi audit after perturbing c_{flow} by $\pm10\%$ on the same frozen bundles. In both cases the behavior changes smoothly rather than collapsing: the labeled-brain TPRs are unchanged

τ_g change	Open-skull TPR@10^{-4}	Structured-clutter TPR@10^{-4}	Shin flow hit@10^{-3}	Gammex across TPR@10^{-3}
-10%	0.869	0.910	0.028	0.184
default	0.869	0.910	0.028	0.150
+10%	0.869	0.910	0.028	0.159

Table 6: Sensitivity of the frozen guard-triggered adaptive threshold τ_g to a $\pm 10\%$ perturbation around the deployed tile-mean guard-energy threshold. Reported endpoints use each regime's headline metric: within-window TPR on the labeled brain stress tests, flow-proxy hit rate on the Shin real-IQ audit, and TPR on the structurally labeled Gammex across-view phantom. The detector behavior changes smoothly rather than collapsing under these perturbations.

c_{flow} change	Active / inert bundles	median p_{shrink}	bg-tail rate / default	bg-tail clusters / default
-10%	50/0	0.097	1.000	1.000
default	50/0	0.097	1.000	1.000
+10%	50/0	0.097	1.000	1.000

Table 7: Sensitivity of the shrink-only regularizer to a $\pm 10\%$ perturbation of the flow-support proxy threshold c_{flow} on the artifact-heavy Twinkling calculi audit. Tail-rate and cluster columns are normalized to the default frozen-bundle audit so that the table emphasizes robustness to threshold perturbation rather than re-reporting the headline calculi numbers. Activation, shrink coverage, and background-tail compression vary modestly rather than collapsing under these perturbations, while the protected-set invariance remains exact to numerical precision.

at the reported precision, the real-IQ and phantom endpoints shift modestly, and the calculi audit preserves both the regularizer activation pattern and exact protected-set invariance.

C.7 Matched-subspace-only ablation and PRF sensitivity sanity check (Gammex)

To isolate the role of the matched-subspace detector core from baseline prefiltering, we run a small matched-subspace-only ablation on the Gammex flow phantom by applying the matched-subspace detector directly to raw (registered) CFM IQ without the SVD band-pass baseline. We also provide a small PRF sensitivity check on the along view to assess whether the reported structural behavior is brittle to modest perturbations of the frozen sequence-level PRF setting. The main text cites the TPR@10^{-3} endpoints from this audit as a bounded fairness sanity check; the full figure below shows the paired first-20-frame comparison and PRF-sensitivity panel.

C.8 Conditional execution leakage ablation (Open-Skull Pilot)

Conditional execution (compute gating) skips matched-subspace detector outside a flow-proxy region and falls back to the baseline score. Because this changes the overall decision rule, we evaluate it explicitly under leakage controls by varying only the *gating* mask: (i) a proxy mask computed from the same scored window, (ii) a proxy mask carried from a disjoint window, and (iii) a randomized mask with matched coverage. Table 8 reports the corresponding change in TPR relative to full matched-subspace detector at fixed low-FPR targets.

This behavior is expected because gating changes the decision rule: leakage-controlled gating produces a small TPR decrease in exchange for a large runtime reduction, making it a practical tradeoff

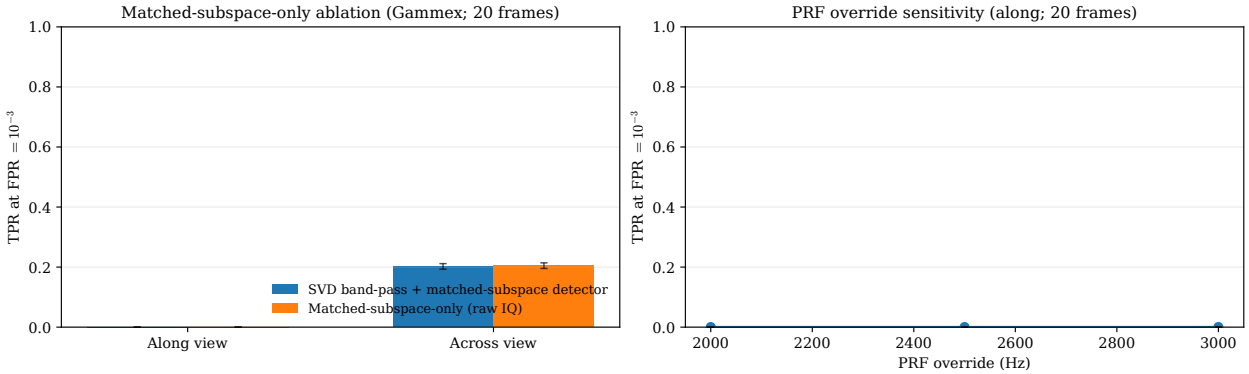

Figure 6: Gammex flow phantom: (left) matched-subspace-only ablation comparing the standard baseline SVD band-pass plus matched-subspace processing pipeline to matched-subspace-only on raw IQ (first 20 cine frames; both views). (right) PRF sensitivity on the along view (first 20 cine frames). Points show TPR at a fixed FPR target of 10^{-3} with 95% frame-bootstrap CIs under the structural tube masks. This is a low-cost robustness check around the frozen PRF/band discretization, not a tuned performance sweep.

Conditional-execution mask	ΔTPR@10^{-4} ($\times 10^{-3}$)	ΔTPR@3×10^{-4} ($\times 10^{-3}$)	ΔTPR@10^{-3} ($\times 10^{-3}$)
Same-window proxy mask	$+0.31\,[0.23, 0.39]$	$+0.81\,[0.54, 1.04]$	$+3.05\,[2.67, 3.43]$
Disjoint-window proxy mask	$-3.05\,[-5.69, -0.92]$	$-2.90\,[-5.34, -0.88]$	$-1.74\,[-4.64, 0.28]$
Random mask (matched coverage)	$-3.05\,[-5.65, -0.92]$	$-2.97\,[-5.68, -0.71]$	$-2.36\,[-4.96, -0.06]$

Table 8: Conditional-execution leakage ablation on the open-skull pilot (five independent realizations; 64-frame window; matched-subspace score). Entries report mean ΔTPR (conditional minus full matched-subspace detector) at fixed FPR targets, with 95% bootstrap CIs over realizations ($n = 5$). This table provides uncertainty details for the conditional-execution tradeoff summarized in the main Results. Evaluation masks are identical across conditions; only the conditional-execution gating mask varies.

rather than a free performance gain.

C.9 Shrink-only regularizer hygiene on artifact-heavy calculi sequences (Twinkling)

To illustrate shrink-only behavior in a strongly artifact-dominated regime, we evaluate Twinkling calculi sequences under a frozen short-ensemble profile. Figure 8 summarizes the shrink-only prior outputs (region and reason) across Gammex and calculi runs, and Figure 9 shows a representative calculi example where the post-regularizer score compresses the background tail without increasing any scores on penalized pixels.

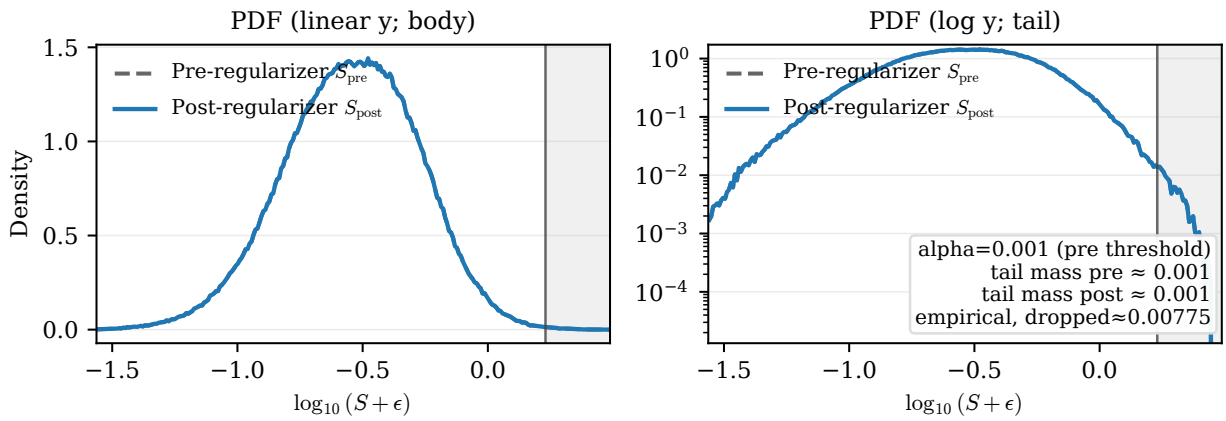

Figure 7: Empirical statistical effect of the strictly bounded, shrink-only penalty on a heavy-tailed background-proxy score distribution from an artificial-calculi acquisition in the open Doppler Twinkling Artifact database. The matched-subspace score before the optional penalty exhibits a heavy right tail characteristic of structured artifacts. Applying the non-increasing weight $w(i) \in (0, 1]$ yields the post-penalty score $S_{\text{det,post}} = w(i)S_{\text{det,pre}}$, compressing the background tail while preserving an explicit do-not-penalize protected set (scores unchanged on high-confidence flow locations). The vertical line indicates the pre-penalty threshold τ_α chosen so that $\mathbb{P}\{S_{\text{det,pre}} \geq \tau_\alpha \mid \text{bg proxy}\} = \alpha$ (here $\alpha = 10^{-3}$); the shaded region highlights the corresponding right tail, and the post-penalty tail mass is evaluated at the same τ_α.

Real-IQ Supporting Audits

This block collects the bounded real-IQ analyses: additional ULM robustness variants, the localization-derived structural audit, and the Shin real-IQ translation and hygiene details.

C.10 ULM 7883227: additional motion variants

In addition to the main `brainlike` motion sweep reported in the main paper, we report an `elastic` (nonrigid-only) motion ladder under the same frozen baseline and matched-subspace profile.

C.11 ULM 7883227: localization-derived structural-label audit on real in-vivo brain IQ

To obtain a stronger anatomical reference on the same open ULM dataset, we use the published PALA rat-brain ULM example rather than the earlier local-density surrogate [38, 39]. The construction is deterministic and frozen. We reproduce the corresponding short-horizon PALA PowDop rendering from the Zenodo IQ archive using the published odd-block selection and SVD cutoff, register that published PowDop crop back to the Zenodo IQ field of view, and then downsample the published super-resolution MatOut vascular rendering onto the short-window evaluation lattice. Vessel pixels are defined as a peak-based core mask on that registered localization map, and background pixels are defined as the immediately surrounding shell. This creates a bounded but genuinely localization-derived vessel-core versus perivascular-shell endpoint on real in-vivo brain IQ, with labels derived from microbubble localization microscopy rather than from the Doppler de-

Regime	Penalty active / inactive bundles	median ρ_{shrink}	background-tail rate at pre-penalty $\tau_{10^{-3}}$ (pre→post)	background-tail clusters at $\tau_{10^{-3}}$ (pre→post)
Twinkling calculi (PRF 500; $L_t = 8$)	50/0	0.102	$1.00 \times 10^{-3} \to 9.38 \times 10^{-4}$	888 → 832

Table 9: Compact shrink-only penalty audit on the artifact-heavy Twinkling calculi regime. Under the prespecified short-ensemble profile, the shrink-only penalty is active on all 50 bundles (49 safety-only, 1 uplift-eligible), and the protected set is preserved exactly (maximum absolute score change = 0 on protected pixels). The fixed-threshold audit uses the pre-penalty threshold $\tau_{10^{-3}}$ chosen so that the background proxy satisfies $\mathbb{P}\{S_{\text{det,pre}} \geq \tau_{10^{-3}}\} = 10^{-3}$; at that same threshold, the shrink-only penalty modestly compresses the background tail without score inflation. This is a quantitative hygiene audit, not a primary ROC claim.

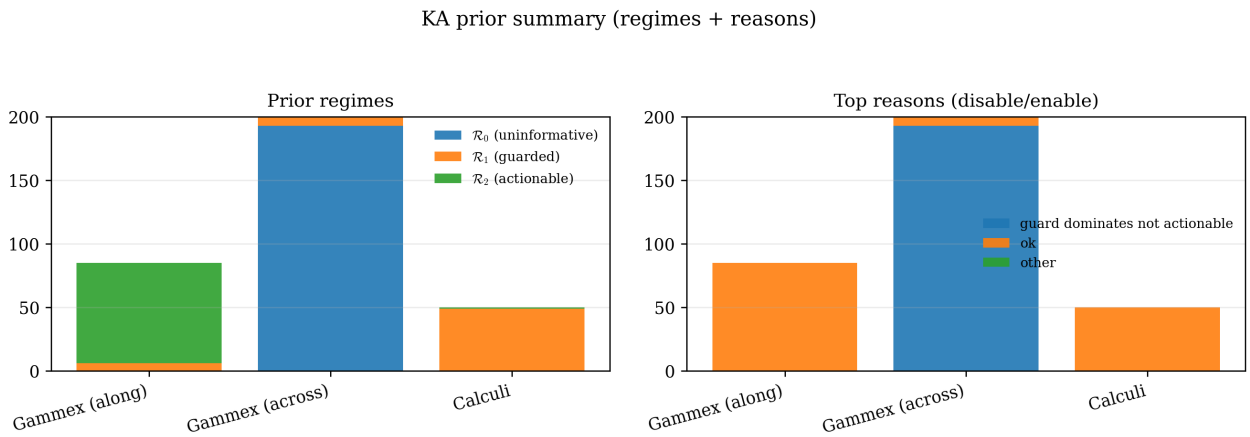

Figure 8: Twinkling shrink-only prior summary panel: prior (region, reason) distributions across Gammex (structural-label) and calculi (artifact-heavy) runs. Reproducibility details are provided in Appendix M.

tector being evaluated. The one-time PALA-to-IQ registration is used only to construct the labels; the frozen short-window detector itself runs without additional frame-to-frame registration.

The completed 10-block / 20-window audit uses the fully whitened matched-subspace specialist because that is the strongest short-window head on the corrected vessel-core versus perivascular-shell endpoint. On short 64-frame windows, the specialist reaches AUC 0.526 [0.514, 0.536], compared with 0.503 [0.493, 0.513] for PD and 0.502 [0.492, 0.512] for Kasai. At matched 70% vessel-core recall, the same specialist slightly lowers perivascular-shell false positives from 0.701 [0.687, 0.715] for PD and 0.704 [0.689, 0.717] for Kasai to 0.690 [0.670, 0.710]. The effect is bounded and modest in absolute magnitude, but it is directionally consistent. This detector choice is also endpoint-specific: the corrected PALA-backed structural audit rewards concentration on the localization-derived vascular core relative to the immediately adjacent shell, whereas the held-out SIMUS nuisance-control benchmark favors the stronger fixed downstream robustness of the simpler matched-subspace default. A direct latency check on the same frozen 64-frame no-registration configuration yields 41.3 ms steady-state total on an RTX 4080 SUPER for a 64 ms acquisition budget. This is still a bounded same-acquisition structural audit using a localization-derived vascular reference rather than independent multimodal validation, and it should be interpreted as a

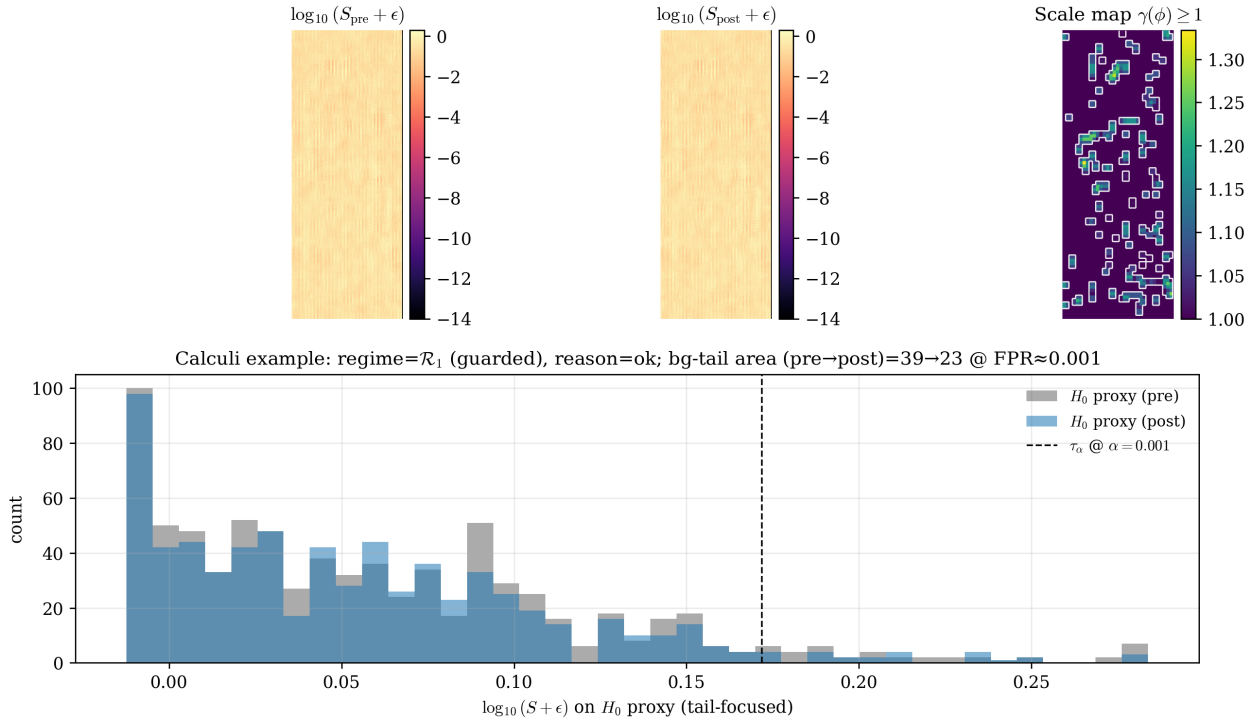

Figure 9: Representative calculi tail example (pre/post shrink-only regularizer) showing shrink-only tail hygiene behavior when the shrink-only prior is active ($w < 1$). Reproducibility details are provided in Appendix M.

structural vascular-fidelity result rather than as a functional-activation or hemodynamic-recovery benchmark.

To reduce overlap between localization-derived label construction and scored windows, we also ran a stricter disjoint-source variant in which odd blocks were used only for localization-derived reference construction and even blocks were used only for scored-window evaluation under the same frozen 64-frame no-registration profile.

As a separate falsification check, we keep the detector outputs fixed and alter only the localization-derived structural masks. A large cyclic translation serves as a misregistration stress test, while a support-matched permutation redraws vessel-core and shell labels at random inside the same anatomical support. The permutation control is the cleaner chance-like test because a translated vascular mask can still land on other vessel-rich anatomy within the same acquisition.

PD-only and Whole-Brain Retrospectives

This block collects the whole-brain Macé/Urban audits, which are useful for band-energy, alias-gating, and threshold-discipline retrospectives but are not treated as primary Doppler-IQ efficacy benchmarks.

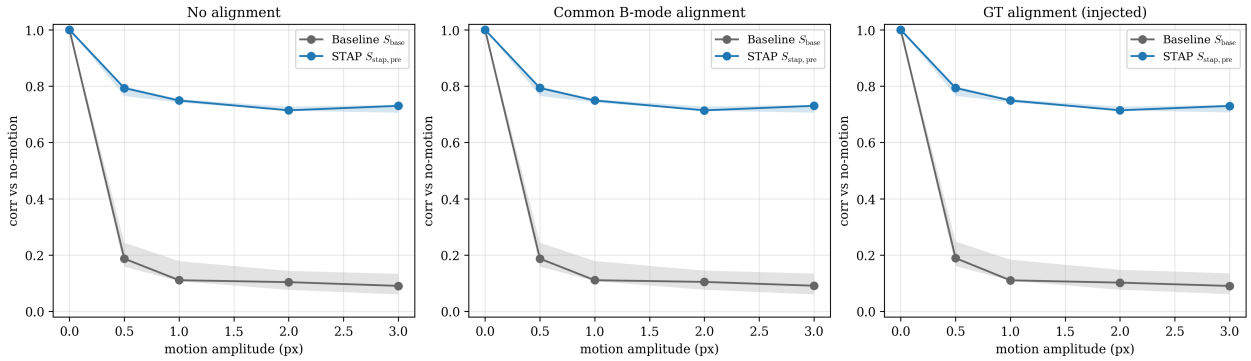

Figure 10: ULM 7883227 motion sweep under `elastic` (nonrigid-only) motion with the same frozen baseline/profile used in the main paper (`blocks 1--3`, frames 0:128, baseline MC–SVD with frozen energy fraction $e = 0.975$). Motion is injected at the beamformed IQ stage as a robustness stress test; it does not model channel-domain decorrelation physics. Reproducibility details are provided in Appendix M.

Score	AUC (H1 vs H0)	TPR@$\alpha = 10^{-1}$	TPR@$\alpha = 10^{-2}$	TPR@$\alpha = 10^{-3}$
Baseline (power Doppler)	0.503 [0.493,0.513]	0.105 [0.097,0.113]	0.009 [0.007,0.011]	0.000 [0.000,0.000]
Baseline (log-power Doppler)	0.503 [0.493,0.513]	0.105 [0.097,0.113]	0.009 [0.007,0.011]	0.000 [0.000,0.000]
Baseline (Kasai lag-1 magnitude)	0.502 [0.492,0.512]	0.106 [0.098,0.114]	0.008 [0.007,0.011]	0.000 [0.000,0.000]
Matched-subspace detector (whitened specialist)	0.526 [0.514,0.536]	0.147 [0.141,0.152]	0.023 [0.018,0.027]	0.003 [0.001,0.007]

Table 10: Structural-label ROC on real in vivo rat-brain IQ from ULM Zenodo 7883227. For each evaluation block, vessel and background masks are derived once from an external super-resolved vascular reference derived by registering the published PALA rat-brain ULM example back onto the Zenodo IQ grid and downsampling the registered MatOut rendering to the short-window evaluation lattice; short 64-frame windows from the held-out block are then scored without additional frame-to-frame registration on the same MC–SVD residual cube using different downstream detector heads. Entries report window-level means with 95% bootstrap CIs over windows. Reproducibility details are provided in Appendix M.

C.12 Macé whole-brain fUS: hemodynamic band-energy statistics and hold-out check

We treat the Macé/Urban whole-brain dataset as a PD-only, open-skull analogue of the open-skull brain pilot and compute tile-level hemodynamic band-energy statistics with fixed bands $P_f = [0, 0.5]$ Hz, guard $[0.5, 1.0]$ Hz, and $P_a = [1.0, 3.0]$ Hz on 8×8 tiles (stride 3). Offline we also compute atlas-derived *region strata* (e.g. VIS/SC/LGd versus CP/CA/DG) for descriptive summaries; these strata are not treated as ground-truth flow labels and are not used as shrink-only regularizer inputs. Figure 13 summarizes a systematic sweep in "Pf occupancy vs alias separation" space. We provide an explicit atlas-alignment sanity check in Appendix G. Here PD-z denotes a per-tile change-from-baseline z-score on the PD time series (distinct from the Doppler/IQ PD-mode score maps S_{PD} exported by the matched-subspace processing pipeline).

Hold-out discipline (avoid plane selection bias and label-tuned thresholds). To address the selection-bias critique and to avoid per-plane PD-z thresholds chosen using atlas strata (e.g.

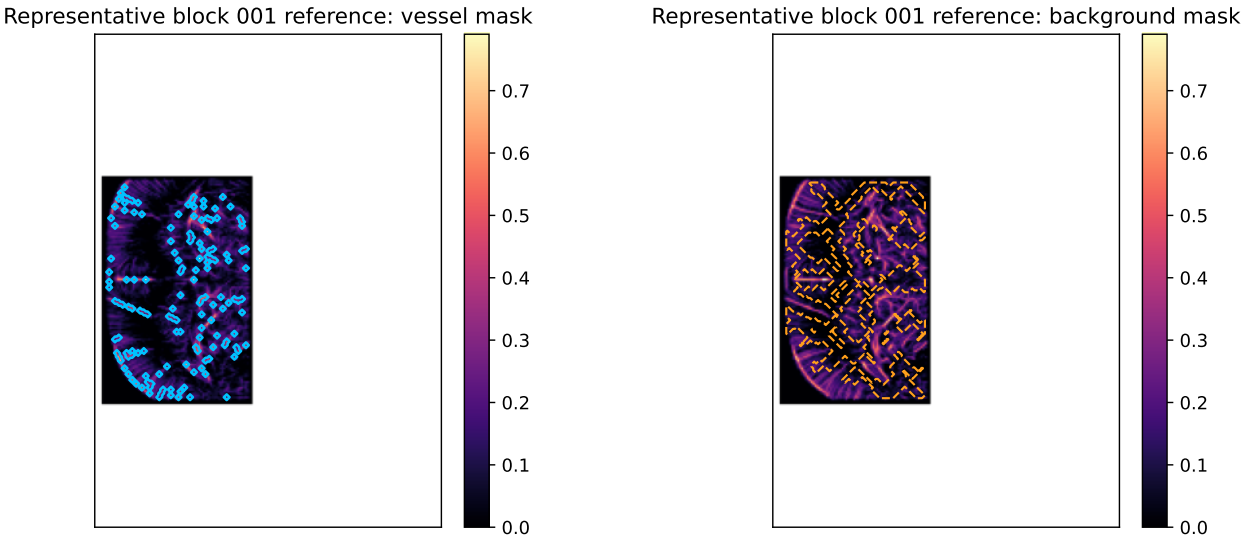

Figure 11: Representative localization-derived structural masks for ULM Zenodo 7883227. The underlying reference is obtained by registering the published PALA rat-brain PowDop example back to the Zenodo IQ field of view and downsampling the corresponding published super-resolution MatOut vascular rendering onto the short-window evaluation lattice. Left: vessel-core mask derived from local maxima of that registered localization map. Right: perivascular-shell mask surrounding the core. These masks are fixed before the short-window detector scores are evaluated.

setting thresholds to force a fixed activation fraction in a preferred stratum), we add a scan-level hold-out evaluation (reproducibility details in Appendix M). We (i) define label-free proxy sets from a mean-PD vascularity mask and set a per-plane PD-z operating point by a fixed background-proxy tail rule $\mathbb{P}\{\text{PD-}z \geq \tau \mid \mathcal{T}_{\text{bg}}\} = \alpha_{\text{pd}}$ (no atlas labels), (ii) define the alias suppression threshold as a fixed background-proxy quantile of $m_{\text{alias}} = \log((E_{\text{Pa}} + \epsilon)/(E_{\text{Pf}} + \epsilon))$, and (iii) select Pf-rich planes in held-out scans using a label-free band-energy statistic (Pf-peak fraction on flow-proxy tiles), with the selection threshold fixed from the training scans. Table 13 summarizes the resulting held-out counts on `scan4`–`scan5` (training: `scan1/2` and `scan3/6`; note `scan2` duplicates `scan1` and `scan6` duplicates `scan3` in this loader). This appendix-only check is intended as a conservative retrospective sanity check on threshold discipline, not as a primary efficacy result.

On held-out scans (`scan4`–`scan5`), this fixed, shrink-only alias suppression reduces proxy activations in the CP/CA/DG stratum while only modestly affecting activations in the VIS/SC/LGd stratum at the chosen label-free operating point. At $\alpha_{\text{pd}} = 0.01$ on the background proxy, aggregate (VIS/SC/LGd / CP/CA/DG hit counts) change from 23/142 to 22/121 (VIS/SC/LGd retention ≈ 0.96, CP/CA/DG reduction = 21 counts). On the Pf-peak-selected subset of held-out planes ($\text{PfPeak}_{\text{flow}} \geq \tau_{\text{sel}}$), only two planes pass selection and the descriptive counts are $19/0 \rightarrow 19/0$; this subset therefore supports the no-peeking selection discipline and perfect retained hits on the selected held-out planes, but it is too small to support a separate false-positive reduction claim. We interpret the hold-out result as a conservative safety-regularization behavior rather than an efficacy claim: Macé is PD-only and provides no Doppler-side flow labels, so its primary role here is to validate the band-energy/proxy machinery and the no-peeking threshold discipline.

Score	AUC (H1 vs H0)	TPR@10^{-3}	shell FPR @70% core recall
Baseline (power Doppler)	0.502 [0.495,0.508]	2.6e-4 [5.9e-5,5.5e-4]	0.704 [0.695,0.713]
Baseline (Kasai lag-1 magnitude)	0.500 [0.494,0.507]	2.0e-5 [0,5.9e-5]	0.709 [0.701,0.718]
Fixed matched-subspace head	0.510 [0.503,0.517]	3.7e-4 [1.8e-4,6.1e-4]	0.709 [0.700,0.719]
Adaptive head	0.512 [0.505,0.519]	3.7e-4 [1.8e-4,6.1e-4]	0.704 [0.693,0.716]
Fully whitened specialist	0.518 [0.511,0.525]	0.005 [0.003,0.006]	0.695 [0.684,0.707]

Table 11: Disjoint-source robustness variant of the ULM localization-derived structural audit. Localization-derived reference construction uses odd source blocks only, while scored windows come from even source blocks only, under the same frozen 64-frame no-registration profile used in the main audit. Entries report window-level means with 95% bootstrap intervals over the 70 evaluated windows. At block level, the fully whitened specialist has higher AUC than PD, Kasai, fixed, and adaptive on 10/10, 10/10, 9/10, and 7/10 evaluation blocks, respectively.

Mask condition	Score	AUC (H1 vs H0)	TPR@10^{-3}	shell FPR @50% core recall
Intended masks	Baseline (power Doppler)	0.504 [0.498,0.510]	7.9e-5 [0,2.4e-4]	0.496 [0.485,0.506]
	Baseline (Kasai lag-1 magnitude)	0.503 [0.496,0.509]	5.9e-5 [0,1.8e-4]	0.496 [0.485,0.506]
	Fully whitened specialist	0.530 [0.524,0.537]	0.004 [0.003,0.006]	0.461 [0.450,0.473]
Large cyclic translation, roll (32, 24)	Baseline (power Doppler)	0.537 [0.531,0.543]	0.004 [0.003,0.005]	0.461 [0.451,0.470]
	Baseline (Kasai lag-1 magnitude)	0.539 [0.532,0.545]	0.002 [0.001,0.003]	0.455 [0.445,0.464]
	Fully whitened specialist	0.509 [0.501,0.517]	0.004 [0.003,0.005]	0.495 [0.482,0.507]
Support-matched permuted masks	Baseline (power Doppler)	0.498 [0.494,0.501]	0.001 [0.001,0.001]	0.503 [0.497,0.510]
	Baseline (Kasai lag-1 magnitude)	0.497 [0.494,0.501]	0.001 [0.001,0.001]	0.504 [0.498,0.510]
	Fully whitened specialist	0.501 [0.499,0.504]	0.002 [0.001,0.002]	0.500 [0.495,0.504]

Table 12: Falsification controls for the ULM localization-derived structural audit. The first block reports the intended vessel-core versus perivascular-shell masks. The second applies a large cyclic translation to both masks without changing their size. The third redraws vessel and background labels at random inside the same support mask while preserving the per-block class counts. All rows use the same frozen 64-frame no-registration whitened profile and the same MC–SVD residual windows; only the structural masks change. The translated-mask block is a misregistration control rather than a pure chance control, because the shifted masks can still land on other vessel-rich anatomy within the same acquisition. The support-matched permutation is therefore the cleaner chance-like falsification for testing whether the reported structural signal depends on vascular alignment rather than generic score morphology.

Pixel-level separability vs. the vascular atlas (independent structural labels). To add one real-data quantitative check with an *independent* label source (not derived from PD), we map the dataset-provided Allen atlas vascular volume `atlas.Vascular` into each Macé scan plane using the affine in `Transformation.mat`. We threshold `atlas.Vascular` at 0.5 to define vascular vs. non-vascular pixels and restrict evaluation to pixels that map inside the atlas and to non-background atlas regions. This yields $n_{\text{neg}} \approx 1.7 \times 10^4$ non-vascular pixels per plane (median $fpr_{\min} \approx 5.9 \times 10^{-5}$), so FPR targets down to 10^{-4} are empirically meaningful at pixel level. As expected for PD-only hemodynamics, extreme-tail separability is weak; the purpose of this check is independent-label auditability and an empirical FPR scale rather than a performance headline.

We then evaluate three families of scalar maps: (i) a structural baseline based on the mean PD image (reported as a right-tail score using a left-tail convention, $S_{\text{inv_meanPD}} = -\log(\overline{P}+\epsilon)$), (ii) PD-z as a per-pixel change-from-baseline z-score on log-PD (the maximum z-score after an early baseline segment), and (iii) a hemodynamic alias band-energy statistic $m_{\text{alias}} = \log((E_a + \epsilon)/(E_f + \epsilon))$ computed from simple FFT band energies on log-PD with fixed hemodynamic bands $P_f = [0, 0.5]$ Hz

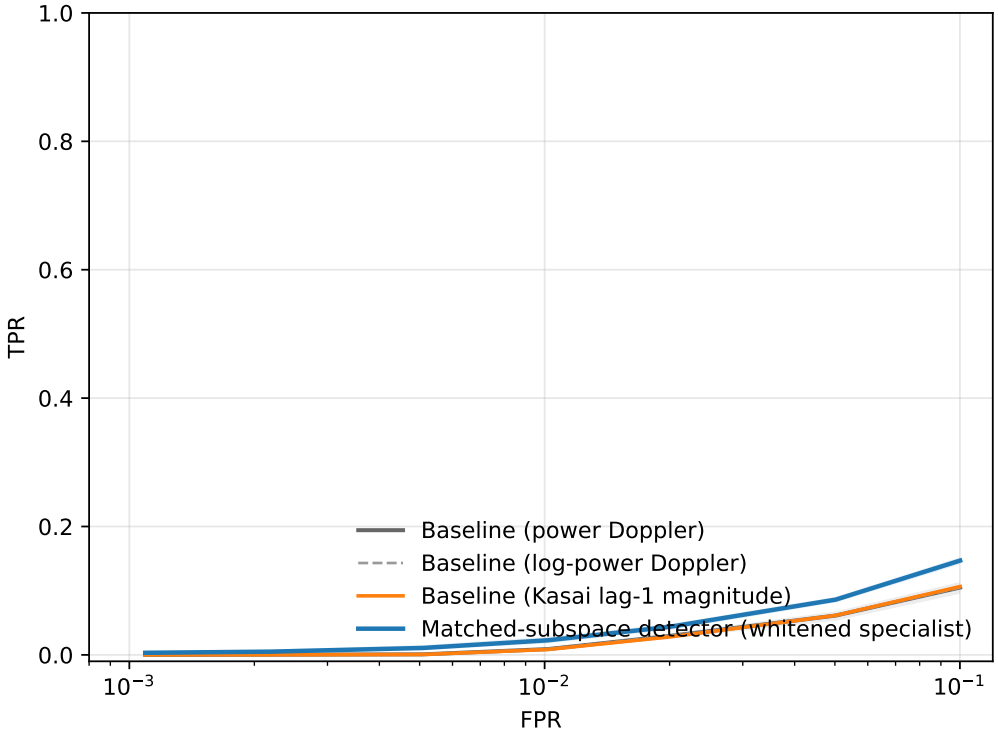

Figure 12: ROC curves for the bounded localization-derived structural audit on ULM Zenodo 7883227 (10 evaluation blocks, 20 short windows). The vessel-core and perivascular-shell labels come from the published PALA localization rendering after registration back to the Zenodo IQ grid and are clipped to anatomical support; the short-window detector scores are evaluated on the identical MC–SVD residual cube. On this corrected endpoint, the fully whitened matched-subspace specialist is modestly but consistently stronger than PD and Kasai.

and $P_a = [1.0, 3.0]$ Hz. Finally, to mirror the conservative posture on real data, we apply a label-free alias penalty as a shrink-only score-space regularizer on PD-z: protect a flow-proxy set defined by a high mean-PD quantile, then down-weight non-proxy pixels whose m_{alias} lies above the 0.90 quantile of m_{alias} on the background proxy. Table 14 summarizes the resulting low-FPR operating points.

PD-only prior summary panel. To keep PD-only Macé reporting aligned with the same audit posture used for Doppler IQ, we summarize the PD-only shrink-only prior outcomes over all planes using a compact summary panel that mirrors the Doppler-IQ summary but operates per plane. The panel reports (region, reason) histograms and shrink-only coverage/strength band-energy summaries under label-free proxy sets (Appendix H), and presents atlas-stratified summaries only as a clearly separated offline/descriptive panel.

Real-IQ Supporting Audits (continued)

The remaining real-IQ material returns to beamformed complex-IQ datasets after the PD-only whole-brain retrospective.

Held-out subset (scan4–scan5)	n	offline VIS/ SC/LGd / CP/ CA/DG hits (pre)	offline VIS/ SC/LGd / CP/ CA/DG hits (post)	VIS/SC/ LGd retention	CP/CA/ DG reduction
All planes	40	23/142	22/121	0.957	21
Selected ($\mathrm{PfPeak}_{\mathrm{flow}} \geq \tau_{\mathrm{sel}}$)	2	19/0	19/0	1.000	0

Table 13: Macé hold-out alias-gate evaluation with label-free thresholding. For each plane we set the PD-z threshold τ by a fixed background-proxy tail rule $\mathbb{P}\{\text{PD-}z \geq \tau \mid \mathcal{T}_{\mathrm{bg}}\} = \alpha_{\mathrm{pd}}$ with $\alpha_{\mathrm{pd}} = 0.01$, and set the alias suppression threshold as the 0.8 quantile of $m_{\mathrm{alias}} = \log((E_{\mathrm{Pa}} + \epsilon)/(E_{\mathrm{Pf}} + \epsilon))$ on the same background proxy set. "Selected" planes are those whose Pf-peak fraction on flow-proxy tiles exceeds a training-derived threshold τ_{sel} (the 0.95 quantile of $\mathrm{PfPeak}_{\mathrm{flow}}$ over the training scans `scan1/2` and `scan3/6`; $\tau_{\mathrm{sel}} \approx 0.055$). Hit counts use atlas strata (VIS/SC/LGd and CP/CA/DG) only for offline descriptive evaluation (not shrink-only regularizer inputs and not a Doppler efficacy claim). Because only two held-out planes satisfy the selection rule in this loader, the selected-row result should be read as a stability/safety check rather than a precise effect-size estimate. Reproducibility details are provided in Appendix M.

C.13 Shin RatBrain Fig3 (LOCA-ULM): real kHz IQ translation and shrink-only regularizer diagnostics

We use the LOCA-ULM RatBrain Fig3 dataset released with Shin et al. as a real kHz IQ audit of end-to-end ingest, band-energy summaries, and shrink-only regularizer behavior [36, 37]. This is a microbubble ULM IQ dataset, so we do not treat it as a physiological fUS benchmark. Instead, we use it to ask a narrower translational question: can the full pipeline run reproducibly on real beamformed IQ, and do the resulting score maps behave sensibly under label-free proxy checks? The Fig3 archive contains 80 clips (`IQData001.dat`–`IQData080.dat`) and a `SizeInfo.dat` file describing the tensor shape. Each clip stores $T = 250$ slow-time frames on a (238×130) spatial grid as little-endian float64 arrays containing I then Q in column-major order (matching the authors' reference MATLAB loader); the effective slow-time sampling is 1 kHz (post-compounding), with center frequency $f_0 \approx 15.6$ MHz [37]. Because there are no independent flow/background labels, we do not report ROC improvement here. Instead, we report label-free proxy separation, prior-regime stability, shrink-only invariants, and motion-robustness summaries.

Processing workflow. We mirror the authors' loader and convert IQ clips into a common analysis format used throughout this work (reproduction details in Appendix M). For a chosen clip and time window we:

- Apply an SVD clutter filter on the Casorati matrix $X \in \mathbb{C}^{T \times HW}$ to form a baseline PD-derived map pd_{base} and a residual IQ cube (the input to matched-subspace detector).
- Construct pixel-level proxy masks from pd_{base} (used here as a conventional PD-brightness proxy $\mathrm{PD}^{\uparrow}$ distinct from the ROC score maps): a flow/vessel proxy (top quantile; minimum pixel count) and a background proxy (complement of the flow mask in a central crop).
- Run the temporal matched-subspace detector core on the residual IQ cube with fixed tile geometry (8×8 tiles, stride 3) and a longer slow-time aperture $L_t = 64$ suited to 1 kHz slow time (bin spacing 15.6 Hz), producing a pre-regularizer PD-mode score map.

Score (vascular vs. non-vascular)	TPR@10^{-4} ($\times 10^{-3}$)	TPR@3×10^{-4} ($\times 10^{-3}$)	TPR@10^{-3} ($\times 10^{-3}$)
$S_{\text{inv_meanPD}} = -\log(\overline{P}+\epsilon)$ (structural baseline)	0.00 (0.00, 0.00)	0.00 (0.00, 1.04)	2.76 (1.84, 3.20)
PD-z (log-PD; max z after baseline)	0.00 (0.00, 0.00)	0.00 (0.00, 0.81)	0.94 (0.00, 1.78)
$m_{\text{alias}} = \log((E_a+\epsilon)/(E_f+\epsilon))$ (hemodynamic alias statistic)	0.00 (0.00, 0.00)	0.00 (0.00, 0.75)	1.05 (0.00, 2.04)
PD-z + label-free alias penalty (shrink-only)	0.00 (0.00, 0.00)	0.00 (0.00, 0.81)	0.92 (0.00, 1.74)

Table 14: Macé pixel-level quantitative evaluation using an independent vascular atlas label source (`atlas.Vascular`; $n = 80$ planes from four deduplicated scans). Entries report median (IQR) TPR at fixed FPR targets for vascular vs. non-vascular pixels. The median negative-pixel count per plane is $\approx 1.69 \times 10^4$ (median $fpr_{\min} \approx 5.9 \times 10^{-5}$). Reproducibility details are provided in Appendix M.

- Apply the shrink-only regularizer (described in the main paper) to produce a region index, reason label, and (when active) a non-increasing post-regularizer score map together with gate/scale maps.
- Record tilewise band-energy statistics (peak frequency, guard fraction, alias/guard risk metrics) and the penalization maps in the same analysis format used throughout this work.

The proxy masks are used only for label-free proxying/protection; prior-regime decisions and hygiene/robustness proxies follow the same right-tail score convention used elsewhere in this manuscript: operating points are expressed as a threshold on a right-tail PD score map, i.e. $S_{\text{PD}} \geq \tau$ at fixed background-proxy tail rate.

Scale-map convention. When the shrink-only prior is active ($w < 1$ on a nontrivial set), we record an equivalent per-pixel scale factor $\gamma(i) \geq 1$ and apply it to the PD score map as

$$\text{PD}_{\text{post}}(i) = \text{PD}_{\text{pre}}(i)/\gamma(i), \qquad \gamma(i) \geq 1,$$

which makes the PD score map non-increasing on penalized regions. For any fixed threshold τ this is a pure shrink-only penalty; when thresholds are recalibrated post-regularizer to match a target background tail rate, any TPR change reflects background-tail suppression rather than score inflation.

Shin operating settings and one-time baseline calibration. To avoid retuning the main brain-Doppler setting to Shin until the results look favorable, we treat Shin as its own prespecified operating regime and calibrate only the baseline clutter filter once. We sweep the MC–SVD energy-fraction cutoff over a calibration subset (10 representative clips; window 0:128) and select a single baseline setting based on band-energy integrity and stability, not on apparent shrink-only regularizer activity. Table 15 reports the corresponding prior-regime counts for transparency, but selection is driven by the label-free integrity and background-tail metrics (Δ_{bf}, Δ_{tail}) together with overall stability. Overly aggressive baseline removal increases Pf peak occupancy but can invert the proxy-separation signal and suppress the more selective prior regime. We therefore freeze MC–SVD with energy fraction ≈ 0.97 as a balanced setting for the Shin-U configuration (reproducibility details in Appendix M).

Energy frac	$\mathcal{R}_1/\mathcal{R}_2$	Δ_{bf} med	Δ_{tail} med	Pf-peak(flow) med	BG clusters Δ	BG area Δ
0.90	8/2	+1.52	+0.65	0.011	−17	−32
0.95	8/2	+1.50	+0.65	0.019	−17	−32
0.97	8/2	+1.49	+0.73	0.050	−19	−45
0.975	9/1	−0.38	+0.77	0.237	−30	−89
0.98	10/0	−0.38	+0.78	0.253	−30	−89
0.99	10/0	−0.47	+0.58	0.510	−6	−59

Table 15: Baseline clutter-filter calibration for the frozen Shin-U profile (10 clips; window 0:128). Δ_{bf} is the median background-vs-flow proxy separation of the risk metric; Δ_{tail} is the background tail association (actionability) metric; Pf-peak(flow) is the fraction of flow-proxy tiles (tiles with high flow-mask coverage) whose PSD peak lies in Pf. BG cluster/area deltas are computed from a fixed background-proxy tail operating point on the PD score map S_{PD} (clusters/area after shrink-only regularizer minus before shrink-only regularizer). We freeze energy fraction ≈ 0.97 because it yields nontrivial Pf realization while preserving positive actionability.

Baseline comparisons (proxy separability on real IQ). Although Shin provides no independent H_1/H_0 labels, it is useful to test whether the matched-subspace detector core improves *proxy* separation on real complex IQ. We run a fixed scenario matrix on 10 clips and three overlapping windows (0:128, 64:192, 122:250), exporting right-tail score maps for MC–SVD, RPCA, HOSVD, matched-subspace detector (on the MC–SVD residual), and an *matched-subspace-only* ablation that runs matched-subspace detector directly on raw/registered IQ. Proxy masks are derived deterministically from the MC–SVD baseline PD map via a `pd_auto` quantile rule (0.995 quantile within depth fraction $[0.25, 0.85]$ with dilation), with the generic default-mask union disabled; masks are then held fixed across methods. We report (i) AUC between flow-proxy and background-proxy score distributions (threshold-free), and (ii) flow-proxy hit rate at matched background tail rates $\alpha \in \{10^{-1}, 10^{-2}, 3\times10^{-3}, 10^{-3}, 3\times10^{-4}, 10^{-4}\}$ (Figure 15; Table 16). In this setup $n_{\text{bg}} \approx 2.6\times10^4$ per scenario ($fpr_{\min} \approx 3.8\times10^{-5}$), so $\alpha = 10^{-4}$ corresponds to roughly 2–3 background pixels.

Across scenarios, matched-subspace detector improves proxy separation (median AUC 0.951 vs 0.922 for MC–SVD) and yields nonzero flow-proxy hit rates at strict tails where MC–SVD is typically zero (e.g. median hit rate 0.070 at $\alpha = 10^{-3}$). The matched-subspace-only ablation is comparable to matched-subspace detector-on-residual, indicating that the localized matched-subspace detector core is the primary driver of the improvement on this dataset. At moderate strictness ($\alpha \approx 10^{-2}$), RPCA/HOSVD can exhibit higher proxy hit rates than matched-subspace detector despite lower AUC, consistent with a broader, lower-contrast score elevation rather than strict-tail selectivity. Finally, a conditioning-target sweep ($\kappa_{\text{target}} = 80$ vs 200) produced identical matched-subspace detector score maps under this frozen profile, indicating that the conditioning target is non-binding in this particular configuration. Figure 17 shows a representative qualitative comparison.

Measured rigid-motion telemetry. Because Shin clips are short (≈ 0.25 s) and there are no external motion sensors, we treat rigid registration telemetry as a descriptive motion proxy rather than a full motion benchmark. Figure 16 summarizes the phase-correlation shifts estimated during baseline preprocessing (RMS shift magnitude per scenario) and relates them to the proxy AUC difference (matched-subspace detector minus MC–SVD). In these windows the estimated rigid motion is typically subpixel, so we do not treat Shin as a stratified motion-stress regime; the

	MC–SVD proxy mask			Consensus proxy mask		
Method	AUC (med [IQR])	hit@$\alpha = 10^{-3}$ (med [IQR])	hit@$\alpha = 10^{-4}$ (med [IQR])	AUC (med [IQR])	hit@$\alpha = 10^{-3}$ (med [IQR])	hit@$\alpha = 10^{-4}$ (med [IQR])
MC–SVD	0.922 [0.916, 0.934]	0.000 [0.000, 0.023]	0.000 [0.000, 0.000]	0.913 [0.908, 0.924]	0.000 [0.000, 0.029]	0.000 [0.000, 0.000]
RPCA	0.902 [0.893, 0.907]	0.024 [0.000, 0.031]	0.008 [0.000, 0.016]	0.909 [0.898, 0.915]	0.028 [0.000, 0.036]	0.010 [0.000, 0.020]
HOSVD	0.881 [0.873, 0.906]	0.040 [0.030, 0.056]	0.008 [0.000, 0.015]	0.883 [0.874, 0.909]	0.055 [0.036, 0.069]	0.010 [0.000, 0.019]
matched-subspace detector	0.951 [0.948, 0.956]	0.070 [0.034, 0.081]	0.014 [0.006, 0.025]	0.944 [0.932, 0.947]	0.051 [0.018, 0.082]	0.009 [0.003, 0.032]
matched-subspace-only	0.948 [0.944, 0.951]	0.065 [0.024, 0.078]	0.020 [0.003, 0.032]	0.939 [0.933, 0.944]	0.065 [0.012, 0.083]	0.021 [0.000, 0.033]

Table 16: Shin RatBrain Fig3 proxy-metric summary over 30 scenarios (10 clips $\times$ 3 windows) under the frozen Shin-U profile (PRF = 1 kHz, $L_t = 64$, MC–SVD energy fraction ≈ 0.97). AUC is computed between flow-proxy and background-proxy score distributions; hit@α is the flow-proxy hit rate when the threshold is chosen to achieve background tail rate α. Metrics are shown under two proxy-mask definitions: an MC–SVD-derived mask and a consensus mask (majority vote over MC–SVD/RPCA/HOSVD masks).

proxy-separation gains are not explained by measured rigid shifts.

Proxy circularity audit. To reduce dependence on any single baseline-derived proxy mask, we report metrics under (i) a flow-proxy mask derived from MC–SVD (pd_{base} via `pd_auto`) and (ii) a consensus proxy mask formed by majority vote over MC–SVD/RPCA/HOSVD `pd_auto` masks; rankings are consistent under this cross-mask audit.

Cross-mask audit (proxy-label sensitivity). Table 16 includes both the baseline-derived mask and the consensus mask results.

Profile variants (Shin-U vs Shin-S/L). In addition to Shin-U, we evaluated two alternative frozen band settings (Shin-S and Shin-L) designed to push the "flow" band Pf toward lower frequencies and increase Pf peak occupancy. Across the same 10-clip calibration subset, these variants typically increase Pf-peak(flow) but weaken the label-free background-tail evidence that would justify a more selective prior response. We therefore treat Shin-S/L as conservative alternatives and use Shin-U as the primary real-IQ audit setting reported here.

Shrink-only regularizer behavior, stability, and hygiene endpoints. Under the frozen Shin-U setting (MC–SVD energy fraction ≈ 0.97, matched-subspace detector $L_t = 64$, fixed Pf/Pa bands, and shrink-only regularizer enabled in `auto` mode), we run the pipeline on 10 pre-selected clips (`IQData001`–`IQData005`, `IQData010`, `IQData020`, `IQData040`, `IQData060`, `IQData080`) and evaluate stability across three overlapping time windows (0:128, 64:192, 122:250). Because Pf realization is weak on this dataset, Shin is not treated as a setting where shrink-only regularizer should deliver measurable performance improvement. Instead, we use it to test determinism, shrink-only invariants, and label-free map-hygiene summaries. Because there are no labels, we report only label-free endpoints (reproducibility details in Appendix M):

- **Regularizer regime stability.** The conservative prior regime dominates in all windows (8–10 of 10 clips), while the more selective regime is rare (0–2 clips); all decisions emit auditable reason labels.
- **Shrink-only and protected invariance.** With the shrink-only regularizer active, the map receives a mild, localized shrink (median $p_{\text{shrink}} \approx 0.09$; median scaled pixel fraction ≈ 0.20) and preserves the protected set exactly (maximum absolute PD change on protected pixels is 0 in all evaluated clips).

Window	$\mathcal{R}_0/\mathcal{R}_1/\mathcal{R}_2$	Pf-peak(flow) med	guard q90 med	p_{shrink} med	scaled-px frac med	scale p90 (γ) med
0:128	0/8/2	0.050	0.104	0.091	0.201	1.167
64:192	0/10/0	0.032	0.110	0.097	0.198	1.168
122:250	0/9/1	0.039	0.108	0.090	0.205	1.167

Table 17: Shin RatBrain Fig3 prior-regime summary band-energy statistics for the frozen Shin-U profile ($e \approx 0.97$, $L_t = 64$) across three overlapping windows (10 clips each).

Window	background clusters (pre→post)	background area (pre→post)
0:128	$125 \to 106$	$263 \to 218$
64:192	$121 \to 104$	$264 \to 212$
122:250	$129 \to 124$	$264 \to 257$

Table 18: Shin RatBrain Fig3 label-free hygiene endpoints under Shin-U ($e \approx 0.97$; $L_t = 64$; 10 clips). "Pre" and "post" denote matched-subspace score-map outputs before and after the non-increasing shrink-only prior penalty. The background-tail metric targets flow-like tail activations under the right-tail score convention: for each clip/window instance we choose τ so that the background proxy satisfies $\mathbb{P}\{S_{\text{PD}} \geq \tau \mid \text{bg proxy}\} = 10^{-3}$ on the corresponding pre-regularizer map, then count connected components (clusters) and pixels (area) with $S_{\text{PD}} \geq \tau$. Reproducibility details are provided in Appendix M.

- **Background-tail hygiene.** A label-free background-tail metric shows consistent tail compression on early windows: for each clip/window we choose a threshold τ so that the background proxy satisfies $\mathbb{P}\{S_{\text{PD}} \geq \tau \mid \text{bg proxy}\} = 10^{-3}$ on the pre-regularizer map, then count clusters/area of pixels satisfying $S_{\text{PD}} \geq \tau$ before vs. after shrink-only regularizer. Total background-tail clusters decrease from $125 \to 106$ (window 0:128) and $121 \to 104$ (window 64:192), while the late window is closer to neutral ($129 \to 124$). We interpret this as improved map hygiene rather than as a detection-improvement claim.

Table 17 summarizes prior-regime counts and key band-energy summaries across windows.

All-clips shrink-only regularizer regime summary (n=80). To reduce dependence on the 10-clip subset, we also run a diagnostics-only sweep over all 80 clips (window 0:128), recording the prior regime outputs. In this sweep we skip the matched-subspace detector pass and do not apply the score-space shrink-only regularizer mapping; we treat the result as an activation-rate audit (region/reason frequencies and band-energy summary statistics), not a performance result. On all clips the prior lies in $\mathcal{R}_1$ for 73/80 and in $\mathcal{R}_2$ for 7/80 (all with `reason=ok`, risk mode `alias`); median Pf-peak(flow)≈ 0.091, guard $q_{0.90} \approx 0.105$, and median predicted $p_{\text{shrink}} \approx 0.095$. Reproducibility details are provided in Appendix M.

Synthetic motion robustness sweeps (label-free). To probe motion robustness on real IQ without ground-truth labels, we implement controlled motion-injection sweeps on Shin IQ cubes (reproduction details in Appendix M). In addition to simple per-frame subpixel translations, global phase modulation, and spatially smooth low-frequency clutter tones, we define a *brain-like* motion model designed to be mechanistically comparable to the structured-clutter pilot: a bulk rigid translation (correctable by phase-correlation registration) plus a smooth, depth-decaying elastic warp with temporally correlated coefficients (not removable by translation-only registration). Motion is applied directly to the beamformed IQ cube (a stress test rather than a full scatterer-level

Window	$\mathrm{corr}_{\mathrm{self}}$ base	$\mathrm{corr}_{\mathrm{self}}$ matched-subspace detector (pre-regularizer)	frac(matched-subspace detector>base) $\mathrm{corr}_{\mathrm{self}}$	hit $\mathrm{rate}_{\mathrm{flow}}$ base @ bg tail $= 10^{-3}$	hit $\mathrm{rate}_{\mathrm{flow}}$ matched-subspace detector @ bg tail $= 10^{-3}$	frac(matched-subspace detector>base) hit $\mathrm{rate}_{\mathrm{flow}}$
0:128	0.188	0.454	1.0	0.000	0.041	1.0
64:192	0.186	0.399	0.9	0.000	0.040	1.0
122:250	0.182	0.364	0.8	0.000	0.041	1.0

Table 19: Shin RatBrain Fig3 motion-sweep endpoints at 2 px amplitude (median over 10 clips) under the frozen Shin-U profile, with rigid registration enabled and shrink-only regularizer disabled (matched-subspace detector reported pre-regularizer). $\mathrm{corr}_{\mathrm{self}}$ is Pearson correlation between each method's PD-mode score map at the given motion amplitude and its own no-motion reference, after phase-correlation alignment and a boundary crop. hit $\mathrm{rate}_{\mathrm{flow}}$ is computed on a fixed, method-independent flow-proxy mask derived from the no-motion baseline PD-derived map pd_{base}: for each run/method we choose τ so that the background proxy achieves a tail rate 10^{-3} under the right-tail score rule $\mathbb{P}\{S_{\mathrm{PD}} \geq \tau \mid \text{bg proxy}\} = 10^{-3}$, then evaluate the fraction of flow-proxy pixels satisfying $S_{\mathrm{PD}} \geq \tau$. frac(matched-subspace detector>base) reports the paired fraction of clips where matched-subspace detector exceeds MC–SVD at 2 px on the corresponding endpoint.

simulator), after which we rerun the same baseline and matched-subspace processing pipeline (the shrink-only regularizer disabled to isolate matched-subspace scoring).

We report two label-free robustness endpoints as functions of motion amplitude: (i) PD-mode score-map correlation versus the no-motion reference (after rigid alignment and a boundary crop), and (ii) a flow-proxy hit rate at a matched background-proxy tail operating point (background-proxy tail rate $= 10^{-3}$), expressed in score space as $\mathbb{P}\{S_{\mathrm{PD}} \geq \tau \mid \text{bg proxy}\} = 10^{-3}$. Concretely, for each clip/window we define a fixed pixel-level flow-proxy mask from the no-motion baseline PD-derived map pd_{base} (top quantile; minimum pixel count) and a background-proxy mask from its complement; for each run/method we choose τ on that run's background proxy and then report the fraction of flow-proxy pixels satisfying $S_{\mathrm{PD}} \geq \tau$. As a degeneracy check (to rule out "stability by collapse"), the sweep also records non-collapse diagnostics (map std/entropy and background variance ratios). A post-hoc sensitivity sweep recomputes correlation under different crop margins and with alignment disabled. Over 10 pre-selected clips and three overlapping time windows (0:128, 64:192, 122:250) under the frozen Shin-U profile (PRF = 1 kHz, $L_t = 64$, baseline MC–SVD with rigid registration enabled), matched-subspace score maps are consistently more stable than MC–SVD as motion increases: at a 2 px motion amplitude, median correlation improves from ≈ 0.18–0.19 (MC–SVD) to ≈ 0.36–0.45 (matched-subspace detector) across windows, and the flow-proxy hit rate improves from 0 to ≈ 0.04 at bg tail rate $= 10^{-3}$. Table 19 summarizes these endpoints; Figures 18–19 show motion curves.

Figure 18 shows the canonical window (0:128) motion curves, while Figure 19 shows the same endpoints across all three overlapping windows; Figure 20 provides a rigid-registration ablation (`reg0`).

Figure 21 shows the corresponding shrink-only prior summary panel for the frozen Shin-U profile. Figure 22 shows a representative per-clip multipanel view separating the baseline PD-derived map, the matched-subspace score map pre-regularizer, the shrink-only regularizer gate/scale, and the final PD-mode score map post-regularizer, along with key band-energy maps and the flow-proxy mask that defines the protected set.

Limitations and claim boundaries (Shin). The Shin RatBrain Fig3 analysis is intended as an end-to-end *real-IQ translation check*, not a clinical detection benchmark. Because there are no task labels and no ground-truth vessel segmentation, quantitative endpoints reported here are label-free proxies: baseline comparisons use deterministic flow/background proxy masks and report both

Method / score	Brain-OpenSkull TPR@10^{-4}	Brain-OpenSkull TPR@3×10^{-4}	Brain-OpenSkull TPR@10^{-3}	Brain-AliasContract TPR@10^{-4}	Brain-AliasContract TPR@3×10^{-4}	Brain-AliasContract TPR@10^{-3}	Brain-SkullOR TPR@10^{-4}	Brain-SkullOR TPR@3×10^{-4}	Brain-SkullOR TPR@10^{-3}
MC–SVD	0	0	0	0	0	0	0	0	0
Adaptive SVD (spatial singular-vector similarity cutoff)	0	0	0	0	0	0	0	0	0
Block-wise local SVD (overlap-add)	0	0	0	0	0	0	0	0	0
RPCA	0.000 (0.000,0.005)	0.002 (0.000,0.009)	0.004 (0.004,0.017)	0	0	0	0	0	0
HOSVD	0.000 (0.000,0.002)	0.002 (0.002,0.002)	0.005 (0.005,0.008)	0.002 (0.002,0.002)	0.002 (0.002,0.002)	0.003 (0.002,0.005)	0.000 (0.000,0.002)	0.000 (0.000,0.002)	0.002 (0.000,0.003)
Matched-subspace detector on MC–SVD residual	0.073 (0.001,0.087)	0.129 (0.001,0.222)	0.222 (0.002,0.437)	0.610 (0.479,0.676)	0.687 (0.609,0.749)	0.830 (0.785,0.882)	0.587 (0.501,0.668)	0.720 (0.605,0.779)	0.873 (0.819,0.876)

Table 20: Brain-* k-Wave low-FPR operating points using flow-evidence detection scores (higher = more likely flow): baseline power Doppler and the matched-subspace detector. Each regime uses five non-overlapping 64-frame windows (offsets 0/64/128/192/256); each window has $n_{\text{neg}} = 50199$ ($fpr_{\min} \approx 1.99 \times 10^{-5}$). At this support, FPR targets 10^{-4}, 3×10^{-4}, and 10^{-3} correspond to approximately 5, 15, and 50 false-positive pixels per window. Entries report median (IQR) TPR across windows at each target FPR. Cells shown as 0 indicate no measurable sensitivity at that supported operating point. Baselines compare clutter suppression strategies (MC–SVD; adaptive SVD with a spatial singular-vector similarity cutoff; block-wise local SVD with overlap-add; RPCA; HOSVD) and are scored by power Doppler; the matched-subspace detector is applied on top of the MC–SVD residual.

threshold-free proxy separation (AUC) and tail-calibrated flow-proxy hit rates, while motion-sweep endpoints measure within-method consistency relative to each method's own no-motion reference and persistence of a fixed flow proxy under a matched background-proxy tail operating point. In addition, the "brain-like" motion stress test is an image-domain warp applied to beamformed IQ (rigid + depth-decaying elastic components) rather than a scatterer-level forward model; it is designed to mimic the specific high-rank, spatially varying residual-motion mismatch represented in the structured-clutter pilot, not to reproduce full ultrasound physics. Accordingly, we interpret Shin results as evidence of (i) deterministic real-IQ ingestion and band-energy summaries, (ii) shrink-only prior auditability and invariants (shrink-only; protected tiles unchanged), and (iii) self-consistency under controlled residual-motion corruption, while leaving ROC improvement claims to labeled simulation regimes and to future raw-IQ datasets with task or landmark structure.

Supplementary Simulation and Held-Out Benchmark Details

This block collects the detailed simulation tables and held-out synthetic benchmark checkpoints that support, but do not replace, the main-text reader path.

C.14 Labeled brain simulation strict-tail operating points

The detailed labeled brain simulation low-FPR operating-point table is placed here so that the main text can foreground the narrower use-case-motivated synthetic *SIMUS-ClinUtility-v2* checkpoint. These simulation pilots remain important as a labeled strict-tail stress test and as the basis for the detector-component ablations that follow.

Secondary Legacy Fairness Checkpoints

These tables remain useful for fairness and version-control context, but they are secondary to the canonical held-out use-case-motivated benchmark in the main paper.

Method	Frozen config	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$
MC–SVD → matched-subspace detector	`Brain-SIMUS-Clin-MotionMidRobust-v0`	0.936	0.997	0.000
MC–SVD	`ef95`	0.807	0.076	0.992
HOSVD	`ef95_ds2_t32`	0.753	0.097	0.973
RPCA	`lam1_it250_ds2_t32_r4`	0.736	0.064	0.992
Local SVD (Fixed Energy)	`tile16_s4_ef95`	0.717	0.426	0.652
Adaptive Local SVD	`tile12_s4_bal_r8`	0.688	0.058	1.000
Adaptive Global SVD	`conservative_r8`	0.651	0.054	0.998

Table 21: Corrected SIMUS/PyMUST moving-scatterer frozen-benchmark summary on held-out corrected actual cases (seeds 23–24). Each method family is given one frozen configuration chosen on the corrected development split (seeds 21–22) using the same selection protocol; labels match the implementation taxonomy used in the benchmark scripts. The matched-subspace detector is reported as the chained pipeline MC–SVD → matched-subspace detector using the pre-regularizer matched-subspace score, while the baseline rows report the corresponding baseline score.

C.15 Corrected SIMUS/PyMUST moving-scatterer benchmark (frozen v1)

To complement the deterministic labeled k-Wave brain simulations with a moving-scatterer forward model, we freeze a corrected SIMUS/PyMUST benchmark after fixing the SIMUS motion-scaling bug uncovered during the motion-ladder audit. At a high level, the bug mis-scaled some simulated motion amplitudes, so the absolute nuisance pressure in earlier exploratory motion-stress runs was not trustworthy. After correcting that path, we refroze the benchmark reported here and the narrower *SIMUS-ClinUtility-v2* benchmark used in the main text. The protocol here is intentionally symmetric across methods: using corrected SIMUS cases from seeds 21–22 as a development split, we select one frozen configuration per method family, then evaluate those fixed choices on held-out corrected cases from seeds 23–24. The matched-subspace detector is scored with its stored pre-regularizer matched-subspace score, while baselines are scored with their native baseline maps. This broader corrected benchmark should be read as a moving-scatterer realism and fairness checkpoint, not as the canonical translational benchmark. The main paper's SIMUS claims are carried instead by the corrected *SIMUS-ClinUtility-v2* results, the corrected held-out stress frontier, and the corrected fixed-calibration transfer audit. Any pre-correction SIMUS numbers from older drafts are superseded and should be ignored.

For readers who prefer a simple decomposition, the pipeline can be viewed in two stages. First, an upstream clutter-suppression method produces a residual slow-time cube. Second, a downstream score is computed on that residual. In this benchmark the upstream methods include global and adaptive-global SVD, local and adaptive-local SVD, RPCA, and HOSVD [11,12,15,31,44,45]. The downstream scores are conventional power Doppler or Kasai on the baseline side, and the matched-subspace score on the proposed side [16, 53, 65]. Table 21 freezes one full processing chain per method family. Table 23 then asks the stricter fairness question: if the upstream residual is held fixed and only the downstream score is changed, does matched-subspace scoring still help?

C.16 Use-case-motivated synthetic SIMUS-ClinUtility-v2 checkpoint: latency versus performance

Tables 21 and 23 establish method-family fairness on the broader corrected moving-scatterer checkpoint. Table 24 reports the narrower use-case-motivated synthetic structural checkpoint used for translational interpretation, and the secondary same-cube functional audit appears later in this appendix. The *SIMUS-ClinUtility-v2* track is intentionally narrower than the corrected frozen-v1

Method	scope	cutoff / estimator	tuned?	score	specified
MC–SVD	global	fixed energy fraction ($e = 0.90$)	tune-once	PD	Appendix M
Adaptive SVD (SV similarity cutoff)	global	rank via spatial singular-vector similarity drop	deterministic	PD	Appendix K.3
Block-wise local SVD (overlap-add)	local	tile-local energy fraction ($e = 0.90$) + overlap-add	deterministic	PD	Appendix K.4
RPCA	global	PCP (IALM) with fixed iterations/tolerance	deterministic	PD (sparse)	Appendix K
HOSVD	global	tensor energy fractions (default $0.99, 0.99, 0.99$)	deterministic	PD (residual)	Appendix K
matched-subspace detector (pre-regularizer)	local	robust covariance → whitening → flow-subspace ratio	prespecified setting	matched-subspace	main Methods

Table 22: Compact family summary for the labeled brain simulation pilot baselines. Baseline rows share the same PD flow-evidence score on their respective residual cubes; matched-subspace detector differs by applying tile-local covariance whitening and a matched-subspace flow-band score on top of the MC–SVD residual.

Residualizer	Frozen config	Native head	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$	matched-subspace detector head	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$
MC–SVD	`ef95`	Kasai	0.865	0.126	0.950	`MotionMidRobust-v0`	0.938	0.997	0.000
Adaptive Global SVD	`conservative_r8`	Kasai	0.694	0.172	0.896	`MotionMidRobust-v0`	0.829	0.991	0.000
Local SVD (Fixed Energy)	`tile16_s4_ef95`	Kasai	0.788	0.611	0.308	`MotionMidRobust-v0`	0.764	0.997	0.000
Adaptive Local SVD	`tile12_s4_bal_r8`	Kasai	0.732	0.148	0.959	`MotionMidRobust-v0`	0.803	0.993	0.001
RPCA	`lam1_it250_ds2_t32_r4`	Kasai	0.777	0.163	0.916	`MotionMidRobust-v0`	0.898	1.000	0.000
HOSVD	`ef95_ds2_t32`	Kasai	0.800	0.342	0.729	`MotionMidRobust-v0`	0.891	1.000	0.000

Table 23: Same-residual score comparison on the corrected SIMUS/PyMUST held-out cases (seeds 23–24). Each upstream residualizer is frozen to the configuration selected on the corrected development split (seeds 21–22). The "native head" is the better of PD and Kasai, chosen on that same development split; in this benchmark Kasai is selected for every residualizer family. The matched-subspace column applies the same frozen matched-subspace score to the same residual cube, rather than changing the upstream residualizer. This is the strict fairness audit: it gives conventional clutter filters their strongest simple Doppler score while asking whether matched-subspace scoring still improves separation on the identical residual. In five of six residualizer families the matched-subspace detector improves both AUC measures and essentially eliminates nuisance-tail activation; for the fixed-energy local SVD residual, the native Kasai score retains slightly higher $\mathrm{AUC}_{\mathrm{main/bg}}$, but the matched-subspace detector still sharply improves nuisance separation ($\mathrm{AUC}_{\mathrm{main/nuis}}$ 0.611→0.997, nuisance FPR 0.308→0.000).

benchmark and uses prespecified nuisance settings informed by reported fUS operating conditions. Competitive structural evaluation is restricted to the mobile and intra-operative parenchymal settings; the more surface-dominated setting remains exploratory. For translational readers, the practical question is not only whether the matched-subspace detector improves nuisance separation, but how much runtime overhead is paid to obtain that improvement. Table 24 therefore summarizes the held-out structural checkpoint on an RTX 4080 SUPER using four representative operating points: the fastest strong non-matched-subspace reference (HOSVD→Kasai), the fastest matched-subspace point on the latency frontier (Adaptive Global SVD→matched-subspace detector), the same-residual native reference (RPCA→Kasai), and the highest-quality matched-subspace setting (RPCA→matched-subspace detector). These are held-out bundle runtimes for the SIMUS structural cases, not the replay-window latency protocol used for the brain simulations, Shin, and Gammex.

Structural stack	Total runtime (s)	matched-subspace detector add-on (s)	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} at $TPR_{main} = 0.5$
HOSVD → Kasai	0.167	–	0.753	0.561	0.453
Adaptive Global SVD → matched-subspace detector	0.342	0.256	0.813	0.874	0.078
RPCA → Kasai	1.783	–	0.837	0.442	0.593
RPCA → matched-subspace detector (headline)	2.155	0.470	0.828	0.920	0.031

Table 24: Clinically anchored *SIMUS-ClinUtility-v2* structural checkpoint on held-out evaluation seeds. "Total runtime" is the mean held-out bundle runtime on the RTX 4080 SUPER. The matched-subspace add-on is the mean extra matched-subspace stage runtime beyond the upstream residualizer. The frontier makes the translational tradeoff easier to read: the same-residual native reference (RPCA→Kasai) preserves slightly higher $AUC_{main/bg}$, but RPCA→matched-subspace detector sharply improves nuisance separation and reduces nuisance-tail activation by nearly an order of magnitude at the matched TPR operating point, while Adaptive Global SVD→matched-subspace detector provides a sub-0.35 s structural operating point that still improves all three structural summary metrics over the fastest strong non-matched-subspace reference (HOSVD→Kasai).

Clinically grounded stress frontier (held-out paper-tier SIMUS confirmation). One remaining fairness concern is that the held-out structural checkpoint may still be too clean: perfect or near-perfect summary metrics can invite the criticism that the task is trivial rather than that the detector is meaningfully better. To address that directly, we first ran a reduced-grid sweep on the same mobile scene family to identify the strongest residualizer families under two clinically motivated stress axes, then reran a held-out paper-tier confirmation restricted to those strongest families (RPCA and adaptive-global SVD). The two stress axes are closer to the nuisance mechanisms emphasized in clinical fUS reports than the earlier generic motion/reflectivity audit. The first is a cardiac-like 1.30 Hz low-frequency tissue-pulsation term whose amplitude is increased across reference, moderate, and hard rows; this is intended to stress the same tissue-motion regime highlighted by Demené *et al.* for SVD clutter filtering and by Baranger *et al.* for neonatal/open-skull Doppler [11, 12]. The second shortens the slow-time support from 64 frames to 48 and 32 frames to mimic clinically constrained acquisitions in which shorter ensembles are used to limit motion corruption or workflow burden [1, 23, 35]. This section asks a narrower question than the main benchmark. It does not replace the full-resolution prespecified benchmark in the main text, and it is not framed as a new universal ranking of all methods. Instead, it asks whether the detector family remains preferable to strong public baseline stacks once the acquisition is pushed toward pulsatile tissue leakage or reduced slow-time support.

Cardiac-like pulsation

Level / perturbation	Best detector-variant pipeline	Metrics (Baseline → Detector)
Ref.	RPCA	$AUC_{main/nuis}$: $0.381^{0.416}_{0.345} \rightarrow 0.911^{0.925}_{0.894}$
1.0x cardiac amplitude at 1.30 Hz	→ Fully whitened variant	FPR_{nuis}@0.5: $0.743^{0.842}_{0.652} \rightarrow 0.017^{0.027}_{0.008}$
Mod.	RPCA	$AUC_{main/nuis}$: $0.388^{0.423}_{0.354} \rightarrow 0.903^{0.920}_{0.886}$
2.0x cardiac amplitude at 1.30 Hz	→ Fully whitened variant	FPR_{nuis}@0.5: $0.668^{0.795}_{0.593} \rightarrow 0.030^{0.043}_{0.019}$
Hard	RPCA	$AUC_{main/nuis}$: $0.399^{0.433}_{0.365} \rightarrow 0.905^{0.921}_{0.890}$
3.0x cardiac amplitude at 1.30 Hz	→ Fully whitened variant	FPR_{nuis}@0.5: $0.609^{0.770}_{0.537} \rightarrow 0.027^{0.041}_{0.017}$

Short ensemble

Stress axis	Supplementary perturbation used here	Literature grounding
Cardiac-like pulsation	Reference: 1.0x cardiac amplitude at 1.30 Hz; Moderate: 2.0x cardiac amplitude at 1.30 Hz; Hard: 3.0x cardiac amplitude at 1.30 Hz	Stress axis motivated by cardiac-like tissue pulsation broadening the low-frequency tissue spectrum in neonatal and open-skull fUS, with Demené et al. identifying roughly 1–10 mm/s tissue motion as the critical operating regime for SVD clutter filtering.
Short ensemble	Reference: 64 frames (42.7 ms); Moderate: 48 frames (32.0 ms); Hard: 32 frames (21.3 ms)	Stress axis motivated by clinically constrained acquisitions in which shorter ensembles are used to limit motion corruption or workflow burden in intra-operative and mobile settings.

Table 25: Literature-grounded stress axes used in the held-out SIMUS stress regime. This analysis fixes attention to the residualizer families that dominated the preliminary reduced-grid sweep (RPCA and adaptive-global SVD), then reruns those clinically motivated perturbations on the held-out mobile setting. These perturbations are intended to approximate clinically relevant nuisance pressure rather than replay any single in vivo acquisition one-to-one.

Level / perturbation	Best detector-variant pipeline	Metrics (Baseline → Detector)
Ref. 64 frames (42.7 ms)	RPCA → Fully whitened variant	$\mathrm{AUC}_{\mathrm{main/nuis}}$: $0.381^{0.415}_{0.345} \rightarrow 0.911^{0.926}_{0.895}$ $\mathrm{FPR}_{\mathrm{nuis}}$@0.5: $0.743^{0.840}_{0.649} \rightarrow 0.017^{0.027}_{0.008}$
Mod. 48 frames (32.0 ms)	RPCA → Fully whitened variant	$\mathrm{AUC}_{\mathrm{main/nuis}}$: $0.390^{0.423}_{0.356} \rightarrow 0.846^{0.868}_{0.824}$ $\mathrm{FPR}_{\mathrm{nuis}}$@0.5: $0.754^{0.814}_{0.708} \rightarrow 0.058^{0.080}_{0.034}$
Hard 32 frames (21.3 ms)	RPCA → Fixed matched-subspace	$\mathrm{AUC}_{\mathrm{main/nuis}}$: $0.426^{0.462}_{0.392} \rightarrow 0.772^{0.796}_{0.746}$ $\mathrm{FPR}_{\mathrm{nuis}}$@0.5: $0.617^{0.727}_{0.536} \rightarrow 0.166^{0.194}_{0.139}$

Table 26: Held-out SIMUS mobile stress regime. Each row compares the detector-family stack with the strongest nuisance-control summary on the displayed endpoints (higher $\mathrm{AUC}_{\mathrm{main/nuis}}$, lower $\mathrm{FPR}_{\mathrm{nuis}}$@0.5) against the best conventional baseline among evaluated methods on the same held-out evaluation seed. On every row in this reduced table, that baseline was RPCA → PD. Point estimates are shown as $x^{\mathrm{hi}}_{\mathrm{lo}}$ with 95% nonparametric bootstrap intervals over the held-out masked score samples. Because this table uses one held-out evaluation seed per axis, these intervals describe within-seed score uncertainty rather than between-seed variation. The held-out reference regime is shown separately to identify the fixed default detector head.

The resulting pattern is stronger than the earlier generic stress audit. On every held-out row in supplementary Table 26, the strongest detector-family stack outperforms the strongest conventional baseline evaluated on that same stressed row. The row-wise winner is not uniform: the fully whitened variant takes all three cardiac-pulsation rows and the reference and intermediate short-ensemble rows, while the fixed matched-subspace head remains strongest only on the hardest 32-frame short-ensemble row. The main scientific use of this table is not to replace the direct per-head stress table in the main paper, but to show that once the nuisance is made more clinically realistic, the detector family still keeps a substantial nuisance-control advantage over the strongest conventional stacks evaluated under the same stress conditions.

Detector-family ablation on a fixed use-case-motivated synthetic residual. The held-out same-residual SIMUS detector-family ablation now appears in the main text as Table **??**. We do not repeat it here. Its role is to hold the MC–SVD residual fixed and change only the downstream head, which sharpens the detector-head claim independently of the end-to-end stack comparisons in Table 24.

Held-out functional same-cube detector-head audit. The held-out functional same-cube audit now appears in the main text as Table **??**. We do not repeat it here. Its role is narrower than the structural SIMUS benchmark: it fixes the residual stream and asks whether the detector head remains directionally positive under a common activation-versus-nuisance readout.

Functional and Translation-Support Audits

This block collects compact real-data score audits and PD-only whole-brain retrospective summaries that support the translational interpretation without serving as primary labeled efficacy benchmarks.

C.17 Compact real-data translation audit

To make the translational story less dependent on single point estimates, we summarize the existing real-data outputs with paired uncertainty and consistency summaries. On the Macé whole-brain phase-2 sweep, label-free alias gating reduces matched-TPR false positives by 18.6% (95% CI 16.8–20.2) while retaining 81.0% of tiles, and the plane-wise false-positive reduction win rate is 118/120. On the appendix-only held-out alias-gate check (`scan4`/`scan5`), false positives are reduced by 14.5% (95% CI 7.9–21.4) with a 97.5% safety win rate across held-out planes. These readout-level results are retrospective and label-free, but they strengthen the claim that nuisance gating can improve whole-brain map hygiene without aggressive hit loss [9, 48].

For the same-residual scoring claim itself, the most direct real-data test is simple: hold the residual cube fixed and compare PD, Kasai, and matched-subspace detector on the same real-IQ bundle. Coverage is intentionally narrow and explicit. On currently available ULM 7883227 bundles, matched-subspace detector is uniformly positive relative to both PD and Kasai under the same residual, whereas the currently available Shin bundles remain mixed and should be read as a low-motion real-IQ audit rather than a superiority dataset [36–39].

Dataset	Residual family	ΔAUC(matched-subspace detector−PD)	ΔAUC(matched-subspace
Shin RatBrain Fig3	raw	−0.012	+0.004
Shin RatBrain Fig3	MC–SVD	−0.134	−0.117
ULM 7883227 brainlike	MC–SVD-style residual	+0.281	+0.261
ULM 7883227 elastic	MC–SVD-style residual	+0.282	+0.251

Table 27: Compact same-residual score comparison on currently available real-IQ bundles. ΔAUC denotes matched-subspace detector minus the named conventional score on the same residual cube. Coverage is intentionally limited to bundles with active matched-subspace detector scores: Shin currently contributes only raw and MC–SVD residual families, while the ULM entries come from the existing MC–SVD-style motion-sweep bundles. The audit therefore supports a strong real-IQ case for matched-subspace detector on ULM, but only a mixed one on the currently available Shin bundles.

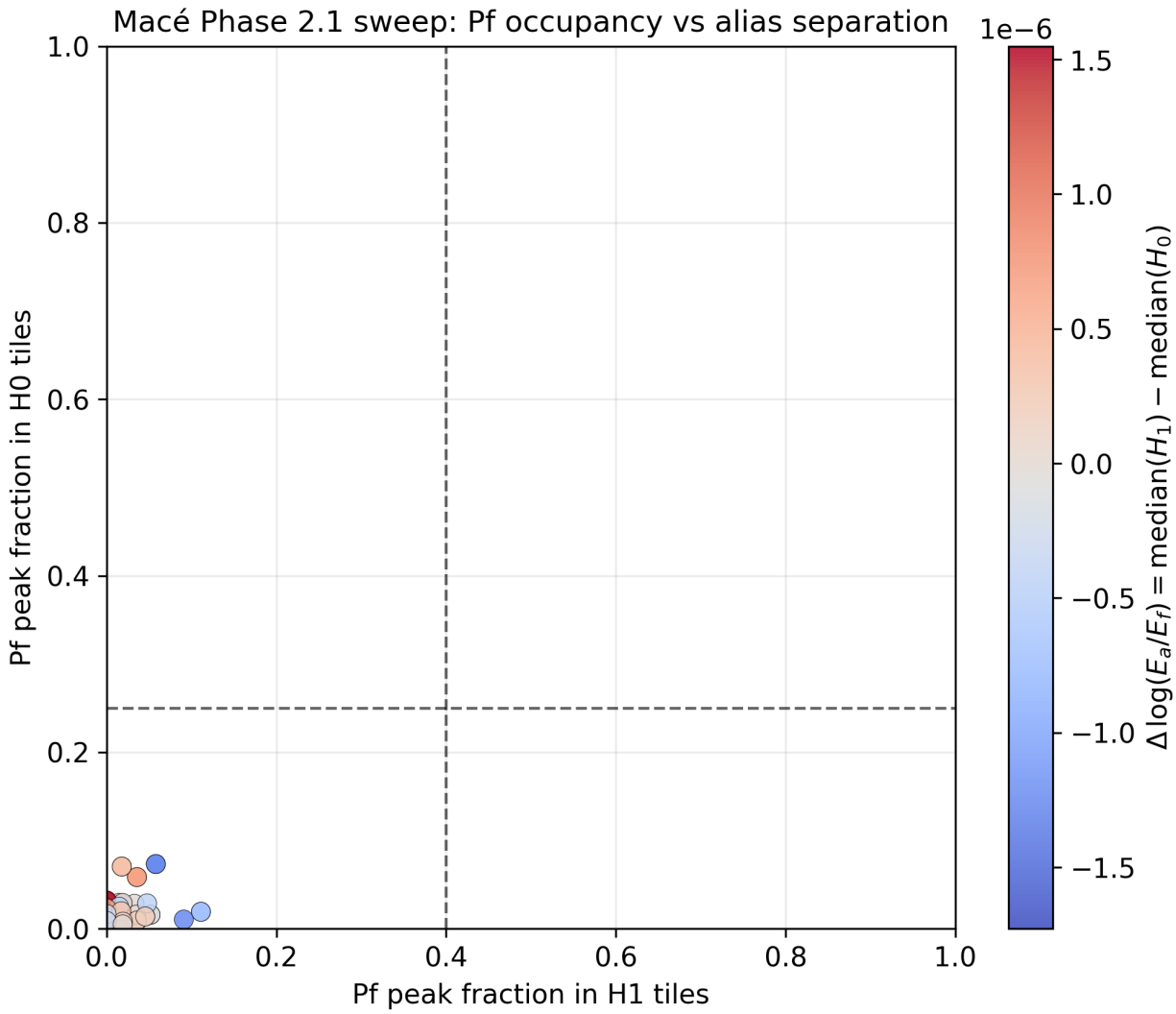

Figure 13: Macé systematic band-energy summary (six files; 84 planes; four unique acquisitions in this loader): Pf peak occupancy vs alias separation under a fixed hemodynamic matched-subspace profile (tiles 8×8, stride 3; $P_f = [0, 0.5]$ Hz, guard $[0.5, 1.0]$ Hz, $P_a = [1.0, 3.0]$ Hz). Each point is one (scan, plane) instance; the x-axis is the Pf peak fraction in the VIS/SC/LGd atlas stratum and the y-axis the Pf peak fraction in the CP/CA/DG stratum (offline, descriptive), and color encodes $\Delta \log(E_a/E_f) = \text{median}(\text{VIS/SC/LGd}) - \text{median}(\text{CP/CA/DG})$ (negative means alias ratios are lower in VIS/SC/LGd). Dashed lines indicate an offline descriptive threshold pair ($x > 0.4$, $y < 0.25$); they are not used for the held-out, label-free operating-point evaluation in Table 13.

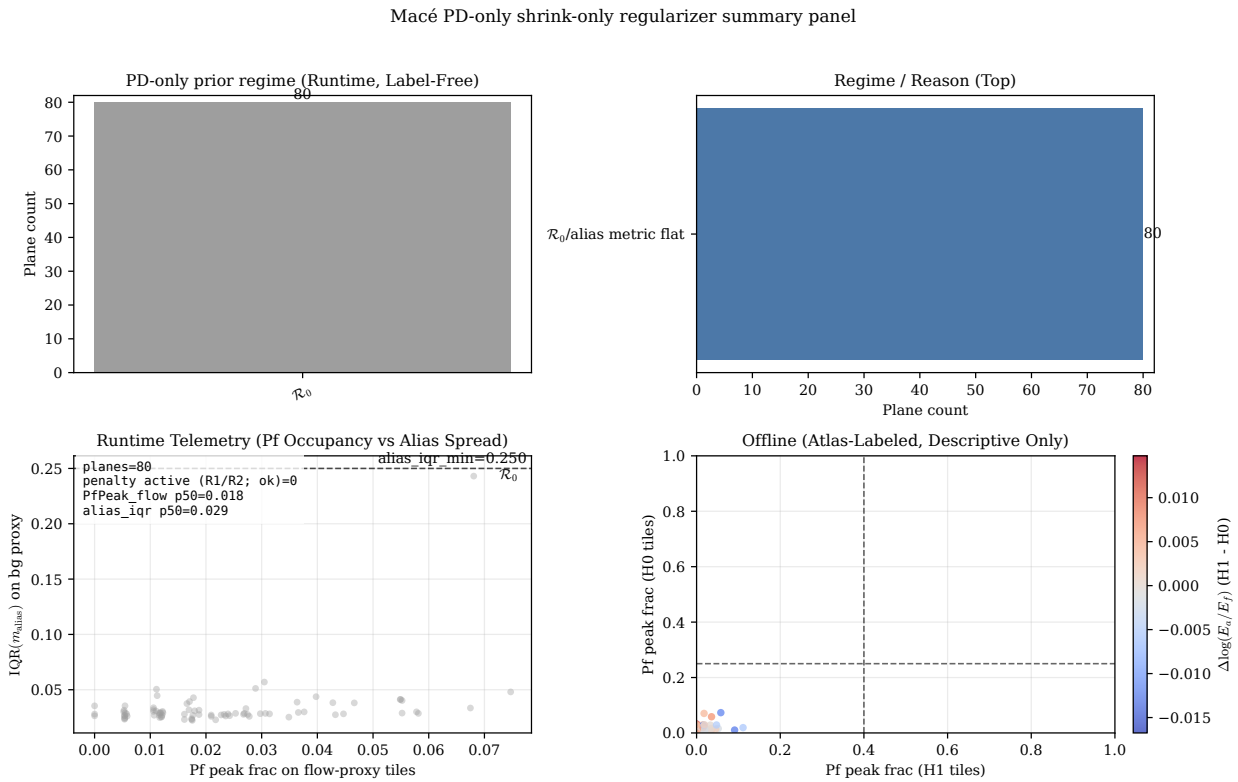

Figure 14: Macé PD-only prior summary panel over all deduplicated scan/plane instances. Top: prior regions and the most common (region, reason) pairs (runtime, label-free). Bottom-left: runtime label-free telemetry summarizing either shrink coverage/strength when penalization is active or, when the contract exits earlier, the upstream Pf-occupancy / alias-spread telemetry that drove the regime assignment. Bottom-right: offline atlas-stratified Pf-peak fractions (VIS/SC/LGd vs CP/CA/DG tiles) colored by $\Delta\log(E_a/E_f)$ (descriptive only; not a shrink-only regularizer input). Reproducibility details are provided in Appendix M.

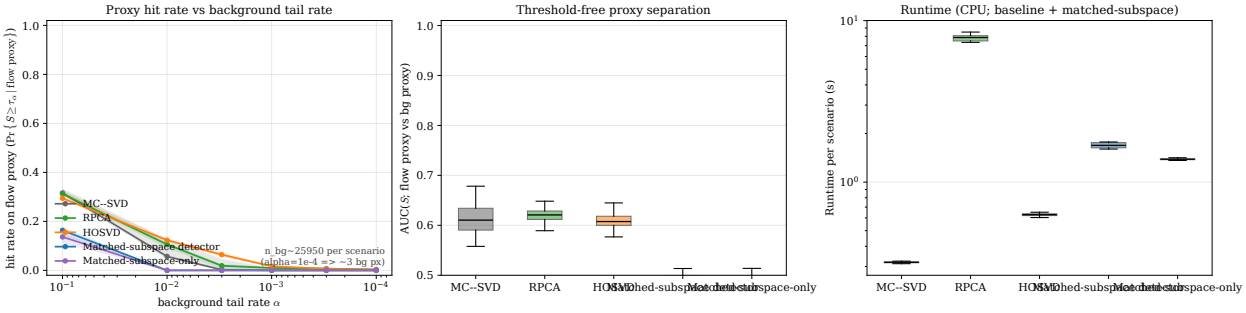

Figure 15: Shin RatBrain Fig3 baseline comparisons under label-free proxy masks (10 clips; three windows; Shin-U profile). Left: flow-proxy hit rate vs background tail rate α (median and IQR over scenarios). Middle: threshold-free AUC between flow-proxy and background-proxy score distributions. Right: runtime distribution per scenario in this CPU-only environment (baseline time for MC–SVD/RPCA/HOSVD; baseline+matched-subspace detector total time for matched-subspace detector variants), using compute-time telemetry captured in bundle metadata (excludes disk I/O and figure rendering).

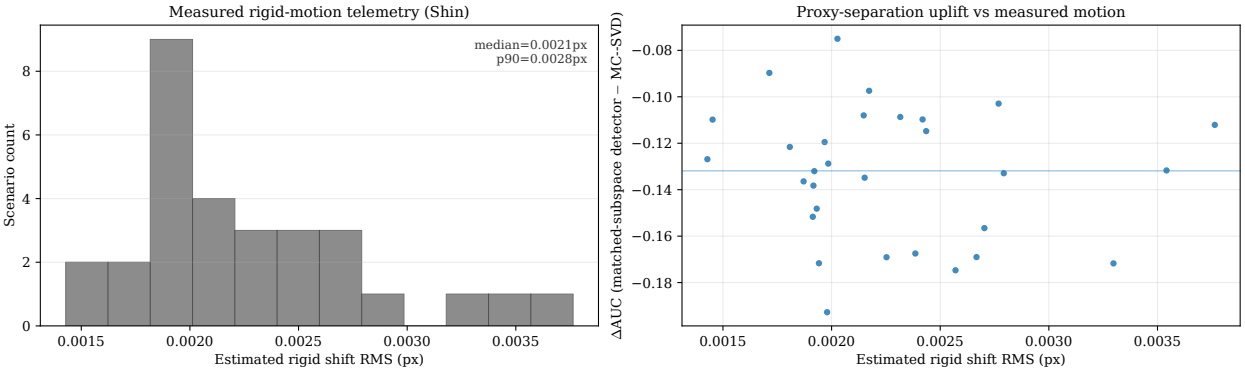

Figure 16: Shin RatBrain Fig3 measured rigid-motion telemetry across the 30 baseline-matrix scenarios (10 clips $\times$ 3 windows). Left: histogram of estimated rigid shift RMS (pixels) from phase-correlation registration used in baseline preprocessing. Right: proxy-separation difference ΔAUC (matched-subspace detector minus MC–SVD) versus the same shift statistic.

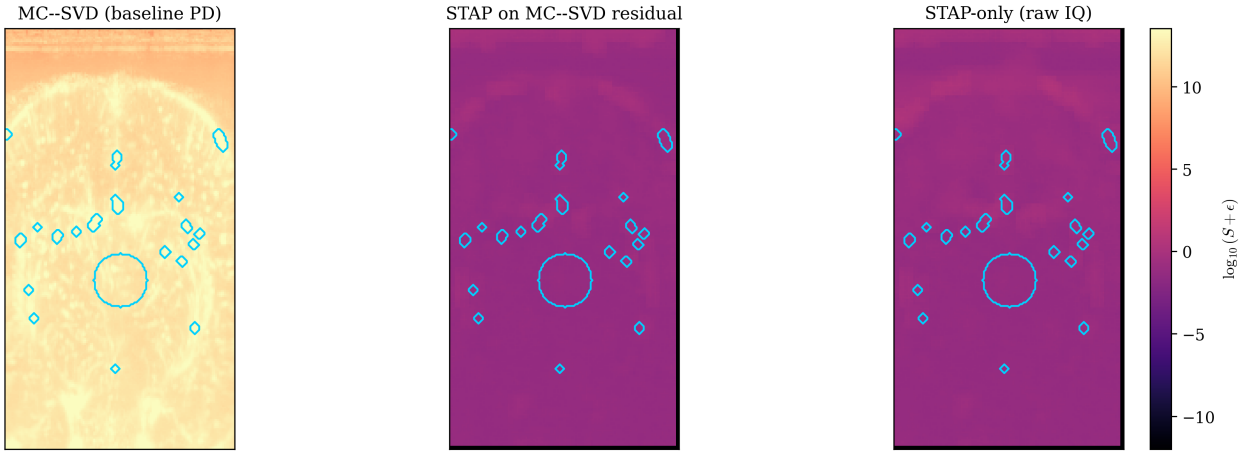

Figure 17: Representative Shin RatBrain Fig3 qualitative comparison (clip `IQData001`, window 0:128). Panels show log-scaled right-tail score maps for MC–SVD baseline PD, matched-subspace detector on the MC–SVD residual, and matched-subspace-only on raw/registered IQ, with the shared flow-proxy mask overlaid (cyan contour).

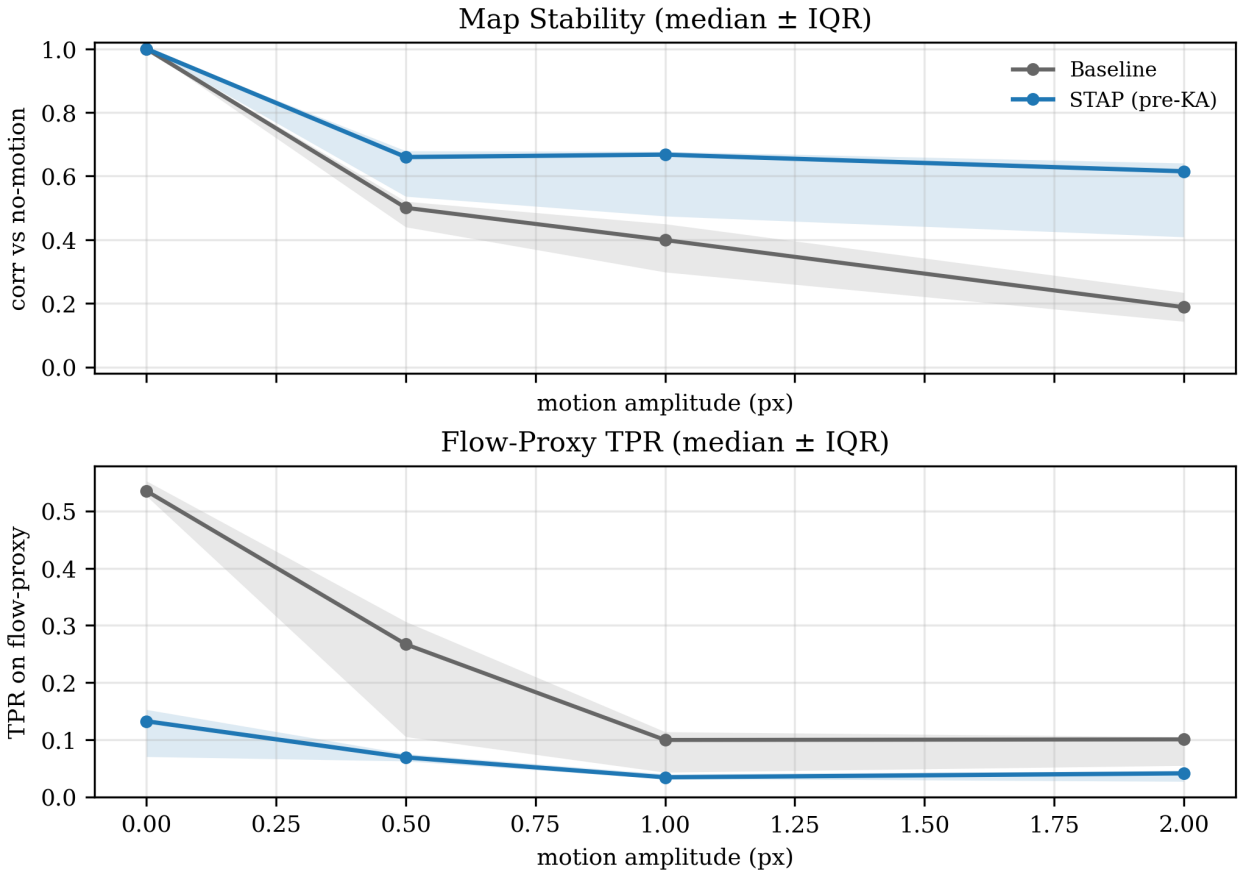

Figure 18: Shin RatBrain Fig3: brain-like residual-motion sweep (canonical window 0:128; 10 clips; median ± IQR) under the frozen Shin-U profile with rigid registration enabled and shrink-only regularizer disabled (matched-subspace detector reported pre-regularizer). Top: PD-mode score-map stability measured as Pearson correlation between each method's PD-mode score map at a given motion amplitude and its own no-motion reference, after phase-correlation alignment and a boundary crop. Bottom: flow-proxy hit rate (median ± IQR) at matched background-proxy tail rate $= 10^{-3}$, computed under the right-tail score rule $\mathbb{P}\{S_{\mathrm{PD}} \geq \tau \mid \text{bg proxy}\} = 10^{-3}$ with a fixed flow-proxy mask derived from the no-motion baseline pd_{base}.

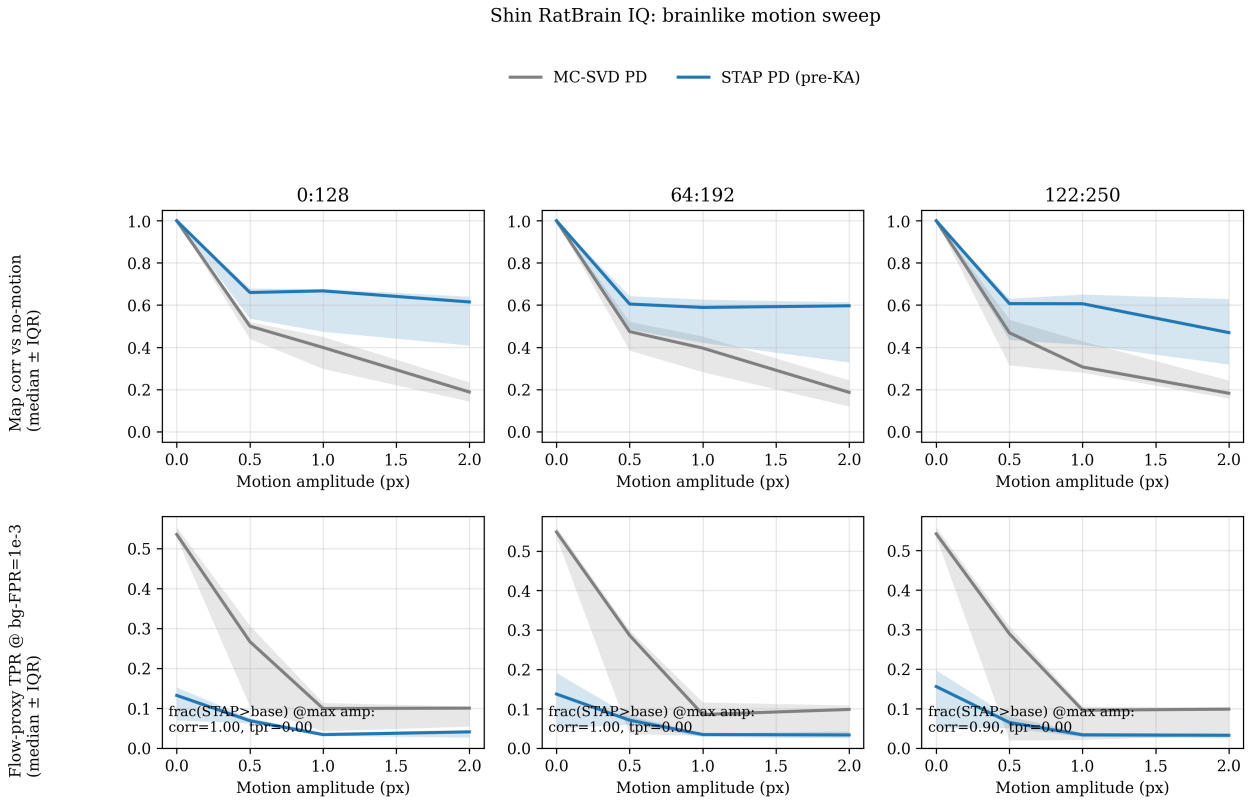

Figure 19: Shin RatBrain Fig3: brain-like residual-motion sweep across three overlapping windows (10 clips; median ± IQR) under the frozen Shin-U profile with rigid registration enabled and shrink-only regularizer disabled (matched-subspace detector reported pre-regularizer). Motion amplitude (x-axis) is the target displacement scale in pixels. "Brain-like" motion is applied directly to the beamformed IQ cube and combines (i) a rigid random-walk translation (70% of the amplitude), (ii) a smooth, depth-decaying elastic warp (30%; spatial smoothing $\sigma \approx 24$ px; depth weight $\propto \exp(-z/0.8)$; random-walk temporal coefficients), and (iii) per-frame micro-jitter (std $\approx 0.08\times$ amplitude). Endpoints match Figure 18. This stress test is image-domain (IQ warp) rather than a scatterer-level simulator; it is intended to mimic a residual rigid-plus-elastic motion mismatch after registration, analogous to the structured-clutter simulation pilot.

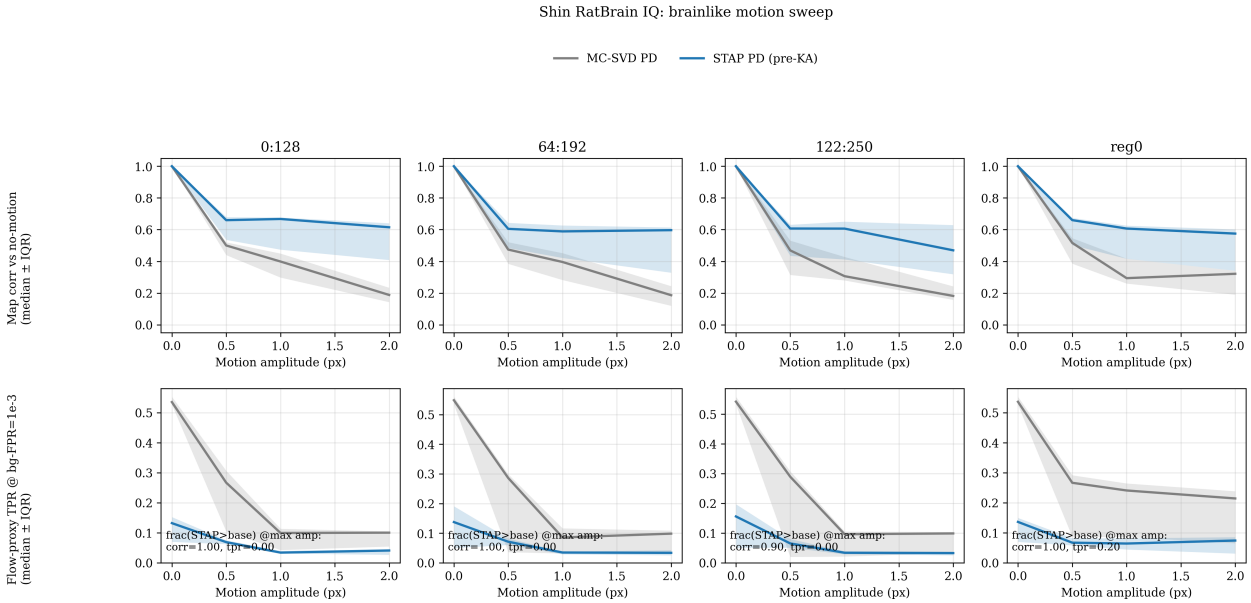

Figure 20: Shin RatBrain Fig3: registration ablation for the brain-like residual-motion sweep. The rightmost column (`reg0`) repeats the canonical window (0:128) with rigid registration disabled in the baseline and matched-subspace processing pipelines, illustrating that the motion stress test contains a substantial bulk rigid component that phase-correlation registration can correct. Even with registration enabled, a smooth depth-decaying elastic warp remains (by design) and drives the robustness gap between MC–SVD and the matched-subspace detector.

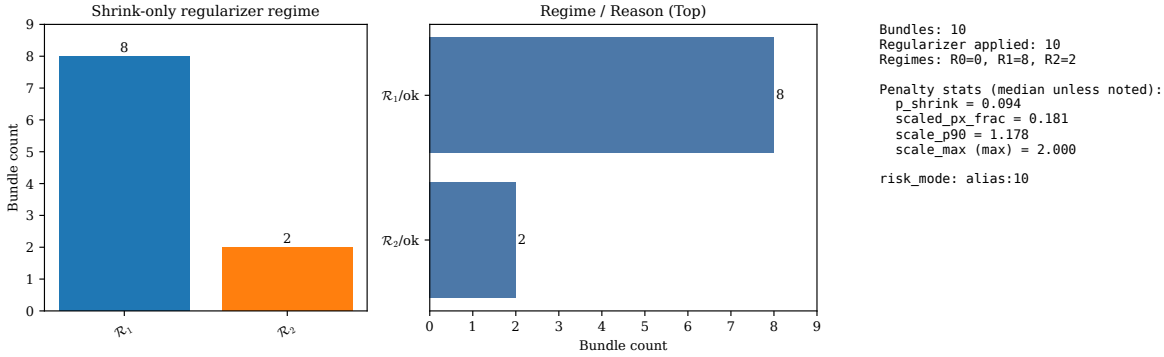

Figure 21: Shin RatBrain Fig3 (10 clips; window 0:128) shrink-only prior summary panel under the frozen Shin-U profile (MC–SVD energy fraction ≈ 0.97, $L_t = 64$). The panel summarizes prior regions and reasons, Pf peak fractions on flow/non-background proxies, and regularizer coverage/strength.

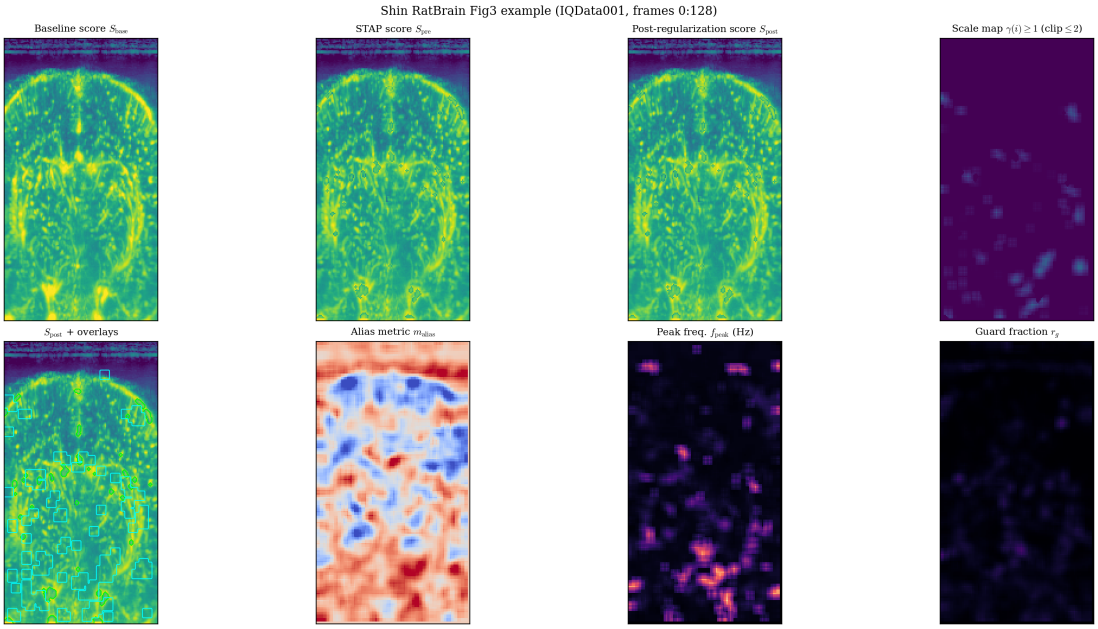

Figure 22: Representative Shin RatBrain Fig3 example (clip `IQData001`; window 0:128) illustrating the end-to-end real-IQ pipeline: baseline PD-derived map, matched-subspace score map (pre-regularizer), shrink-only regularizer gate/scale overlay, and final PD-mode score map (post-regularizer), together with band-energy maps and the flow-proxy mask that defines protected (do-not-penalize) tiles. The shrink-only regularizer never increases scores and leaves protected tiles unchanged. Reproducibility details are provided in Appendix M.

D Practical safeguards and audit checks

We implement a reproducible processing pipeline (decode $\rightarrow$ baseline $\rightarrow$ detector score $\rightarrow$ spatial-spectral features $\rightarrow$ optional shrink-only penalty layer) and treat tail-regime validity as part of the experimental design. This appendix lists practical failure modes that matter for interpretation, together with deterministic label-free checks that either record provenance or force conservative behavior. When a safeguard triggers, we either proceed with explicit provenance or disable the optional penalty layer ($w \equiv 1$) with an explicit reason label, so that "penalty layer off" is an auditable success mode under flat or non-actionable evidence.

Pitfall (what can go wrong)	Why it matters	Mitigation / where it is logged
Vendor raw-container decode ambiguity (header sizing, line ordering, beam/shot layout)	A mis-decode can silently invalidate structural masks and invalidate Doppler scores and band-energy statistics.	Decoder sanity checks reconstruct B-mode and compare to dataset screenshots; ordering/coherence checks validate CFM layout. Summaries are provided in Appendix F.
Missing or incorrect Doppler PRF (Twinkling metadata does not reliably store PRF)	PRF determines discrete band binning and peak-occupancy statistics; wrong PRF can make Pf/Pa empty or overlapping under short ensembles, creating misleading "actionability".	PRF is fixed once per sequence family, using the source-paper value where it is explicitly documented and a recorded sequence-level override otherwise (Appendix F). If binned Pf/Pa are empty/overlap, we force $\phi_i \in \mathcal{R}_0$ with an explicit reason.
Short-ensemble band discretization (coarse $\Delta f = \mathrm{PRF}/N$)	Hz-narrow bands can collapse to DC-only or become indistinguishable with $N \in \{5, 9, 17\}$.	Band operators for short ensembles are constructed as exact orthogonal DFT-bin projectors with non-overlapping bin partitions; bin membership and overlap counts are exported per run and audited by the band-sanity sentinel (the main paper).
Tail sample insufficiency ($n_{\mathrm{tail}} < n_{\mathrm{tail,min}}$)	Tail association metrics (e.g. Δ_{tail}) become undefined or noisy if the background tail is too small, which can spuriously toggle $\mathcal{R}_2$ eligibility.	Tail sizing is made explicit: we record tail sizes and use a size-aware tail fraction so that Δ_{tail} is only computed when supported; otherwise we remain conservative ($\mathcal{R}_0/\mathcal{R}_1$ with an explicit reason).
Proxy masks misread as ground truth (real data)	Proxy-derived labels can create circular evaluation or leakage.	We restrict quantitative ROC to regimes with explicit labels: simulator-truth masks from the labeled brain simulations and B-mode structural masks from the Gammex flow phantom sequences. Proxy masks are used only for band-energy statistics/protected sets and for conditional execution, never for detection efficacy claims.
Conditional execution (compute gating) changes the decision rule	Same-window proxy gating can measurably change operating characteristics (including improving them), so it cannot be presented as a cost-free speedup.	Conditional execution is treated as a separate algorithm variant; any performance reporting is accompanied by leakage controls (same-window vs disjoint-window vs random-mask). Reproducibility details are provided in Appendix M.

Table 28: Practical safeguards and audit checks. Each check is deterministic and label-free; it either records provenance/QC metrics or forces conservative penalty-layer behavior (e.g. $\phi_i \in \mathcal{R}_0$, $w \equiv 1$) with an explicit reason label.

E Practical Band Design for Brain fUS (PRF 1.5–5 kHz, T 8–16)

We provide practical band guidelines for selecting Pf/guard/Pa with explicit physical anchors.

For brain fUS at center frequencies $f_0 \approx 6$–15 MHz and PRF ≈ 1.5–5 kHz [1, 3, 6, 7, 9]:

- Microvascular brain flows (arterioles/venules) in humans and rodents are typically in the range $v \approx 3$–20 mm/s for the vessels that dominate functional responses [1, 3]. At $f_0 = 7.5$ MHz and $c \approx 1540$ m/s, this corresponds to Doppler shifts $f_D \approx 30$–200 Hz.
- Low–frequency tissue motion (respiration, cardiac pulsation) produces strong spectral content below ≈ 20 Hz; this should be contained in Po and largely removed by high–pass filtering and/or SVD.
- Large–vessel flows (e.g. MCA–like arteries) reach $v \approx 0.3$–0.8 m/s, corresponding to f_D in the kHz range [4, 5]. Depending on PRF and angle, a portion of this energy appears in high Doppler bands and can alias into the >400 Hz region corresponding to Pa [19, 23].

Thus a reasonable fixed band design for brain fUS with $T \approx 8$–16 and PRF ≈ 1.5–3 kHz is:

- Flow band $\mathcal{B}_f$: $[30, 250]$ Hz, covering microvascular flows ≈ 3–20 mm/s.

- Guard band $\mathcal{B}_g$: $[250, 400]$ Hz, used to protect against band-edge leakage.
- Alias band $\mathcal{B}_a$: $[400, \text{Nyquist}]$ Hz, capturing high Doppler where large-vessel or aliased flows and certain motion artifacts live.

In discrete terms, for $T = 8$ at PRF=1.5 kHz (bin spacing 187.5 Hz), this maps to:

- P_f on first off–DC bin (with DPSS leakage covering $\approx$ 30–250 Hz),
- guard around the second bin,
- P_a on higher bins with bin–level guard as per BG–1/BG–2.

F Twinkling PRF provenance and decoder sanity

The Twinkling RawBCF sidecars do not reliably store Doppler PRF at the file level, but the source paper does report PRF for several acquisition families. Leonov *et al.* explicitly describe the linear-probe transverse Gammex acquisition with $N = 17$ pulses at PRF= 2.5 kHz, while also noting that PRF information is not always available across the database as a whole [43]. We therefore freeze `prf_hz` once per sequence family and record that value in each bundle's `meta.json`: published values are used where the paper states them directly, and screenshot- or sequence-level overrides are used where the sidecars remain incomplete. Table 29 lists the values used for the Gammex structural-label experiments and the calculi hygiene experiments; a full extracted table from bundle metadata is written to `reports/twinkling_prf_override_table.csv` / `reports/twinkling_prf_override_table.json`.

Sequence family (RawBCF)	N shots	PRF used (Hz)	Provenance in this manuscript
Flow in Gammex phantom (along – linear probe; `RawBCFCine.par`)	17	2500	Frozen sequence-level setting recorded in bundle metadata; retained as an explicit override because the file-level sidecars do not preserve PRF reliably.
Flow in Gammex phantom (across – linear probe; `RawBCFCine_*_17.par`)	17	2500	Directly aligned with the transverse linear-probe Gammex acquisition described by Leonov *et al.* at PRF= 2.5 kHz.
Twinkling/Flash artifacts on artificial calculi (`calcifications.par`)	9	500	Screenshot-encoded PRF (`picture (prf=500HZ).png`); consistent with the Blue Phantom PRF range reported by Leonov *et al.* and used only for shrink-only penalty hygiene audits.
Twinkling artifact on artificial calculi (`RawBCFCine_29062016_174207.par`)	9	500	Screenshot-encoded PRF (`picture (prf=500HZ).png`); consistent with the Blue Phantom PRF range reported by Leonov *et al.* and used only for shrink-only penalty hygiene audits.

Table 29: Frozen PRF settings for Twinkling RawBCF sequences used in this manuscript. The across-view linear-probe Gammex PRF is directly supported by Leonov *et al.*, while the along-view and calculi settings remain explicit frozen sequence-level overrides because the RawBCF sidecars do not preserve PRF reliably. For the calculi sequences, the 500 Hz setting is consistent with screenshot labels and with the PRF range reported for the Blue Phantom microcalcification acquisitions in [43]. Values are consistent with `reports/twinkling_prf_override_table.csv`.

Sequence family	B-mode vs PNG corr (best)	coherence ratio (beam/shot)
Gammex along – linear17 (`RawBCFCine.par`)	0.737	2.31
Gammex across – linear17 (`RawBCFCine_*_17.par`)	0.610	2.46
Calculi calcifications (`calcifications.par`)	0.382	2.92
Calculi artificial (`RawBCFCine_29062016_174207.par`)	0.415	1.08

Table 30: Twinkling RawBCF decoder sanity metrics. "B-mode vs PNG corr" is the correlation between a decoded B-mode envelope image (log-compressed) and a dataset-provided screenshot (after bilinear resize; max over left/right flips). "Coherence ratio" is the ratio of mean temporal coherence under beam-major ordering versus shot-major ordering; values > 1 support the beam-major assumption used throughout. Per-sequence decode reports are written to `reports/twinkling_decode_sanity/*/decode_report.json`; the aggregate summary is written to `reports/twinkling_decode_sanity_summary.csv` / `reports/twinkling_decode_sanity_summary.json`.

G Macé atlas alignment sanity check

To reduce the risk that offline atlas-derived summaries are dismissed as misregistered, Figure 23 overlays the Allen atlas alignment inputs used throughout the Macé section: atlas region masks (`atlas.Regions`) and the vascular atlas volume (`atlas.Vascular`) mapped into scan planes using the provided affine transform (`Transformation.mat`).

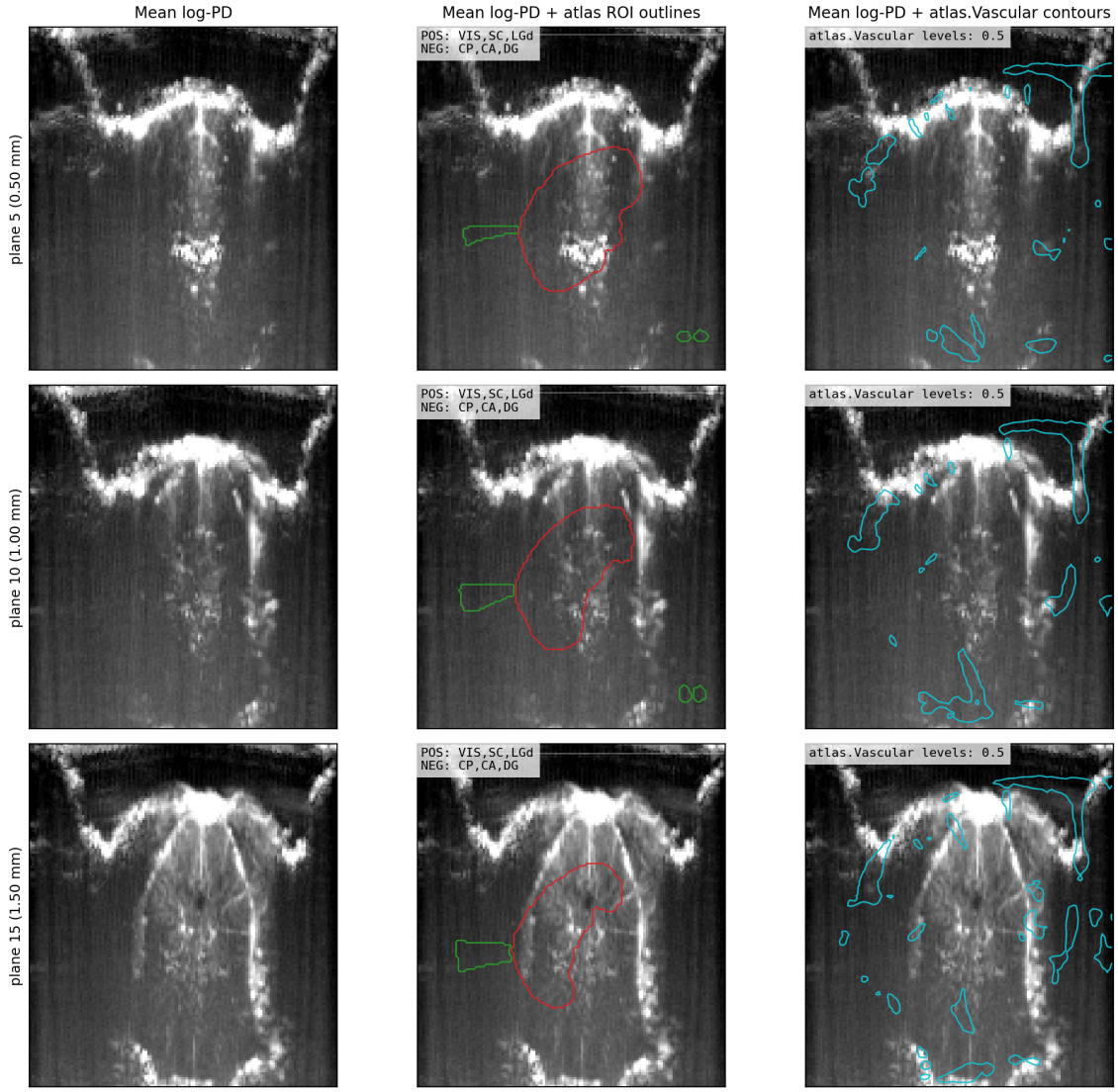

Figure 23: Macé atlas-alignment sanity figure (representative planes from the anatomical reference volume `scan_anatomy.mat`). Left: anatomical scan-space image. Middle: the same image with outlines of selected atlas ROIs mapped into the scan plane using `Transformation.mat` (VIS/SC/LGd in green; CP/CA/DG in red). Right: the anatomical image with contours of `atlas.Vascular` (cyan; level 0.5) mapped into the same plane with the same affine. This figure is generated by `scripts/mace_atlas_overlay_fig.py` and written to `figs/paper/mace_atlas_overlay.png`.

H PD-only shrink-only prior summary panel (hemodynamic): band-energy statistics and aggregation

This appendix specifies a PD-only instantiation of the shrink-only prior and the associated summary-panel aggregation used for systematic plane sweeps (e.g. Macé/Urban whole-brain mouse fUS). The key discipline is unchanged: *runtime* prior selection is label-free (proxy sets), while atlas-defined H_1/H_0 labels are used only *offline* for descriptive reporting and are never used as prior inputs.

A) PD-only band-energy feature definitions (math)

PD movie and tiling. Fix one coronal plane (one "configuration") with a PD movie

$$P[t, y, x] \in \mathbb{R}_{\geq 0}, \qquad t = 0, \ldots, T-1,$$

sampled at rate $f_s = 1/dt$ (Macé: $f_s \approx 10$ Hz). Tile each frame into overlapping spatial tiles Ω_i of size $h \times w$ (Macé: 8×8) with stride s (Macé: $s = 3$).

Tile PD time series. Define the tile-averaged PD time series

$$p_i[t] := \frac{1}{|\Omega_i|} \sum_{(y,x)\in\Omega_i} P[t, y, x].$$

For robustness across tiles, we optionally use a log-stabilized, mean-removed series

$$u_i[t] := \log(p_i[t] + \epsilon_{\mathrm{pd}}), \qquad \tilde{u}_i[t] := u_i[t] - \frac{1}{T}\sum_{t'=0}^{T-1} u_i[t'].$$

Sliding-window temporal embedding (Hankel snapshots) and whitening. Fix a slow-time aperture L_t (tens of frames in PD-only). Define $K = T - L_t + 1$ and form multiple slow-time snapshots using a sliding-window temporal embedding (Hankel matrix construction):

$$x_{i,k} := \begin{bmatrix} \tilde{u}_i[k] \\ \tilde{u}_i[k+1] \\ \vdots \\ \tilde{u}_i[k+L_t-1] \end{bmatrix} \in \mathbb{R}^{L_t}, \qquad k = 0, \ldots, K-1.$$

Estimate a per-tile covariance (SCM or robust variant)

$$\widehat{R}_i := \frac{1}{K}\sum_{k=0}^{K-1} x_{i,k} x_{i,k}^{\mathrm{H}},$$

apply conditioning/diagonal loading to obtain R_i, and whiten

$$y_{i,k} := R_i^{-1/2} x_{i,k}.$$

We pool whitened snapshots to a single per-tile slow-time vector

$$z_i := \frac{1}{K}\sum_{k=0}^{K-1} y_{i,k} \in \mathbb{R}^{L_t}.$$

Hemodynamic band projectors and energies. Let $\mathcal{B}_f, \mathcal{B}_g, \mathcal{B}_a$ denote Pf/guard/Pa hemodynamic bands (Macé: $\mathcal{B}_f = [0, 0.5]$ Hz, $\mathcal{B}_g = (0.5, 1.0]$ Hz, $\mathcal{B}_a = (1.0, 3.0]$ Hz). Build fixed temporal projectors (P_f, P_g, P_a) onto the span of discrete Fourier tones whose frequencies lie in these bands, and define $P_o := I - P_f - P_g - P_a$ ("other"/out-of-band). Define band energies

$$E_f(i) := z_i^{\mathrm{H}} P_f z_i, \quad E_g(i) := z_i^{\mathrm{H}} P_g z_i, \quad E_a(i) := z_i^{\mathrm{H}} P_a z_i, \quad E_o(i) := z_i^{\mathrm{H}} P_o z_i,$$

and $E_{\mathrm{tot}}(i) := E_f + E_g + E_a + E_o + \epsilon$.

Risk metrics and Pf-peak statistics. We use the same band-energy feature primitives as in the Doppler regime:

$$m_{\mathrm{alias}}(i) := \log \frac{E_a(i) + \epsilon}{E_f(i) + \epsilon}, \qquad m_{\mathrm{guard}}(i) := \log \frac{E_g(i) + \epsilon}{E_f(i) + \epsilon}, \qquad r_g(i) := \frac{E_g(i)}{E_{\mathrm{tot}}(i)}.$$

Define a whitened periodogram of the pooled vector z_i,

$$\widehat{z}_i[n] := \sum_{\ell=0}^{L_t-1} z_i[\ell]\, e^{-2\pi i n \ell / L_t}, \qquad S_i[n] := |\widehat{z}_i[n]|^2, \qquad n = 0, \ldots, \lfloor L_t/2 \rfloor,$$

and the (non-DC) peak frequency $f_{\mathrm{peak}}(i) = \frac{n_{\mathrm{peak}}}{L_t dt}$ where $n_{\mathrm{peak}}(i) := \arg\max_{n \geq 1} S_i[n]$. The Pf-peak indicator and occupancy on a set $\mathcal{T}$ are

$$\pi_f(i) := \mathbf{1}\{f_{\mathrm{peak}}(i) \in \mathcal{B}_f\}, \qquad \mathrm{PfPeakFrac}(\mathcal{T}) := \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \pi_f(i).$$

B) PD-only proxy sets and prior inputs (math)

The PD-only prior uses label-free proxy sets based on a PD-derived flow-coverage proxy, analogous to the IQ flow-mask coverage c_f. Let $\bar{P}[y, x] := \frac{1}{T} \sum_t P[t, y, x]$ be the mean PD image (optionally restricted to a brain mask). Define a high-quantile flow proxy mask

$$M_{\mathrm{flow}}(y, x) := \mathbf{1}\{\bar{P}[y, x] \geq Q_{q_{\mathrm{flow}}}(\bar{P})\},$$

and per-tile flow coverage

$$c_f(i) := \frac{1}{|\Omega_i|} \sum_{(y,x) \in \Omega_i} M_{\mathrm{flow}}(y, x) \in [0, 1].$$

With frozen design thresholds c_{bg} and c_{flow} (typical $c_{\mathrm{bg}} \approx 0.05$, $c_{\mathrm{flow}} \approx 0.20$), define

$$\mathcal{T}_{\mathrm{bg}} := \{i : \ c_f(i) \leq c_{\mathrm{bg}}\}, \quad \mathcal{T}_{\mathrm{flow}} := \{i : \ c_f(i) \geq c_{\mathrm{flow}}\}, \quad \mathcal{T}_{\mathrm{nonbg}} := \{i : \ c_f(i) > c_{\mathrm{bg}}\}.$$

If $\mathcal{T}_{\mathrm{flow}}$ is too small (insufficient proxy support), a conservative fallback is to define proxy sets via tilewise mean-$\bar{P}$ quantiles and record this fallback choice in the exported metadata.

C) Prior-regime selection for PD-only (math)

Baseline score for PD-only prior. For Macé, the baseline scalar detector is PD-z on tile time series: choose a baseline segment B (e.g. first third of frames) and define

$$s_{\text{base}}(i) := \max_{t \notin B} \frac{u_i[t] - \mu_i^{(B)}}{\sigma_i^{(B)} + \epsilon_z},$$

with $\mu_i^{(B)}, \sigma_i^{(B)}$ the mean and standard deviation of $u_i[t]$ over $t \in B$. Higher s_{base} means more activation.

Risk-mode selection and integrity checks. The PD-only prior follows the same posture as the Doppler-IQ prior: if risk statistics are flat (e.g. $\text{IQR}(m_{\text{alias}} \mid \mathcal{T}_{\text{bg}})$ too small) or proxy support is insufficient, we default to the uninformative regime ($\mathcal{R}_0$; $w \equiv 1$) and record an explicit disablement reason in the metadata. If guard dominates (e.g. $Q_{0.90}(r_g \mid \mathcal{T}_{\text{bg}})$ above a frozen threshold), we disallow the experimental alias-dominant regime $\mathcal{R}_2$ and either remain in a guard-risk safety lane ($\mathcal{R}_1$; requiring tail association of r_g) or default to $\mathcal{R}_0$ as out of regime.

Actionability checks (experimental alias-dominant regime $\mathcal{R}_2$, label-free). The experimental alias-dominant regime $\mathcal{R}_2$ (research-only eligibility; reporting-only) requires additional evidence that risk statistics explain high-score background tails. In the selected risk mode ($m_{\text{risk}} = m_{\text{alias}}$ or m_{guard}), define

$$\Delta_{bf} := \text{median}(m_{\text{risk}} \mid \mathcal{T}_{\text{bg}}) - \text{median}(m_{\text{risk}} \mid \mathcal{T}_{\text{flow}}),$$

and within $\mathcal{T}_{\text{bg}}$ define a tail subset $\mathcal{T}_{\text{tail}}$ as the top 1% of s_{base} and a mid subset $\mathcal{T}_{\text{mid}}$ as a central quantile band; require

$$\Delta_{\text{tail}} := \text{median}(m_{\text{risk}} \mid \mathcal{T}_{\text{tail}}) - \text{median}(m_{\text{risk}} \mid \mathcal{T}_{\text{mid}}) \geq \Delta_{\text{tail,min}}.$$

Finally, we enforce a Pf-realization eligibility condition for $\mathcal{R}_2$:

$$\text{PfPeakFrac}(\mathcal{T}_{\text{flow}}) \geq p_{f,\text{min}}.$$

If these checks fail, the prior remains in $\mathcal{R}_0$ or $\mathcal{R}_1$ (guarded only) and does not enter $\mathcal{R}_2$.

Coverage sentinels. As in the Doppler-IQ prior, penalization must be neither inert nor global: after defining protected/candidate sets (below), the penalized fraction

$$p_{\text{shrink}} := \mathbb{P}\{g = 1 \mid \mathcal{T}_{\text{cand}}\}$$

must lie in frozen bounds (e.g. $p_{\text{min}} \leq p_{\text{shrink}} \leq p_{\text{max}}$), otherwise we default to $\mathcal{R}_0$ ($w \equiv 1$) and record a coverage-out-of-bounds sentinel.

D) PD-only shrink-only prior mapping (optional; summary-panel parity)

The PD-only instantiation supports both a hard suppression used in the Macé systematic-sweep reporting and a soft shrink mapping for summary-panel parity. Define a protected set $\mathcal{T}_{\text{prot}}$ (tiles with high flow coverage and/or very high baseline scores) and a candidate set $\mathcal{T}_{\text{cand}}$ (non-protected tiles in an uncertain score window). Using a background-calibrated risk threshold $\tau_{\text{risk}} = Q_{q_{\text{risk}}}(m_{\text{risk}} \mid \mathcal{T}_{\text{bg}})$, define the penalization indicator

$$g(i) = \mathbf{1}\{i \in \mathcal{T}_{\text{cand}},\ m_{\text{risk}}(i) \geq \tau_{\text{risk}}\}.$$

Hard suppression (Macé sweep): apply the binary indicator g as a hard suppression on positive activations,

$$s_{\text{shrink}}(i) = s_{\text{base}}(i) - g(i)\max(s_{\text{base}}(i), 0),$$

which is equivalent to setting positive penalized tiles to zero while leaving negative scores unchanged. *Soft shrink (summary-panel parity):* set $w(i) = \max(w_{\min}, \exp(-k\,u(i)))$ with a robust normalized exceedance $u(i) \geq 0$ and apply $s_{\text{shrink}}(i) = s_{\text{base}}(i) - (1 - w(i))\max(s_{\text{base}}(i), 0)$. In both cases, $g(i) = 0$ (and $w(i) = 1$) on $\mathcal{T}_{\text{prot}}$ enforces the do-not-penalize constraint (protected-set scores unchanged) and $s_{\text{shrink}}(i) \leq s_{\text{base}}(i)$ enforces shrink-only behavior.

E) Summary-panel schema and aggregation (PD-only)

For PD-only datasets the summary panel is computed per plane (configuration) and written as a CSV with one row per (scan, plane). Each row records: (i) identifiers and frozen settings (scan/plane, T, f_s, L_t, tile geometry, band edges), (ii) proxy support counts ($|\mathcal{T}_{\text{bg}}|$, $|\mathcal{T}_{\text{flow}}|$), (iii) prior outputs (region $\in \{\mathcal{R}_0, \mathcal{R}_1, \mathcal{R}_2\}$, reason label, risk_mode; exported in code as `state/reason`), (iv) key band-energy statistics (PfPeakFrac($\mathcal{T}_{\text{flow}}$), $Q_{0.90}(r_g \mid \mathcal{T}_{\text{bg}})$, IQR($m_{\text{risk}} \mid \mathcal{T}_{\text{bg}}$), Δ_{bf}, Δ_{tail}), and (v) regularizer coverage/strength when active (p_{shrink} and scale/shrink quantiles). Atlas-labeled descriptive fields are prefixed `offline_` (e.g. `offline_pf_peak_frac_H1`, `offline_pf_peak_frac_H0`, `offline_hits_pre/post`) to make clear that they are not prior inputs.

Summary-panel aggregation mirrors the Doppler-IQ diagnostic panel: histograms of prior regions and (region, reason) pairs, a risk-mode histogram, distributions of key band-energy statistics (Pf peak fraction, guard q90, risk IQR), and (for active planes) shrink coverage/strength summaries. Any atlas-stratified Macé plots (e.g. PfPeak(VIS/SC/LGd) vs PfPeak(CP/CA/DG), alias separation) are presented as a separate, explicitly offline/descriptive panel.

F) "Do not overclaim" messaging (PD-only summary panel)

When reporting a PD-only prior summary panel, we use the following posture in captions and limitations:

- PD-only band-energy statistics operate on low-rate PD time series and are an auditability check; they do not validate Doppler-side uplift on raw IQ.
- The uninformative regime ($\mathcal{R}_0$; $w \equiv 1$) is an intended success outcome when feature evidence is flat or non-actionable; the summary panel is designed to make these cases explicit via reason labels.
- Atlas-stratified results on Macé are descriptive; prior-positive slices identified by scanning the same dataset are not treated as held-out confirmation.

- The experimental alias-dominant regime ($\mathcal{R}_2$) is a reporting discipline requiring additional label-free actionability evidence; $\mathcal{R}_1$ is guarded only and should not be presented as a clinical performance guarantee.

I Algorithmic details (supplementary)

I.1 Shrink-only prior regime selection (frozen defaults)

The shrink-only prior consumes per-tile arrays S_{base}, m_{alias}, r_g, c_f, and PfPeak and returns a compact regime report (region label, a disablement reason when inert, and diagnostic metrics). The reference implementation is provided with the accompanying code. The frozen default thresholds used throughout this repository are:

- **Proxy thresholds:** $c_{\text{bg}} = 0.05$, $c_{\text{flow}} = 0.20$.
- **Candidate/protected score quantiles:** candidate window $q_{\text{lo}} = 0.30$, $q_{\text{hi,cand}} = 1.0$; optional high-score protection $q_{\text{hi,prot}} = 0.99999$ when the protected set includes extreme-score tiles.
- **Risk threshold:** $q_{\text{risk}} = 0.90$ applied to m_{alias} on $\mathcal{T}_{\text{bg}}$.
- **Feature integrity (flatness + guard):** $\text{IQR}(m_{\text{alias}} \mid \mathcal{T}_{\text{bg}}) \geq 0.25$; actionable eligibility disabled if $Q_{0.90}(r_g) > 0.25$; "guard safety" lane allowed up to $Q_{0.90}(r_g) \leq 0.70$ if guard is actionable, requiring $\text{IQR}(r_g \mid \mathcal{T}_{\text{bg}}) \geq 0.03$ and $\Delta_{\text{tail,guard}} \geq 0.01$.
- **C2 actionability thresholds:** $\Delta_{bf} \geq 0.30$ and $\Delta_{\text{tail}} \geq 0.20$ (computed on $\mathcal{T}_{\text{bg}}/\mathcal{T}_{\text{flow}}$ proxies as in the main paper).
- **C2 Pf-realization eligibility:** $\text{PfPeakFrac}(\mathcal{T}_{\text{flow}}) \geq 0.05$ with $|\mathcal{T}_{\text{flow}}| \geq 50$.
- **Coverage bounds:** safety lane $p_{\text{shrink}} \in [0.02, 0.25]$ (guard lane $[0.02, 0.15]$), actionable lane $p_{\text{shrink}} \in [0.05, 0.40]$.
- **Invariance sentinel:** if $\text{IQR}(\log w \mid \text{penalized}) < 0.02$ then return the off state with an invariance/no-effect sentinel.
- **Sample support:** $n_{\min} = 500$ valid tiles, $n_{\text{bg,min}} = 200$ background-proxy tiles, $n_{\text{cand,min}} = 200$ candidate tiles, $n_{\text{tail,min}} = 20$, $n_{\text{mid,min}} = 20$.

The report exports one of three states (off / safety / actionable), corresponding to feature-space regimes $\mathcal{R}_0/\mathcal{R}_1/\mathcal{R}_2$ in the main text. When the prior is forced inert, the report includes an explicit reason category (e.g. missing inputs, insufficient proxy support, feature flatness, invariance/no-effect sentinels, or invalid band discretization in short ensembles). Reason labels are enumerated in the code and logged in the exported metadata.

I.2 Score-space shrink mapping and exported maps

When the score-space shrink-only regularizer is enabled, per-tile shrink factors $w(i) \in (0, 1]$ are derived deterministically from the regime report and applied in score space as

$$S_{\text{post}}(i) = w(i)\, S_{\text{pre}}(i), \qquad w(i) \in (0, 1].$$

For convenience (and compatibility with legacy `pd` arrays), bundles store an equivalent per-pixel *scale* map $\gamma(i) = 1/w(i) \geq 1$, averaged over overlapping tiles and applied only on the union of penalized tiles as

$$\text{PD}_{\text{post}}(i) = \text{PD}_{\text{pre}}(i)/\gamma(i).$$

A binary gate map stores the penalized region. Protected pixels are enforced by setting $\gamma = 1$ on the protected set (flow mask and extreme-score pixels as in the main paper).

Implementation score conventions. Throughout the manuscript, quantitative low-FPR operating points are reported using flow-evidence score maps under a right-tail convention (higher = more flow-like; thresholds calibrated on the high-score tail) intended for labeled ROC and robustness reporting (Table 2). Exported bundles include the baseline score and the matched-subspace score before the optional shrink-only penalty, along with optional conventional baseline comparators (e.g. log-PD and Kasai) and diagnostic/proxy maps used for gating and audits. Proxy maps are not used as the primary detector scores for labeled ROC reporting. When the shrink-only penalty is enabled, the exported maps enforce a shrink-only mapping on the chosen detector score: in this right-tail score space, $S_{\text{post}} = w\, S_{\text{pre}}$ with $w \in (0, 1]$; for convenience, bundles also store the equivalent per-pixel scale map $\gamma = 1/w \geq 1$ applied on the penalized region.

J Simulation details (supplementary)

J.1 k-Wave brain pilots (profile definitions)

The k-Wave brain pilots used in this repository are generated using our simulation pipeline (see Appendix M for the exact commands). The shared acquisition/geometry settings are:

- Grid: $N_x = N_y = 240$ with $dx = dy = 90$ μm; PML size 16; $c \approx 1540$ m/s; $f_0 = 7.5$ MHz; 3 cycles.
- Angles: $(-12, -6, 0, 6, 12)^\circ$; ensembles: 5; pulses per ensemble: 64 (so $T = 320$ compounded frames); PRF = 1500 Hz.
- Tiling: 8×8 tiles, stride 3; $L_t = 8$ (conservative profile temporal aperture).

Synthetic flow/alias/clutter overlays. To make Doppler/alias structure explicit and reproducible, the pilots apply deterministic slow-time overlays on the beamformed cube (code: `sim/kwave/common.py`). The canonical open-skull pilot overlays (internally, profile `Brain-OpenSkull`) in the Makefile are:

- Flow Doppler on the flow mask: `--flow-doppler-min-hz 60`, `--flow-doppler-max-hz 180`.
- Background alias tone: `--bg-alias-hz 650` with fraction 0.3 in depth $[0.30, 0.70]$ and jitter ± 50 Hz.
- Colored temporal clutter: `--clutter-beta` 1.0 with target SNR 20 dB over depth $[0.20, 0.95]$.
- Aperture phase screen: `--aperture-phase-std 0.8` rad, correlation length 14 elements.

The shallow-pial variant (internal profile `Brain-Pial128`) uses 4 ensembles with 32 pulses each ($T = 128$ compounded frames) and concentrates shallow alias in depth $[0.15, 0.30]$ with a stronger alias fraction and a vibration tone. The alias-dominant and structured-clutter variants (internal profiles `Brain-AliasContract` and `Brain-SkullOR`) use additional alias/vibration settings and skull/aberration tagging; all simulation parameters are fixed per profile and are specified in the reproducibility manifest (Appendix M).

J.2 Replay/analysis pipeline used for ROC and latency

Downstream processing is run on the saved pilot angle data using the standard replay script, which loads angle directories (or ensembles of angle directories), forms the slow-time cube, applies the

chosen baseline, runs the matched-subspace detector, and writes a standard output bundle (`sim/kwave/common.py:write_acceptance_bundle`). Low-FPR ROC points and feasibility telemetry are extracted using `scripts/hab_contract_check.py`.

For the conservative-profile matched-subspace detector runs used in the latency suite and robustness sweeps, we additionally use `scripts/run_brain_replay_sweep.py`, which:

- Reuses a fixed pilot per (regime, PRF) and replays the baseline plus matched-subspace detector under a frozen profile on 64-frame windows (`--time-window-length 64`).
- Evaluates the detector variants used in the latency suite, including conditional execution and a reduced-support mode that caps effective training support by subsampling training snapshots. Exact runtime settings are recorded in bundle metadata and listed in the reproducibility manifest (Appendix M).
- Uses a frozen MC–SVD baseline and simulator-truth evaluation masks. Conditional execution is driven by a deterministic, label-free proxy flow mask derived from baseline PD brightness.

Latency reporting protocol (steady-state, CUDA graphs). When GPU acceleration is enabled, the first replay window can include one-time overheads such as CUDA context initialization, kernel compilation, and optional CUDA-graph capture. For any GPU latency numbers reported from this codebase, we therefore use a steady-state replay protocol: `scripts/latency_rerun_check.py` replays N windows in a single process (`--steady-windows` N or `--window-offsets`) and reports per-window timing, with the publishable latency taken as the mean over windows $2..N$ (window 1 treated as cold-start overhead).

K Implementation details (supplementary)

K.1 RPCA baseline (full)

The RPCA baseline used in this repository is implemented as `sim/kwave/common.py:_baseline_pd_rpca`. It performs a windowed Principal Component Pursuit (PCP) using an inexact augmented Lagrangian method (IALM) on a downsampled Casorati matrix:

- Spatial downsampling: complex average pooling by factor 2 when $\max(H, W) > 128$.
- Temporal windows: default subwindow length $t_{\text{sub}} = 80$ (or shorter if $T < 80$); windows are processed sequentially.
- Optimization: IALM with at most 80 inner iterations per window (`max_iters_inner=min(max_iters,80)`), tolerance 10^{-4}.
- Shrinkage: $\lambda = 1/\sqrt{\max(m, n)}$ when not supplied, where $m = T_w$ and $n = H_d W_d$.
- Low-rank step: singular value thresholding using either full SVD or a randomized SVD depending on matrix size, with a per-window rank heuristic $k = \min(8, \max(2, \lfloor T_w/10 \rfloor))$.
- Output PD: sum of sparse-component energy over windows on the downsampled grid, then nearest-neighbor upsampling to the original resolution.

K.2 HOSVD baseline (full)

The tensor HOSVD baseline is implemented as `sim/kwave/common.py:_baseline_pd_hosvd`. The cube $X \in \mathbb{C}^{T \times H \times W}$ is optionally spatially downsampled (complex average pooling). Mode-0/1/2 SVDs are computed on the unfoldings and ranks are chosen either explicitly (`--hosvd-ranks`)

or by energy fractions (`--hosvd-energy-fracs`, default $(0.99, 0.99, 0.99)$). The low-rank approximation is reconstructed by projecting onto the multilinear subspace, and PD is the time-averaged residual energy $\frac{1}{T}\sum_t |X(t) - L(t)|^2$. In the brain baselines used in this repository, the Makefile target `r4c-replay-hosvd` runs HOSVD with `--hosvd-spatial-downsample 2` and `--hosvd-energy-fracs 0.99,0.99,0.99`.

K.3 Adaptive SVD baseline (spatial singular-vector similarity cutoff)

To preempt the common critique that global SVD baselines are sensitive to an arbitrary fixed rank/energy threshold, we include an adaptive spatiotemporal SVD baseline inspired by Baranger *et al.* [12]. For a windowed slow-time cube $X \in \mathbb{C}^{T\times H\times W}$ we form the Casorati matrix $C \in \mathbb{C}^{T\times N}$ with $N = HW$ and compute its SVD $C = U\Sigma V^{\mathrm{H}}$. Let $v_k \in \mathbb{C}^N$ denote the kth *spatial* singular vector (column of V). Because singular vectors are defined up to an arbitrary complex phase, we measure spatial coherence on magnitudes and define

$$s_k := \frac{\langle |v_k|,\ |v_{k+1}|\rangle}{\||v_k|\|\ \||v_{k+1}|\|}, \qquad k = 1, \ldots, T-1.$$

We smooth $\{s_k\}$ by a short median filter and choose the clutter rank r as the first index where s_k falls below a robust threshold $\tau = \mathrm{median}(s) - \kappa \cdot \mathrm{MAD}(s)$, with r clamped to a fixed range $[r_{\min}, r_{\max}]$ (all hyperparameters frozen per regime). For the labeled brain simulation pilots ($T = 64$) we use median-filter width 7, $\kappa = 2.5$, and clamp $[r_{\min}, r_{\max}] = [1, 32]$. The filtered residual is

$$C_{\mathrm{res}} := \left(I - U_r U_r^{\mathrm{H}}\right) C,$$

and the baseline score is the same PD flow-evidence statistic used elsewhere, $S_{\mathrm{base}}(y, x) = \frac{1}{T}\sum_t |X_{\mathrm{res}}(t, y, x)|^2$ after reshaping C_{res} back to cube form. This baseline is implemented as `sim/kwave/common.py:_baseline_pd_svd_similarity`.

K.4 Block-wise local SVD baseline (overlap-add reconstruction)

To address reviewer objections that *local* adaptive SVD clutter suppression is already standard in small-vessel ultrasound imaging, we also include a block-wise local SVD clutter filter in the style of Song *et al.* [15]. We tile the spatial grid into overlapping blocks $\{\Omega_i\}$ (block size and stride frozen per regime) and form local Casorati matrices $C_i \in \mathbb{C}^{T\times(hw)}$ by stacking slow-time vectors within each block. For each tile, we compute $C_i = U_i \Sigma_i V_i^{\mathrm{H}}$ and select a tile-local clutter rank r_i using a deterministic, frozen energy-fraction rule:

$$r_i := \min\left\{ r : \frac{\sum_{k=1}^{r} \sigma_{i,k}^2}{\sum_{k=1}^{T} \sigma_{i,k}^2} \geq e \right\},$$

where $e \in (0, 1)$ is fixed. For the labeled brain simulation pilots we use blocks 8×8, stride 3, $e = 0.90$, and Hann overlap-add weighting. The local residual is $C_{i,\mathrm{res}} = (I - U_{i,r_i} U_{i,r_i}^{\mathrm{H}}) C_i$, reshaped back to a block cube. We reconstruct a full-resolution residual cube by overlap-add with a separable Hann weight inside each block and compute the baseline score as the PD statistic on the reconstructed residual. This baseline is implemented as `sim/kwave/common.py:_baseline_pd_local_svd`.

K.5 Adaptive-local SVD baseline (tilewise similarity-drop rank selection)

For the stricter SIMUS fairness study we additionally include an adaptive-local SVD baseline that combines the two literature ideas above rather than treating them as mutually exclusive. As in the Song-style local baseline, we tile the spatial grid into overlapping blocks $\{\Omega_i\}$, form local Casorati matrices $C_i \in \mathbb{C}^{T\times(hw)}$, and reconstruct by overlap-add with optional Hann weighting [15]. However, instead of a fixed tilewise energy threshold, each tile chooses its clutter rank from the same Baranger-style spatial singular-vector similarity-drop heuristic used by the adaptive-global baseline [12]. Concretely, for each tile we compute $C_i = U_i \Sigma_i V_i^{\mathrm{H}}$, form tile-local spatial singular-vector similarities

$$s_{i,k} := \frac{\langle |v_{i,k}|,\ |v_{i,k+1}| \rangle}{\||v_{i,k}|\|\ \||v_{i,k+1}|\|},$$

smooth $\{s_{i,k}\}$ with the same short median filter used in the global adaptive baseline, and choose

$$r_i := \min\{k : s_{i,k} < \mathrm{median}(s_i) - \kappa \cdot \mathrm{MAD}(s_i)\},$$

with r_i clamped to a frozen range $[r_{\min}, r_{\max}]$. The local residual remains $C_{i,\mathrm{res}} = (I - U_{i,r_i} U_{i,r_i}^{\mathrm{H}}) C_i$, followed by overlap-add reconstruction and PD scoring on the residual cube. This baseline is implemented as `sim/kwave/common.py:_baseline_pd_adaptive_local_svd`. In the benchmark tables we report the three SVD-family baselines separately as *Adaptive Global SVD*, *Local SVD (Fixed Energy)*, and *Adaptive Local SVD*.

K.6 Matched-subspace detector temporal core (full)

For k-Wave pilots and replay bundles, the reference temporal matched-subspace detector implementation follows the same conceptual decomposition as the main text: tile extraction and optional sliding-window temporal embedding (Hankel construction) in slow time; per-tile covariance estimation with a robust estimator (Tyler-type or Huber); diagonal loading to enforce conditioning targets; whitening; and band-energy scoring on a fixed Doppler tone grid. Deployment-oriented variants can (i) cap effective training support by subsampling training snapshots and (ii) reduce computation by skipping the matched-subspace stage on tiles with negligible proxy flow support, reverting to the baseline score there. Exact run settings and exported metadata used for the experiments in this manuscript are listed in the reproducibility manifest (Appendix M).

CUDA acceleration strategy (batched GPU implementation). For the latency results reported here we implement the matched-subspace stage as a GPU-batched pipeline: tiles are extracted on device using tensor *unfold* operators, processed in batches to amortize per-tile overhead, and stitched back to full-resolution maps by overlap-add. This hardware-aware formulation aligns with prior work on GPU eigen-based clutter filtering and randomized/approximate SVD acceleration for real-time Doppler processing [17, 18]. Within each batch, the temporal core is expressed as batched linear algebra (sliding-window temporal embedding; Hankel matrix construction; robust covariance estimation; whitening; and band-energy scoring), with rearrangements that favor GEMM-dominated workloads on GPUs. For fixed input shapes, the batched core can optionally be captured as a CUDA graph and replayed to reduce kernel-launch overhead; the first processed window/frame therefore includes one-time GPU overheads and is excluded from reported steady-state means.

Algorithmic choices behind the reported replay budget. The reported replay numbers are not obtained by changing the detector into a different approximate method after the fact. Three choices matter most. First, the default deployment path remains the fixed statistic unless guard telemetry justifies stronger whitening, so the measured workloads match the detector family actually reported in the paper rather than a universally whitened worst case. Second, when whitening is active, computation is organized around tile-batched GPU linear algebra with cached overlap geometry and optional CUDA-graph replay, which reduces launch overhead without changing the exported detector scores. Third, the final latency configurations use only exact-output-preserving bookkeeping reductions: redundant raw-baseline PD generation is skipped in the short-ensemble phantom path, nonessential per-tile diagnostics are omitted after extracting the quantities needed by the reported detector hierarchy, and zero-shift replay windows bypass registration work exactly when the estimated motion is zero.
The final latency configuration reported in the main paper uses only exact-output-preserving implementation optimizations. In particular, the implementation reuses cached geometry-only overlap counts, defers redundant raw-baseline PD generation in the short-ensemble phantom path, omits nonessential per-tile diagnostics after extracting the quantities required by the reported detector hierarchy, and applies an exact zero-shift registration bypass when the replayed window has no estimated motion. These changes reduce bookkeeping and launch overhead without changing exported detector scores, routing masks, or baseline maps. Exact commands and runtime settings used for the CUDA latency tables are listed in the reproducibility manifest (Appendix M).

Execution architecture behind replay feasibility. The practical replay story can be decomposed into five questions: what is computed on every support, what is computed only on promoted supports, what is cached and reused, what is skipped without changing outputs, and what belongs to the measurement protocol rather than to the detector itself. In the reported adaptive and fully whitened paths, all supports still receive the fixed branch quantities needed for detector output formation and branch telemetry, but covariance estimation, whitening, and covariance-adaptive scoring are only executed on supports selected by the branch policy. Reusable geometry-dependent terms are cached across repeated windows, and the batched GPU workload is kept shape-stable so that fixed-shape replay mechanisms such as CUDA-graph capture can reduce launch overhead. Table 31 summarizes the exact-output-preserving design choices, and Algorithm 1 gives the execution flow in compact form.

Algorithm 1 (Branch-aware execution path for replay-feasible detector evaluation)**.** *1. Form fixed-shape tile batches and any fixed geometry-dependent support operators.*

2. *Compute fixed-branch band summaries and branch-telemetry features on all supports.*
3. *Apply the branch policy to mark promoted supports for covariance-adaptive processing.*
4. *For non-promoted supports, emit the fixed statistic directly.*
5. *For promoted supports only, estimate local covariance, apply conditioning and whitening, and compute the covariance-adaptive score.*
6. *Stitch support-level outputs back to the image lattice using cached overlap-add bookkeeping.*
7. *Export detector outputs and required telemetry; omit nonessential diagnostics in deployment-facing replay runs.*

Design choice	Runtime effect	Exact-output-preserving?
Fixed-first staged execution with promotion only on selected supports	Avoids covariance estimation and whitening on supports that remain on the fixed branch	Yes
Fixed-shape tile batching and temporal embedding	Amortizes per-support overhead and enables repeated-workload replay	Yes
Cached overlap counts, support geometry, and reusable band operators	Avoids recomputing geometry-only terms across repeated windows	Yes
Optional CUDA-graph replay after first-window setup	Reduces kernel-launch overhead on repeated fixed-shape workloads	Yes
Zero-shift registration bypass when estimated motion is exactly zero	Skips unnecessary registration work in replay windows with no motion	Yes
Omission of nonessential diagnostics after extracting reported telemetry	Reduces bookkeeping and export overhead for deployment-facing runs	Yes
Steady-state timing over windows/frames 2..N, excluding disk I/O	Measurement protocol only; removes cold-start and I/O artifacts from reported replay times	Not part of detector output

Table 31: Execution-architecture choices behind the reported replay-feasibility numbers. The key distinction is between exact-output-preserving runtime reductions, which leave exported detector scores and routing decisions unchanged, and the separate timing protocol used to measure steady-state replay.

K.7 Covariance blending constraints and shrink-only regularizer modes

We permit anisotropic shrinkage, but only *within* bands. In practice we consider two shrink-only regularizer paths:

- **Updated (commuting band–scalar).** $Q^{\mathrm{H}}Q$ is forced to be bandwise scalar with target gains (s_f, s_a, s_o) (typical targets: $s_f = s_o \approx 1$, $s_a \approx 1.5$), small mixing, and no off–block priors; the content of R_0 beyond (P_f, P_a) is ignored.
- **Blend.** A trace–normalized blend $R_\beta = (1-\beta)R_t^{(n)} + \beta R_0^{(n)}$ with a small diagonal lift. In practice R_0 is overridden to a band–scalar form (Pf weight 1, Pa weight $\gamma_a > 1$, Po weight 1) so the geometry is explicit; β is treated as a fixed hyperparameter per regime.

Both modes enforce bandwise structure; there are no off–block priors.

K.8 CB–1: Band–Block Blending

We construct

$$R_\beta := (1-\beta)\,\widehat{R} + \beta\big(R_{0,f} \oplus R_{0,a} \oplus R_{0,o}\big) + \delta I,$$

where $R_{0,f} = P_f R_0 P_f$, etc., for some prior R_0. There are no off–block priors; $\beta \in [0,1)$ and $\delta \geq 0$.

Interpretation: this structure makes Q nearly block–diagonal in (P_o, P_f, P_g, P_a) while still allowing bandwise control to realize OC–2.

K.9 CB–2: Bandwise Spectral Targets

We choose (β, δ) to enforce OC–2 explicitly:

$$\lambda_{\min}(Q^{\mathrm{H}} Q|_{P_f}) \in [s_f^2(1-\delta_f), s_f^2(1+\delta_f)], \quad \text{etc.}$$

with design tolerances $\delta_\bullet \leq 0.1$ and separation $\Lambda = s_a/s_f \geq 1.25$.

K.10 CB–3: Bandwise Condition Number Safeguard

We constrain bandwise conditioning:

$$\kappa(R_\beta|_{P_f}) \leq \kappa_f^{\max}, \qquad \kappa(R_\beta|_{P_a}) \leq \kappa_a^{\max},$$

avoid an incompatibility between conditioning and preserving flow Rayleigh quotients.

K.11 Localization (restricting shrink-only regularizer in score space)

A global band–constant transform with Q cannot improve ROC if applied everywhere (it is close to a scalar reweighting of BR/GLRT). To create re–ranking, the shrink-only regularizer must be *localized* to tiles where:

- scores are dominated by alias/undesired components,
- positives are unlikely to lie.

We define localization based on a combination of alias metrics, flow coverage, depth, PD, and registration. This supplements G–1/G–2 with a generalized alias score m_{alias} from Section B.2. In plain language, the optional penalty layer is allowed to act only in tiles that look clutter-heavy and not vessel-dominant; otherwise it stays off and leaves the detector score unchanged.

K.12 G–1: Alias–dominant tile selection

Given a baseline tile z, compute per–band energies (E_o, E_f, E_g, E_a) and a generalized alias score $m_{\text{alias}}(z)$ satisfying R–2.

Data–adaptive threshold. For BG tiles (or their proxy), collect $m_{\text{alias}}(z)$ and define

$$\tau_{\text{alias}} := Q_{q_{\text{alias}}}\big(m_{\text{alias}} \mid H_0\big),$$

where Q_q is the q–quantile and $q_{\text{alias}} \in [0.7, 0.9]$ is a design choice (typical: 0.8 or 0.9).

Define an alias gate

$$g_{\text{alias}}(z) = \mathbf{1}\left\{m_{\text{alias}}(z) \geq \tau_{\text{alias}}\right\}.$$

Band–edge purity. Additionally require that guard energy is small relative to Pf+Pa:

$$\frac{E_g}{E_o + E_f + E_a + E_g} \leq \eta_{\text{edge}}, \quad \eta_{\text{edge}} \leq 0.1.$$

Interpretation: this gate selects the top tail of alias–heavy tiles among negatives, while protecting against band–edge contamination. The distribution of m_{alias} across classes is governed by R–2; empirical validation (Section B.2) must show a nontrivial difference between H_0 and H_1.

K.13 G–2: Flow coherence protection

We can optionally compute a flow coherence metric FPC (first principal component fraction) within the flow band:

$$\mathrm{FPC} := \frac{\lambda_1(\widehat{S}_f)}{\mathrm{tr}(\widehat{S}_f)}, \qquad \widehat{S}_f := \frac{1}{K}\sum_{k=1}^{K} u_k u_k^{\mathrm{H}}, \quad u_k := P_f z^{(k)},$$

where $z^{(k)}$ are projected slow–time vectors across DPSS tapers or within a local ensemble.

Gate *may* require

$$\mathrm{FPC} \le c_{\mathrm{pc}}, \qquad c_{\mathrm{pc}} \in [0.4, 0.6].$$

Interpretation: flow tiles tend to exhibit higher coherence in Pf; when enabled, we suppress penalization when FPC is large, thereby reducing positive penalization probability p_+. Alias/motion tiles have lower flow coherence and pass this condition. In the current runs this check is typically left off; coverage and depth localization carry most of the class separation.

K.14 G–2': Flow coverage protection

Let c_f be the tile–wise flow coverage fraction:

$$c_f := \frac{\#\{\text{pixels labeled as flow}\}}{\#\{\text{pixels in tile}\}},$$

using a label–free proxy such as a PD–derived flow mask.

We require

$$c_f \le c_{f,\max}, \qquad c_{f,\max} \in [0.1, 0.2],$$

and optionally $c_f \ge c_{f,\min}$ to avoid pure background noise tiles (typical $c_{f,\min} \approx 0$).

Interpretation: we prefer penalization on tiles with *low to moderate* flow coverage. This is the regime where alias/motion dominates and where down–scaling scores is most beneficial. Dense flow tiles are either left to the matched-subspace detector alone or are protected via the flow coherence check. (In the current implementation, coverage is treated as an *upper* bound; missing coverage defaults to "pass" in this test.)

K.15 G–3: Baseline score quantile window

Let $S_{\mathrm{base}}(z)$ be the baseline BR/GLRT score. Within a given depth stratum, compute score quantiles $q_{0.3}, q_{0.95}$ over BG tiles. Gate requires

$$S_{\mathrm{base}}(z) \in [q_{0.3}, q_{0.95}].$$

Interpretation: avoid penalization on the noise floor (very low scores) and on already high–confidence positives (very high scores). This focuses the shrink-only regularizer on the part of the negative tail that matters near the target FPR.

K.16 G–4: Registration quality (optional)

If motion compensation / registration is used, include a registration quality metric (e.g. PSR) and require

$$\mathrm{PSR}_{\mathrm{reg}} \le \mathrm{PSR}_{\max},$$

or any similar condition ensuring that tiles with gross registration failure are handled separately.

K.17 Offline validation of gating

On a held–out labeled configuration, define

$$p_+ := \mathbb{P}\{g = 1 \mid H_1\}, \qquad p_- := \mathbb{P}\{g = 1 \mid H_0\},$$

for the combined gate g that incorporates G–1, G–2, G–2', G–3, G–4. Require:

$$p_- \geq 0.2, \quad p_- \leq 0.5, \quad p_+ \leq 0.1 p_-.$$

Interpretation: a nontrivial fraction of negatives are gated (but not too many), while positives are rarely gated. This condition is central to the SL–2 quantile shift argument.

K.18 Summary: Compact Feasibility Checklist (Updated)

The combination of matched-subspace scoring and the shrink-only regularizer can be expected to outperform MC–SVD (or matched-subspace-only scoring) in the low-FPR regime if and only if:

1. **Regime preconditions.** Band occupancy R–1, alias separation R–2, and baseline headroom R–3 are satisfied on a held–out validation set.
2. **Band design.** BG–1/BG–2/BG–3 hold for the chosen PRF/T/DPSS; Pf/Pa are physically meaningful (e.g. Pf $\approx$ 30–250 Hz, Pa $\gtrsim$ 400 Hz in brain fUS).
3. **Operator geometry.** OC–1 with $\varepsilon_{\text{mix}} \leq 0.05$ and OC–2 with $\Lambda = s_a/s_f \geq 1.25$ and $\delta_\bullet \leq 0.1$.
4. **Blending realizable.** CB–1/CB–2/CB–3 satisfied with chosen (β, δ).
5. **Gating conservative and class–differential.** G–1/G–2/G–2'/G–3/G–4 implemented; offline, $p_- \in [0.2, 0.5]$ and $p_+ \leq 0.1 p_-$.
6. **Regime sentinel.** RS–1 does not trigger (the shrink-only regularizer is not out of regime).
7. **Invariance sentinel.** $\widehat{s}_a/\widehat{s}_f \notin [1-\epsilon, 1+\epsilon]$ with $\epsilon \leq 0.02$; otherwise shrink-only regularizer is effectively scalar and can be skipped.
8. **Safety guard.** Safety–1 (monotonicity on gated flow tiles) passes.
9. **Evaluation discipline.** GLRT controls and tail CI practices (see the main paper) in place for fair low–FPR comparison.

When all of the above hold, the shrink-only regularizer applies a non–mixing, alias–inflating band transform chiefly to alias–dominant tiles, pushing negative scores down while largely leaving positives unchanged. This lowers the negative α–quantile and increases TPR at fixed small FPR in a manner consistent with the SL–2 quantile shift analysis.

L Bundle artifacts and naming conventions

This appendix summarizes the public bundle artifact categories and the mapping between the mathematical score notation in the main text and the stored arrays. Exact legacy compatibility filenames remain documented in the machine-readable reproducibility manifest; they are omitted here because they are not needed to follow the method or interpret the results.

Artifact class	Meaning	Primary use / direction
Baseline PD map; detector PD map	PD-mode maps for the baseline and detector pipelines.	Deterministic preprocessing (proxy masks) and legacy visualization/compatibility; not used as the primary detector score for quantitative reporting in this manuscript.
Baseline PD score; detector PD score	Legacy right-tail PD score maps under a fixed sign convention.	Optional legacy PD summaries used for compatibility and auditing rather than for headline ROC.
Detector PD map before regularizer; detector PD score before regularizer (optional)	Pre-regularizer detector-side exports when the shrink-only regularizer is enabled.	Pre/post regularizer auditing under the same sign convention.

Table 32: Bundle PD naming/sign schema.

Artifact class	Meaning	Primary use / direction
Baseline power Doppler score	Baseline power Doppler score (right tail).	Structural ROC; empirical tail calibration; proxy masks.
Log-power baseline score (optional)	$\log(S_{\text{base}} + \epsilon)$ (right tail).	Baseline fairness / strawman avoidance.
Kasai score (optional)	$\log(\lvert \sum_t y_t \overline{y_{t-1}} \rvert + \epsilon)$ (Kasai lag-1 power; right tail).	Short-ensemble Doppler baseline fairness.
Matched-subspace detector score before regularizer; after regularizer	Detector score maps before and after the shrink-only regularizer (right tail).	Structural ROC; within-ensemble motion ladders; motion robustness; shrink-only regularizer auditing.

Table 33: Bundle score-map naming schema for labeled ROC and label-free reporting (right-tail score convention).

Exported artifact class	Property / monotonicity	Allowed claim scope (this manuscript)
Matched-subspace detector score before regularizer	Primary matched-subspace detector score (right tail). Not a shrink-only reweighting of baseline power Doppler.	Quantitative ROC on labeled simulations and structurally labeled phantoms; label-free motion robustness endpoints on unlabeled IQ.
Baseline power Doppler score	Baseline power Doppler flow-evidence score (higher = more flow-like).	Quantitative ROC as baseline comparator (same labels/masks as the matched-subspace detector).
Log-power baseline score	Monotone transform of the baseline power Doppler score (ordering preserved).	Baseline-score fairness check; does not change within-method rankings.
Kasai score	Lag-1 autocorrelation magnitude score (Kasai power; higher = more flow-like).	Short-ensemble baseline-score fairness check (Twinkling).
Matched-subspace detector score after regularizer	Post-regularizer score map. By construction, shrink-only relative to the pre-regularizer detector score on gated regions ($S_{\text{post}} \leq S_{\text{pre}}$).	Shrink-only regularizer auditing and tail-hygiene reporting; not used as the primary detector score unless stated explicitly.
Baseline PD map	Conventional PD brightness proxy $\text{PD}^{\uparrow}$ (nonnegative).	Deterministic preprocessing (proxy masks) and legacy visualization; quantitative reporting uses detector *scores*.
Detector PD map	PD-mode detector map exported for legacy compatibility/visualization. In PD-mode it is a non-increasing reweighting of the baseline PD map.	Visualization / compatibility output; can suppress background tails but cannot recover flow removed by baseline clutter suppression.

Table 34: Score taxonomy and claim scope. Quantitative detection results are reported using flow-evidence detection scores (baseline power Doppler versus the matched-subspace detector; higher = more likely flow). PD-mode maps are treated as legacy/visualization artifacts with explicit monotonicity constraints.

M Repository Reproducibility Manifest

This appendix is repository-facing archival material rather than part of the main scientific reading path. It records a minimal reproduction recipe: data locations, artifact outputs, and a pinned repository state. The machine-readable manifest `repro_manifest.json` stores the exact command blocks; this PDF appendix lists the corresponding artifacts and entry points. Run `PYTHONPATH=.condarun-nfus-detectorspythonscripts/generate_repro_manifest.py` to refresh this manifest.

Repository state. Public repository: `git@github.com:ArthurShune/fus-detectors.git`. Release tag: (not recorded; set `FUS_DETECTORS_RELEASE_TAG` or create a git tag). Commit: `601966c` (full: `601966c3b25b33a0ac2018e5c41facdf0dd984b1`); branch: `main`; dirty: `false`. Dirty details: non-ignored paths=0, ignored build products=0. Note: dirty status ignores common LaTeX build products (e.g., `*.aux, *.log, *.out, *.fls, *.fdb_latexmk, *.synctex.gz, *.pdf`). For archival reproducibility, generate this manifest from a clean, tagged release commit (`dirty=false`) and record the public repository URL and tag.

Environment. Conda spec: `environment.yml` (sha256: `b50293345709179b8df97e68b2886dda6a91d39b68cdc7c6b6caa670490cb194`). Create env: `condaenvcreate-fenvironment.yml`. Docker: `Dockerfile` (sha256: `0746a21bcc0ecc75a67ac5f84c155a5185522b831eac4cdf3da05b777bf581ac`). Build container: `dockerbuild-tfus-detectors-fDockerfile.`. Runtime (conda env `fus-detectors`): Python: `3.10.19|packagedbyconda-forge|(main,Oct222025,22:29:10)[GCC14.3.0]`. Platform: `Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-with-glibc2.39`. Packages: `numpy`=1.26.4, `scipy`=1.14.1, `torch`=2.1.2, `matplotlib`=3.10.0. Torch CUDA: available=`true`, cuda=`11.8`, device0=`NVIDIAGeForceRTX4080SUPER`.

Datasets. Paths below are the expected on-disk locations used by the scripts in this repository. Large outputs under `runs/` and `reports/` are not tracked.

- **Brain-* k-Wave pilots (simulated IQ)**: Labeled low-FPR ROC + ablations.
  Local: `runs/pilot/`.
  Source: generated locally via sim/kwave/pilot_motion.py (Makefile targets r4c-*).
- **Mace/Urban whole-brain mouse fUS (PD-only)**: PD-only telemetry + hold-out alias-gate evaluation (no Doppler IQ).
  Local: `data/whole-brain-fUS`.
  Source: `https://doi.org/10.5281/zenodo.4905862`.
- **Shin RatBrain Fig3 (LOCA-ULM) beamformed IQ**: Real-IQ ingestion, telemetry, contract + label-free robustness sweeps.
  Local archive: `data/shin_zenodo_10711806/RatBrain_Fig3.zip`.
  Extracts to: `data/shin_zenodo_10711806/ratbrain_fig3_raw`.
  Source: `https://doi.org/10.5281/zenodo.10711806`.
- **Twinkling artifact dataset (RawBCF phantoms)**: Structurally labeled phantom ROC (Gammex) + within-ensemble motion ladders + shrink-only regularizer hygiene (calculi).
  Local: `data/twinkling_artifact/`.
  Source: `https://doi.org/10.17816/DD76511`.

- **ULM Zenodo 7883227 (rat brain kHz IQ)**: Motion robustness, same-residual score audit, one-time baseline calibration sweep, and the single paper-facing same-acquisition structural audit using a published PALA localization-derived vascular reference.
  Local archive: `data/ulm_zenodo_7883227/IQ_*.zip`.
  Extracts to: `tmp/ulm_zenodo_7883227/`.
  Source: `https://doi.org/10.5281/zenodo.7883227`.

Pre-registered selections used in this manuscript.

- **Brain low-FPR matrix (fixed-profile pilots)**: OpenSkull seed1; AliasContract seed2; SkullOR seed2; 64-frame windows at offsets 0,64,128,192,256.
- **Shin all-clips telemetry**: IQData001-IQData080 (n=80), window 0:128.
- **Shin motion subset**: IQData001-005, IQData010, IQData020, IQData040, IQData060, IQData080 (n=10).
- **Mace holdout split (deduplicated)**: train: scan1/2 + scan3/6; test: scan4 + scan5.
- **Gammex flow phantom structural ROC (along-linear17)**: frames 0:85 (n=85), PRF=2500 Hz, N=17 shots.
- **Gammex flow phantom structural ROC (across-linear17)**: frames 0:200 (n=200), PRF=2500 Hz, N=17 shots.
- **Gammex flow phantom within-ensemble motion ladder**: along-linear17, frames 0:85, amps 0..2 px, kinds rw,step.
- **Twinkling calculi shrink-only regularizer hygiene**: calcifications sequence, frames 0:50 (n=50), PRF approx 500 Hz, N=9 shots.
- **ULM 7883227 baseline sweep**: blocks 1-3, frames 0:128, MC-SVD energy-frac sweep (label-free).
- **ULM 7883227 motion sweeps**: blocks 1-3, frames 0:128, frozen baseline e=0.975, motion kinds brainlike,elastic.
- **ULM 7883227 localization-derived structural audit (published PALA reference)**: reference/eval blocks 1-10, 64-frame windows at offsets 0,128, max 2 windows/block, published PALA localization-derived vessel-core versus perivascular-shell reference, no-registration fully whitened matched-subspace variant. This supersedes the earlier 128-frame local-density surrogate and is the only paper-facing ULM structural endpoint.

ULM structural endpoint note. The only paper-facing ULM structural benchmark in this appendix is the 64-frame same-acquisition structural audit that uses the published PALA localization-derived vascular reference and the no-registration fully whitened matched-subspace variant. Earlier 128-frame local-density surrogate runs were exploratory, are superseded, and do not support any manuscript claim.

Artifact commands. Each entry below lists the paper-facing artifact outputs and their repository entry points. The exact machine-readable command blocks are stored in `repro_manifest.json`.

(R1) **RTX 4080 SUPER audit logs (nvidia-smi + fus-detectors conda env versions)**
(paper refs: Table: latency_summary_4080super). Outputs:
`reports/hw/4080super_nvidia_smi.txt`
`reports/hw/4080super_env.txt`

Notes: These logs are hardware-dependent and are captured once per machine/driver stack. Exact command block: `repro_manifest.json`.

(R2) **RTX 4080 SUPER refactor gates (quick/phase/full)** (paper refs: Reproducibility gates (Phase 7)). Outputs:
`reports/refactor/4080super_refactor_quick.log`
`reports/refactor/4080super_refactor_phase.log`
`reports/refactor/4080super_refactor_full.log`
Notes: The 'full' gate is optional on constrained machines; see Makefile targets. Exact command block: `repro_manifest.json`.

(R3) **PD-mode sanity checks (PD-mode map and score S=PD)** (paper refs: Notation (PD convention)). Outputs:
`reports/pd_mode_sanity/(optionalfigs)`
Notes: Run on any generated acceptance bundle root containing pw_* directories. Exact command block: `repro_manifest.json`.

(R4) **Conditional matched-subspace leakage ablation (full vs conditional masks)** (paper refs: Results (Brain-k-Wave): conditional matched-subspace ablation). Outputs:
`reports/condstap_leakage.csv`
`reports/condstap_leakage.json`
Exact command block: `repro_manifest.json`.

(R5) **CUDA latency replay (Brain-* k-Wave; steady-state windows 2..5)** (paper refs: Table: latency_summary). Outputs:
`runs/latency_s12_publish_offsets/(Brain-OpenSkull;full)`
`runs/latency_s13_publish_offsets_cond/(Brain-OpenSkull;conditional)`
`runs/latency_s14_aliascontract_full/(Brain-AliasContract;full)`
`runs/latency_s14_aliascontract_cond/(Brain-AliasContract;conditional)`
`runs/latency_s15_skullor_full/(Brain-SkullOR;full)`
`runs/latency_s15_skullor_cond/(Brain-SkullOR;conditional)`
Notes: GPU latency is reported as the steady-state mean over windows 2..5 (window1 is cold: CUDA init, Triton JIT, and CUDA-graph capture). Use the script's 'optimized steady(avg win2..5)' line. Exact command block: `repro_manifest.json`.

(R6) **CUDA latency replay (real data: Shin + Gammex; steady-state frames 2..N)** (paper refs: Table: latency_summary). Outputs:
`runs/latency_s17_realdata_cuda_unfold/(Shin+GammexCUDAlatencyruns)`
Notes: The script prints cold(win1) and steady(avg win2..N); the latency table uses the steady-state mean and excludes window/frame 1 (graph capture + Triton JIT). Exact command block: `repro_manifest.json`.

(R7) **RTX 4080 SUPER latency reruns + consolidated summaries (legacy vs optimized; steady-state windows/frames 2..N)** (paper refs: Table: latency_summary_4080super). Outputs:
`reports/hw/4080super_nvidia_smi.txt`
`reports/hw/4080super_env.txt`
`runs/latency_4080super_brain_openskull_w64_tb192_off0_64_128_192_256/`
`runs/latency_4080super_brain_openskull_w64_tb192_off0_64_128_192_256_cond/`
`runs/latency_4080super_brain_aliascontract_w64_tb192_off0_64_128_192_256/`
`runs/latency_4080super_brain_aliascontract_`

`w64_tb192_off0_64_128_192_256_cond/`
`runs/latency_4080super_brain_skullor_w64_tb192_off0_64_128_192_256/`
`runs/latency_4080super_brain_skullor_w64_tb192_off0_64_128_192_256_cond/`
`runs/latency_4080super_shin_fig3_w128_tb192/`
`runs/latency_4080super_gammex_linear17_f0_6_tb512/`
`reports/latency/4080super_latency_summary.csv`
`reports/latency/4080super_latency_summary.json`
Notes: The replay scripts print cold(win1) and steady(avg win2..N); the paper tables report steady-state means. For Brain k-Wave, scripts/latency_rerun_check.py forces legacy vs optimized CUDA-graph settings internally (legacy graph capture off; optimized graph capture on). Exact command block: `repro_manifest.json`.

(R8) **RTX 4080 SUPER CUDA latency replay (Brain-* k-Wave; legacy vs optimized; steady-state windows 2..5)** (paper refs: Table: latency_summary_4080super). Outputs:
`runs/latency_4080super_brain_openskull_w64_tb192_off0_64_128_192_256/`
`runs/latency_4080super_brain_openskull_w64_tb192_off0_64_128_192_256_cond/`
`runs/latency_4080super_brain_aliascontract_w64_tb192_off0_64_128_192_256/`
`runs/latency_4080super_brain_aliascontract_`
`w64_tb192_off0_64_128_192_256_cond/`
`runs/latency_4080super_brain_skullor_w64_tb192_off0_64_128_192_256/`
`runs/latency_4080super_brain_skullor_w64_tb192_off0_64_128_192_256_cond/`
Notes: The script writes both legacy/ and optimized/ subtrees per out-root and reports cold(win1) and steady(avg win2..N). Publishable latency uses windows 2..N. Exact command block: `repro_manifest.json`.

(R9) **RTX 4080 SUPER CUDA latency replay (real data: Shin + Gammex; legacy vs optimized; steady-state frames 2..N)** (paper refs: Table: latency_summary_4080super). Outputs:
`runs/latency_4080super_shin_fig3_w128_tb192/legacy/`
`runs/latency_4080super_shin_fig3_w128_tb192/optimized/`
`runs/latency_4080super_gammex_linear17_f0_6_tb512/legacy/`
`runs/latency_4080super_gammex_linear17_f0_6_tb512/optimized/`
Notes: These are latency-only replay settings with one Tyler iteration and snapshot stride 4 to isolate compute throughput; see per-bundle meta.json for exact telemetry. Exact command block: `repro_manifest.json`.

(R10) **RTX 4080 SUPER latency summary aggregation (CSV/JSON)** (paper refs: Table: latency_summary_4080super). Outputs:
`reports/latency/4080super_latency_summary.csv`
`reports/latency/4080super_latency_summary.json`
Notes: Reads existing runs/latency_4080super_* folders and writes steady-state windows 2..N summaries. Exact command block: `repro_manifest.json`.

(R11) **Brain-* baseline fairness: MC–SVD(e) tune-once sweep** (paper refs: Baseline fairness (Brain-*)). Outputs:
`reports/brain_mcsvd_energy_sweep_seed1.csv`
`reports/brain_mcsvd_energy_sweep_seed1.json`
Exact command block: `repro_manifest.json`.

(R12) **Brain-* low-FPR baseline matrix (vnext; includes adaptive/local SVD baselines)**

Paper refs:
Table: brain_kwave_vnext_baselines
Figure: `brain_kwave_roc_curves.pdf`
Outputs:
`runs/pilot/fair_filter_matrix_pd_r3_localbaselines/(generatedbundles)`
`reports/fair_matrix_vnext_r3_localbaselines.csv`
`reports/fair_matrix_vnext_r3_localbaselines.json`
`reports/companion/brain_kwave_vnext_baselines_table.tex`
Notes: Generates per-window acceptance bundles and writes a vnext-style strict-tail report, then renders the LaTeX table used in the main Brain-* baseline table; this run root is also consumed by the ROC and cross-window calibration scripts. Exact command block: `repro_manifest.json`.

(R13) **Brain-* ROC curve figure (median+IQR over disjoint windows)** (paper refs: Figure: `brain_kwave_roc_curves.pdf`). Outputs:
`figs/paper/brain_kwave_roc_curves.pdf`
Notes: Reads per-window score/mask arrays under the Brain-* pilot run root. Exact command block: `repro_manifest.json`.

(R14) **Brain-* cross-window threshold-transfer audit (matched-subspace detector)** (paper refs: Appendix: cross-window threshold calibration audit (Brain-*)). Outputs:
`reports/brain_crosswindow_calibration.csv`
`reports/brain_crosswindow_calibration_summary.json`
`reports/brain_crosswindow_calibration_table.tex`
Notes: Calibrates thresholds on one 64-frame window's negatives and evaluates on disjoint windows (ordered pairs). Exact command block: `repro_manifest.json`.

(R15) **Brain-* baseline sanity table (strict vs relaxed operating points)** (paper refs: Appendix: baseline sanity checks at relaxed operating points (Brain-*)). Outputs:
`reports/brain_baseline_sanity_relaxed.csv`
`reports/brain_baseline_sanity_relaxed.json`
`reports/brain_baseline_sanity_relaxed_table.tex`
Notes: Reads per-window score/mask arrays under the Brain-* pilot run root. Exact command block: `repro_manifest.json`.

(R16) **Brain-* detector-component ablation (whitening vs band selectivity)** (paper refs: Table: brain_detector_ablation (detector-component ablation)). Outputs:
`runs/pilot/fair_filter_matrix_pd_r3_localbaselines/*_mcsvd_det_whitened_power/(generatedbundles)`
`runs/pilot/fair_filter_matrix_pd_r3_localbaselines/*_mcsvd_det_unwhitened_ratio/(generatedbundles)`
`reports/fair_matrix_detector_ablations.csv`
`reports/fair_matrix_detector_ablations.json`
`reports/brain_detector_ablation.csv`
`reports/brain_detector_ablation.json`
`reports/brain_detector_ablation_table.tex`
Notes: Ablates detector components on the identical MC–SVD residual: whitened total power (no band partition) and unwhitened matched-subspace ratio (R=I). Exact command block: `repro_manifest.json`.

(R17) **Brain-* detector-swap fairness check (matched-subspace scoring on competing residuals)** (paper refs: Table: brain_detector_swap (paired residual detector swap)).
Outputs:
`runs/pilot/fair_filter_matrix_pd_r3_localbaselines/*_hosvd_stap_full/(generatedbundles)`
`runs/pilot/fair_filter_matrix_pd_r3_localbaselines/*_rpca_stap_full/(generatedbundles)`
`reports/fair_matrix_detector_swap.csv`
`reports/fair_matrix_detector_swap.json`
`reports/brain_detector_swap.csv`
`reports/brain_detector_swap.json`
`reports/brain_detector_swap_table.tex`
Notes: Paired-residual comparison: each run generates both PD (score_base.npy) and matched-subspace detector scores (score_stap_preka.npy) on the same RPCA/HOSVD residual for each window. Exact command block: `repro_manifest.json`.

(R18) **Brain-* covariance-training trim ablation (self-training sensitivity)** (paper refs: Results: covariance-training contamination ablation (Brain-*)). Outputs:
`runs/pilot/fair_filter_matrix_pd_r3_localbaselines/*_mcsvd_covtrim_q05/(generatedbundles)`
`reports/fair_matrix_covtrim.csv`
`reports/fair_matrix_covtrim.json`
`reports/brain_cov_train_ablation.csv`
`reports/brain_cov_train_ablation.json`
`reports/brain_cov_train_ablation_table.tex`
Notes: Enables the ablation via replay_stap_from_run.py –stap-cov-trim-q (plumbed through fair_filter_comparison method key stap_covtrim_q05). This excludes the top 5% highest-energy Hankel snapshots per tile when estimating R_i (guard/trim style training contamination check). Exact command block: `repro_manifest.json`.

(R19) **Brain-OpenSkull fixed-profile parameter sensitivity sweep (diag-load, tile size, Lt)** Paper refs:
Appendix: fixed-profile parameter sensitivity (Brain-OpenSkull)
Results: covariance-training contamination ablation (Brain-*)
Outputs:
`runs/pilot/brain_openskull_profile_sensitivity_r1/(generatedbundles)`
`reports/brain_openskull_profile_sensitivity.csv`
`reports/brain_openskull_profile_sensitivity.json`
`figs/paper/brain_openskull_profile_sensitivity.pdf`
Notes: Runs 1D sweeps (vary one parameter at a time) around the frozen Brain-OpenSkull operating profile, reporting within-window strict-tail TPR medians/IQR over five disjoint windows. Sweep ranges are configurable via –diag-loads/–tile-sizes/–lts. Exact command block: `repro_manifest.json`.

(R20) **Brain-* strict-tail collapse visual (representative window; baseline versus matched-subspace detector)** (paper refs: Appendix: baseline sanity checks at relaxed operating points (Brain-*)). Outputs:
`figs/paper/brain_tail_collapse_visual.pdf`

Notes: Uses one representative Brain-OpenSkull window to visualize background-tail dominance (baseline PD) and tail reshaping under matched-subspace scoring. Exact command block: `repro_manifest.json`.

(R21) **Shrink-only regularizer falsifiability ablation (matched-subspace-only vs contract vs forced)** (paper refs: Shrink-only regularizer evaluation discipline). Outputs:
`reports/ka_v2_ablation.csv`
`reports/ka_v2_ablation.json`
Exact command block: `repro_manifest.json`.

(R22) **Shrink-only regularizer positive-control ablation (OpenSkull shallow-alias + high-rank ghosts)** (paper refs: Shrink-only regularizer evaluation discipline (positive control)). Outputs:
`reports/ka_v2_ablation_openskull_shallowalias_e50.csv`
`reports/ka_v2_ablation_openskull_shallowalias_e50.json`
Notes: Synthetic, replay-only positive-control regime used as a contract-mechanics check; not a realism benchmark. Exact command block: `repro_manifest.json`.

(R23) **Mace PD-only holdout alias-gate evaluation (label-free thresholds)** (paper refs: Table: Mace hold-out alias gate). Outputs:
`reports/mace_alias_gate_holdout.csv`
`reports/mace_alias_gate_holdout.json`
Exact command block: `repro_manifest.json`.

(R24) **Mace pixel-level vascular-atlas check (independent structural labels)** (paper refs: Mace section: vascular atlas pixel-level check). Outputs:
`reports/mace_vascular_pixel_eval.csv`
`reports/mace_vascular_pixel_eval.json`
Notes: Maps atlas.Vascular into each scan plane using Transformation.mat; evaluates vascular vs non-vascular pixels (structural, not clinical efficacy). Exact command block: `repro_manifest.json`.

(R25) **Mace atlas-alignment sanity overlay figure** (paper refs: Appendix: Mace atlas alignment sanity check). Outputs:
`figs/paper/mace_atlas_overlay.png`
Notes: Overlays atlas ROIs and atlas.Vascular contours on the anatomical reference volume for representative planes (sanity check of Transformation.mat alignment). Exact command block: `repro_manifest.json`.

(R26) **Telemetry regime comparison (sim vs real; contract-v2 scalars)** (paper refs: Figure: `telemetry_regime_compare.png`). Outputs:
`figs/paper/telemetry_regime_compare.png`
Notes: Overlays PfPeakFrac(flow), guard_q90, and iqr_alias_bg across Brain-* simulations, Shin real IQ, and Mace PD-only planes; intended as telemetry anchoring (not a claim that sims match real data). Exact command block: `repro_manifest.json`.

(R27) **Mace PD-only prior dashboard (plane sweep + paper figure)** (paper refs: Figure: `mace_pdonly_contract_v2_dashboard.png`). Outputs:
`reports/mace_pdonly_contract_v2.csv`
`figs/paper/mace_pdonly_contract_v2_dashboard.png`
Exact command block: `repro_manifest.json`.

(R28) **Shin all-clips prior-regime telemetry (n=80; telemetry-only)** (paper refs: Shin

section: all-clips telemetry audit). Outputs:
`reports/shin_ratbrain_shinU_e970_telemetry_only_0_128_all80.csv`
`reports/shin_ratbrain_shinU_e970_telemetry_only_0_128_all80_summary.json`
Exact command block: `repro_manifest.json`.

(R29) **Shin brainlike motion sweep (10 clips) + aggregate plots** Paper refs:
Table: Shin motion endpoints
Figure: `shin_motion_brainlike_batch_U_agg.png`
Figure: `shin_motion_brainlike_compare_reg1.png`
Figure: `shin_motion_brainlike_compare.png`
Outputs:
`reports/shin_motion_brainlike_batch_U/*.csv`
`reports/shin_motion_brainlike_batch_U_agg.csv`
`figs/paper/shin_motion_brainlike_batch_U_agg.png`
`figs/paper/shin_motion_brainlike_compare_reg1.png`
`figs/paper/shin_motion_brainlike_compare.png`
Notes: Window-specific batch runs (64:192, 122:250) follow the same pattern. Exact command block: `repro_manifest.json`.

(R30) **Shin motion crop/alignment sensitivity (post-hoc)** (paper refs: Shin motion section: crop sensitivity note). Outputs:
`reports/shin_motion_brainlike_batch_U_crop_sensitivity.csv`
Exact command block: `repro_manifest.json`.

(R31) **Leading structural-fidelity figure (Gammex phantom; matched-FPR decision differences)** Paper refs:
Introduction: worked Gammex structural-fidelity example at matched 1e-2 background FPR
Twinkling section: representative structural-fidelity panel
Outputs:
`figs/paper/leading_structural_fidelity_gammex.pdf`
Notes: Generates a single-frame qualitative panel using the structurally labeled tube mask; the script writes a temporary one-frame acceptance bundle under runs/_tmp_leading_structural_fidelity/.
Exact command block: `repro_manifest.json`.

(R32) **Twinkling decode sanity (RawBCF -¿ B-mode/CFM)** (paper refs: Twinkling section: RawBCF decode sanity). Outputs:
`reports/twinkling_decode_sanity/*/decode_report.json`
Exact command block: `repro_manifest.json`.

(R33) **Gammex flow phantom structural ROC (along-linear17, PRF=2500)** Paper refs:
Twinkling section: Figure (structural evaluation masks)
Figure: `twinkling_gammex_along_mask_overlay.png`
Outputs:
`runs/real/twinkling_gammex_alonglinear17_prf2500_str6_msd_ka/`
`reports/twinkling_gammex_alonglinear17_prf2500_str6_msd_ka_with_baselines.csv`
`reports/twinkling_gammex_alonglinear17_prf2500_str6_msd_ka_with_baselines_summary.json`
`figs/paper/twinkling_gammex_along_mask_overlay.png`
Notes: Along-linear uses stride 6; this matches the fixed geometry-only stride policy (largest

stride with ¿=500 tiles). Exact command block: `repro_manifest.json`.

(R34) **Gammex flow phantom structural ROC (across-linear17, PRF=2500)** Paper refs: Twinkling section: Figure (structural evaluation masks)
Figure: `twinkling_gammex_across_mask_overlay.png`
Outputs:
`runs/real/twinkling_gammex_across17_prf2500_str4_msd_ka/`
`reports/twinkling_gammex_across17_prf2500_str4_msd_ka_with_baselines.csv`
`reports/twinkling_gammex_across17_prf2500_str4_msd_ka_with_baselines_summary.json`
`figs/paper/twinkling_gammex_across_mask_overlay.png`
Notes: Across-linear uses stride 4; this matches the fixed geometry-only stride policy (largest stride with ¿=500 tiles). Exact command block: `repro_manifest.json`.

(R35) **Gammex same-residual detector-component sweep (along/across; no shrink-only regularizer)** (paper refs: Twinkling section: Table (structural ROC)). Outputs:
`runs/real/twinkling_gammex_alonglinear17_prf2500_str6_msd_ratio_fast/`
`runs/real/twinkling_gammex_alonglinear17_prf2500_str6_whitened_power/`
`runs/real/twinkling_gammex_alonglinear17_prf2500_str6_unwhitened_ratio/`
`runs/real/twinkling_gammex_across17_prf2500_str4_msd_ratio_fast/`
`runs/real/twinkling_gammex_across17_prf2500_str4_whitened_power/`
`runs/real/twinkling_gammex_across17_prf2500_str4_unwhitened_ratio/`
`reports/twinkling_gammex_alonglinear17_prf2500_str6_msd_ratio_fast_structural_summary.json`
`reports/twinkling_gammex_alonglinear17_prf2500_str6_whitened_power_structural_summary.json`
`reports/twinkling_gammex_alonglinear17_prf2500_str6_unwhitened_ratio_structural_summary.json`
`reports/twinkling_gammex_across17_prf2500_str4_msd_ratio_fast_structural_summary.json`
`reports/twinkling_gammex_across17_prf2500_str4_whitened_power_structural_summary.json`
`reports/twinkling_gammex_across17_prf2500_str4_unwhitened_ratio_structural_summary.json`
`reports/twinkling_gammex_detector_ablation.csv`
`reports/twinkling_gammex_detector_ablation.json`
`reports/paper/twinkling_gammex_detector_ablation_table.tex`
Notes: Same-residual fairness audit for the main Gammex table. Detector-variant ablations require the batched fast matched-subspace core; only the detector head changes across rows. Exact command block: `repro_manifest.json`.

(R36) **Gammex within-ensemble motion ladder (along-linear17; PRF=2500)** Paper refs: Twinkling section: Figure (within-ensemble motion ladder)
Figure: `twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_rw_ci_v2.png`
Figure: `twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_step_ci_v2.png`
Outputs:

`runs/real/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_rw_v2/`
`runs/real/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_step_v2/`
`reports/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_rw_ci_v2.json`
`reports/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_step_ci_v2.json`
`figs/paper/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_rw_ci_v2.png`
`figs/paper/twinkling_gammex_alonglinear17_prf2500_within_ensemble_motion_ladder_step_ci_v2.png`
Exact command block: `repro_manifest.json`.

(R37) **Twinkling calculi shrink-only regularizer hygiene + contract telemetry (calcifications; PRF approx 500)** Paper refs:
Twinkling section: shrink-only regularizer hygiene on calculi
Figure: `twinkling_calculi_tail_example.png`
Outputs:
`runs/real/twinkling_calculi_calcifications_prf500_str4_Lt8_msd_ka_scm_dl015_f050/`
`reports/twinkling_calculi_calcifications_prf500_str4_Lt8_msd_ka_scm_dl015_f050_roc.csv`
`reports/twinkling_calculi_calcifications_prf500_str4_Lt8_msd_ka_scm_dl015_f050_summary.json`
Notes: Calculi runs use proxy masks (not tube structural masks). Used for contract/tail-hygiene instrumentation (not clinical detection claims). Exact command block: `repro_manifest.json`.

(R38) **Twinkling contract figure pack (states/reasons + calculi tail example)** Paper refs:
Twinkling section: Figure (twinkling_contract_states_reasons.png)
Twinkling section: Figure (twinkling_calculi_tail_example.png)
Outputs:
`figs/paper/twinkling_contract_states_reasons.png`
`figs/paper/twinkling_calculi_tail_example.png`
`reports/twinkling_contract_figpack.json`
Exact command block: `repro_manifest.json`.

(R39) **ULM 7883227 one-time baseline sweep (MC-SVD energy fraction)** (paper refs:
ULM section: baseline calibration (frozen svd_energy_frac=0.975)). Outputs:
`runs/real/ulm7883227_baseline_mcsvd_energy_sweep_ULM_0000_0128/`
`reports/ulm7883227_baseline_mcsvd_energy_sweep_ULM_0000_0128.csv`
`reports/ulm7883227_baseline_mcsvd_energy_sweep_ULM_0000_0128.json`
Exact command block: `repro_manifest.json`.

(R40) **ULM 7883227 motion sweeps (brainlike + elastic; fairness alignment + degeneracy)** Paper refs:
ULM section: Figure (ulm7883227_motion_sweep_*_brainlike_e975.png)
ULM section: Figure (ulm7883227_motion_sweep_*_elastic_e975.png)

Outputs:
`reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_brainlike_e975.csv`
`reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_brainlike_e975.json`
`figs/paper/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_brainlike_e975.png`
`reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_elastic_e975.csv`
`reports/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_elastic_e975.json`
`figs/paper/ulm7883227_motion_sweep_ULM_blocks1_3_0000_0128_elastic_e975.png`
Exact command block: `repro_manifest.json`.

(R41) **ULM 7883227 localization-derived structural audit (published PALA reference)**
Paper refs:
Front matter: ULM structural-audit hero figure (ulm7883227_pala_headline_hero.pdf)
ULM section: localization-derived structural audit on the same dataset
ULM section: Table (ulm7883227_pala_structural_roc_table.tex)
Supplement: Figure (ulm7883227_pala_structural_roc_curves.pdf)
Discussion: RTX 4080 latency note for the same 64-frame no-registration fully whitened variant
Outputs:
`reports/ulm7883227_pala_structural_roc.csv`
`reports/ulm7883227_pala_structural_roc.json`
`reports/companion/ulm7883227_pala_structural_roc_table.tex`
`reports/ulm7883227_pala_structural_roc_short32.json`
`reports/ulm7883227_pala_structural_roc_brainlike_motion.json`
`reports/ulm7883227_pala_latency_rtx4080.json`
`figs/paper/ulm7883227_pala_headline_hero.pdf`
`figs/paper/ulm7883227_pala_headline_hero.png`
`figs/paper/ulm7883227_pala_structural_masks.pdf`
`figs/paper/ulm7883227_pala_structural_roc_curves.pdf`
`figs/paper/ulm7883227_pala_structural_roc_curves.png`
Notes: This is the only paper-facing ULM structural endpoint: a bounded same-acquisition structural audit on real in vivo IQ using the published PALA localization rendering registered back onto the Zenodo IQ grid. The vessel-core and perivascular-shell masks are clipped to anatomical support before evaluation. It supersedes the earlier 128-frame local-density surrogate audit, which should be ignored for manuscript claims. It is stronger than a label-free audit but still not an independent multimodal task benchmark. The short-window detector itself runs without additional frame-to-frame registration in the frozen headline configuration because re-enabling registration changed the structural ROC only at the fourth decimal place while pushing latency well above the 64-frame acquisition budget. Exact command block: `repro_manifest.json`.

(R42) **Fixed-calibration threshold-transfer audits (SIMUS/PyMUST + PALA-backed ULM)** Paper refs:
Thresholding/evaluation protocol section: fixed-calibration transfer rationale
SIMUS results: Table (simus_fixed_calibration_transfer_table.tex)
ULM section: Table (ulm7883227_pala_fixed_calibration_transfer_table.tex)
Discussion: deployment-thresholding paragraph

Outputs:
`reports/fixed_calibration_transfer.json`
`reports/simus_v2/simus_fixed_calibration_transfer.csv`
`reports/paper/simus_fixed_calibration_transfer_table.tex`
`reports/ulm7883227_pala_fixed_calibration_transfer.csv`
`reports/paper/ulm7883227_pala_fixed_calibration_transfer_table.tex`
Notes: This is the paper's non-retrospective threshold-transfer check. On SIMUS/PyMUST, thresholds are learned once from separate calibration seeds 125–126 and applied unchanged to held-out seeds 127–128. On the bounded PALA-backed ULM endpoint, complementary 5-block calibration banks are used so that thresholds are fixed before evaluation on the held-out blocks. These are deployment-relevant transfer checks, not per-window oracle calibrations. Exact command block: `repro_manifest.json`.

(R43) **SIMUS/PyMUST moving-scatterer sanity link + bundle contract (micro + alias; non-performance claims)** Paper refs:
Simulation spec: moving-scatterer physical Doppler regime (SIMUS)
Appendix (QC): physical-doppler sanity-link telemetry
Outputs:
`reports/physdoppler_sanity_link/simus_micro_paper_`
`seed0_vs_shin_gammex_summary.json`
`reports/physdoppler_sanity_link/simus_micro_paper_`
`seed0_vs_shin_gammex_table.json`
`reports/physdoppler_sanity_link/simus_alias_paper_`
`seed0_vs_shin_gammex_summary.json`
`reports/physdoppler_sanity_link/simus_alias_paper_`
`seed0_vs_shin_gammex_table.json`
`reports/simus_baseline_compare/simus_pymust_paper_seed0_compare.csv`
`reports/simus_baseline_compare/simus_pymust_paper_seed0_compare.json`
`reports/simus_contract/simus_pymust_paper_micro_seed0_hab_contract.txt`
`reports/simus_contract/simus_pymust_paper_alias_seed0_hab_contract.txt`
`runs/sim/simus_pymust_paper_micro_`
`seed0/bundle/simus_pymust_paper_micro_seed0/`
`runs/sim/simus_pymust_paper_alias_`
`seed0/bundle/simus_pymust_paper_alias_seed0/`
`runs/sim_eval/simus_baseline_compare_r1/`
Notes: SIMUS/PyMUST runs are used as a moving-scatterer credibility anchor. Sanity-link summaries compare PSD-band / coherence / low-rank proxies against open real IQ; contract checks verify bundle integrity and regularizer-friendly regime telemetry. No performance claims are made from these comparisons. Exact command block: `repro_manifest.json`.

(R44) **Paper build modes (paper vs supplement vs full)** (paper refs: Build packaging (Phase 7)). Outputs:
`paper/preprint.pdf`
`paper/supplement.pdf`
`paper/methods_companion.pdf`
Exact command block: `repro_manifest.json`.